



Office of Sponsored Projects
and Research Compliance
Award Data Sheet

SUBCONTRACT OUT

☒ YES ☐ NO

Current Action: Basic Award

BASIC INFORMATION

PI Name: Han Xiao

☐ Check if title has changed

CONTACTS

SPARC Contact Stacy Ridling- 6173

Additional Notice Recipients: OCFR ☐ OTT ☐ Property ☐ Others ☐

PROCESSING

Parent Proposal # 23-0381

Supplement Proposal #

Rfund # New

F&A Rate 56.5 %

Pass-through award #

IMPORTANT DATES

Award Notice Received Date: 7/3/23

Official Report Date: 7/3/23

AWARDS

NUMBER	STATUS
Award #1 A24-0032-001	Obligated
Award #2 A24-0032-002	Anticipated
Award #3 A24-0032-003	Anticipated
Award #4 A24-0032-004	Anticipated
Award #5 A24-0032-005	Anticipated

COMPLIANCE

TYPE	PROTOCOL#	DATES	SPENDING RESTRICTION
IACUC	IACUC-23-061	Approved on 5/17/2023 Expires on 5/17/2026	☐ No expenditures until approved
		Approved on Expires on	☐ No expenditures until approved
		Approved on Expires on	☐ No expenditures until approved
		Approved on Expires on	☐ No expenditures until approved

OTHER INFORMATION

☐ CDI/CUI	☐ China Restriction	☐ FFATA Sub	☐ Cost sharing	☐ e-Verify*
☐ Export Control	☐ SBIR/STTR	☐ Other notes		

REPORTS

☐ No changes ☒ Annual Technical (Numbers of days prior to anniversary) 60
☒ Final Technical (Numbers of days after termination) 120

PROCESSING

Completed by SR 7/3/23 Notice Sent CLT 7/5/23

COMMENTS

Award received through 6/30/2024. Sub-out to BCM.

INFORMATION: In accordance with the National Cancer Institute's (NCI's) current fiscal year (FY) funding policies, this award has been issued at 83% of the adjusted requested level*.



Recipient Information

1. Recipient Name

WILLIAM MARSH RICE UNIVERSITY
6100 MAIN ST
HOUSTON, TX 77005

2. Congressional District of Recipient

02

3. Payment System Identifier (ID)

1741109620A1

4. Employer Identification Number (EIN)

741109620

5. Data Universal Numbering System (DUNS)

050299031

6. Recipient's Unique Entity Identifier

K51LECU1G8N3

7. Project Director or Principal Investigator

Han Xiao, PHD (Contact)

hx17@rice.edu

713-348-8239

8. Authorized Official

Ms. Stacy Lynn Ridling

Federal Agency Information

9. Awarding Agency Contact Information

RICKEYA NOEL Franklin

NATIONAL CANCER INSTITUTE
rickeya.franklin@nih.gov
(240) 276-6908

10. Program Official Contact Information

Sudhir B Kondapaka
Program Director
NATIONAL CANCER INSTITUTE
kondapas@mail.nih.gov
240-276-5365

Federal Award Information

11. Award Number

1R01CA277838-01A1

12. Unique Federal Award Identification Number (FAIN)

R01CA277838

13. Statutory Authority

42 USC 241 42 CFR 52

14. Federal Award Project Title

Modulation of Epigenetic Target in the Bone to Treat Breast Cancer Metastasis

15. Assistance Listing Number

93.395

16. Assistance Listing Program Title

Cancer Treatment Research

17. Award Action Type

New Competing

18. Is the Award R&D?

Yes

Summary Federal Award Financial Information

19. Budget Period Start Date 07/03/2023 – End Date 06/30/2024

20. Total Amount of Federal Funds Obligated by this Action

\$666,169

20 a. Direct Cost Amount

\$521,241

20 b. Indirect Cost Amount

\$144,928

21. Authorized Carryover

22. Offset

23. Total Amount of Federal Funds Obligated this budget period

\$666,169

24. Total Approved Cost Sharing or Matching, where applicable

\$0

25. Total Federal and Non-Federal Approved this Budget Period

\$666,169

26. Project Period Start Date 07/03/2023 – End Date 06/30/2028

27. Total Amount of the Federal Award including Approved Cost

\$666,169

Sharing or Matching this Project Period

28. Authorized Treatment of Program Income

Additional Costs

29. Grants Management Officer - Signature

Shane Woodward

30. Remarks

Acceptance of this award, including the "Terms and Conditions," is acknowledged by the recipient when funds are drawn down or otherwise requested from the grant payment system.



RESEARCH
Department of Health and Human Services
National Institutes of Health

NATIONAL CANCER INSTITUTE

Notice of Award



SECTION I – AWARD DATA – 1R01CA277838-01A1

Principal Investigator(s):

Han Xiao (contact), PHD

Xiang Zhang, PHD

Award e-mailed to: osrrice@rice.edu

Dear Authorized Official:

The National Institutes of Health hereby awards a grant in the amount of \$666,169 (see “Award Calculation” in Section I and “Terms and Conditions” in Section III) to WILLIAM MARSH RICE UNIVERSITY in support of the above referenced project. This award is pursuant to the authority of 42 USC 241 42 CFR 52 and is subject to the requirements of this statute and regulation and of other referenced, incorporated or attached terms and conditions.

Acceptance of this award, including the "Terms and Conditions," is acknowledged by the recipient when funds are drawn down or otherwise requested from the grant payment system.

Each publication, press release, or other document about research supported by an NIH award must include an acknowledgment of NIH award support and a disclaimer such as “Research reported in this publication was supported by the National Cancer Institute of the National Institutes of Health under Award Number R01CA277838. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.” Prior to issuing a press release concerning the outcome of this research, please notify the NIH awarding IC in advance to allow for coordination.

Award recipients must promote objectivity in research by establishing standards that provide a reasonable expectation that the design, conduct and reporting of research funded under NIH awards will be free from bias resulting from an Investigator’s Financial Conflict of Interest (FCOI), in accordance with the 2011 revised regulation at 42 CFR Part 50 Subpart F. The Institution shall submit all FCOI reports to the NIH through the eRA Commons FCOI Module. The regulation does not apply to Phase I Small Business Innovative Research (SBIR) and Small Business Technology Transfer (STTR) awards. Consult the NIH website <http://grants.nih.gov/grants/policy/coi/> for a link to the regulation and additional important information.

If you have any questions about this award, please direct questions to the Federal Agency contacts.

Sincerely yours,

Shane Woodward
Grants Management Officer
NATIONAL CANCER INSTITUTE

Additional information follows

Cumulative Award Calculations for this Budget Period (U.S. Dollars)

Salaries and Wages	\$119,668
Fringe Benefits	\$14,142
Personnel Costs (Subtotal)	\$133,810
Materials & Supplies	\$42,367
Travel	\$1,660
Other	\$37,964
Subawards/Consortium/Contractual Costs	\$305,440

Federal Direct Costs	\$521,241
Federal F&A Costs	\$144,928
Approved Budget	\$666,169
Total Amount of Federal Funds Authorized (Federal Share)	\$666,169
TOTAL FEDERAL AWARD AMOUNT	\$666,169

AMOUNT OF THIS ACTION (FEDERAL SHARE)	\$666,169
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SUMMARY TOTALS FOR ALL YEARS (for this Document Number)		
YR	THIS AWARD	CUMULATIVE TOTALS
1	\$666,169	\$666,169
2	\$652,042	\$652,042
3	\$652,042	\$652,042
4	\$652,042	\$652,042
5	\$652,042	\$652,042

Recommended future year total cost support, subject to the availability of funds and satisfactory progress of the project

Fiscal Information:

Payment System Identifier: 1741109620A1
Document Number: RCA277838A
PMS Account Type: P (Subaccount)
Fiscal Year: 2023

IC	CAN	2023	2024	2025	2026	2027
CA	8479567	\$666,169	\$652,042	\$652,042	\$652,042	\$652,042

Recommended future year total cost support, subject to the availability of funds and satisfactory progress of the project

NIH Administrative Data:

PCC: OKBP / **OC:** 41021 / **Released:** Woodward, Shane 06/25/2023
Award Processed: 07/03/2023 12:05:28 AM

SECTION II – PAYMENT/HOTLINE INFORMATION – 1R01CA277838-01A1

For payment and HHS Office of Inspector General Hotline information, see the NIH Home Page at <http://grants.nih.gov/grants/policy/awardconditions.htm>

SECTION III – STANDARD TERMS AND CONDITIONS – 1R01CA277838-01A1

This award is based on the application submitted to, and as approved by, NIH on the above-titled project and is subject to the terms and conditions incorporated either directly or by reference in the following:

- a. The grant program legislation and program regulation cited in this Notice of Award.
- b. Conditions on activities and expenditure of funds in other statutory requirements, such as those included in appropriations acts.
- c. 45 CFR Part 75.
- d. National Policy Requirements and all other requirements described in the NIH Grants Policy Statement, including addenda in effect as of the beginning date of the budget period.
- e. Federal Award Performance Goals: As required by the periodic report in the RPPR or in the final progress report when applicable.
- f. This award notice, INCLUDING THE TERMS AND CONDITIONS CITED BELOW.

(See NIH Home Page at <http://grants.nih.gov/grants/policy/awardconditions.htm> for certain references cited above.)

Research and Development (R&D): All awards issued by the National Institutes of Health (NIH) meet the definition of "Research and Development" at 45 CFR Part§ 75.2. As such, auditees should identify NIH awards as part of the R&D cluster on the Schedule of Expenditures of Federal Awards (SEFA). The auditor should test NIH awards for compliance as instructed in Part V, Clusters of Programs. NIH recognizes that some awards may have another classification for purposes of indirect costs. The auditor is not required to report the disconnect (i.e., the award is classified as R&D for Federal Audit Requirement purposes but non-research for indirect cost rate purposes), unless the auditee is charging indirect costs at a rate other than the rate(s) specified in the award document(s).

An unobligated balance may be carried over into the next budget period without Grants Management Officer prior approval.

This grant is subject to Streamlined Noncompeting Award Procedures (SNAP).

This award is subject to the requirements of 2 CFR Part 25 for institutions to obtain a unique entity identifier (UEI) and maintain an active registration in the System for Award Management (SAM). Should a consortium/subaward be issued under this award, a UEI requirement must be included. See <http://grants.nih.gov/grants/policy/awardconditions.htm> for the full NIH award term implementing this requirement and other additional information.

This award has been assigned the Federal Award Identification Number (FAIN) R01CA277838. Recipients must document the assigned FAIN on each consortium/subaward issued under this award.

Based on the project period start date of this project, this award is likely subject to the Transparency Act subaward and executive compensation reporting requirement of 2 CFR Part 170. There are conditions that may exclude this award; see <http://grants.nih.gov/grants/policy/awardconditions.htm> for additional award applicability information.

In accordance with P.L. 110-161, compliance with the NIH Public Access Policy is now mandatory. For more information, see NOT-OD-08-033 and the Public Access website: <http://publicaccess.nih.gov/>.

Recipients must administer the project in compliance with federal civil rights laws that prohibit discrimination on the basis of race, color, national origin, disability, age, and comply with applicable conscience protections. The recipient will comply with applicable laws that prohibit discrimination on the basis of sex, which includes discrimination on the basis of gender identity, sexual orientation, and pregnancy. Compliance with these laws requires taking reasonable steps to provide meaningful access to persons with limited English proficiency and providing programs that are accessible to and usable by

persons with disabilities. The HHS Office for Civil Rights provides guidance on complying with civil rights laws enforced by HHS. See <https://www.hhs.gov/civil-rights/for-providers/provider-obligations/index.html> and <https://www.hhs.gov/>.

- Recipients of FFA must ensure that their programs are accessible to persons with limited English proficiency. For guidance on meeting the legal obligation to take reasonable steps to ensure meaningful access to programs or activities by limited English proficient individuals, see <https://www.hhs.gov/civil-rights/for-individuals/special-topics/limited-english-proficiency/fact-sheet-guidance/index.html> and <https://www.lep.gov>.
- For information on an institution's specific legal obligations for serving qualified individuals with disabilities, including providing program access, reasonable modifications, and to provide effective communication, see <http://www.hhs.gov/ocr/civilrights/understanding/disability/index.html>.
- HHS funded health and education programs must be administered in an environment free of sexual harassment; see <https://www.hhs.gov/civil-rights/for-individuals/sex-discrimination/index.html>. For information about NIH's commitment to supporting a safe and respectful work environment, who to contact with questions or concerns, and what NIH's expectations are for institutions and the individuals supported on NIH-funded awards, please see <https://grants.nih.gov/grants/policy/harassment.htm>.
- For guidance on administering programs in compliance with applicable federal religious nondiscrimination laws and applicable federal conscience protection and associated anti-discrimination laws, see <https://www.hhs.gov/conscience/conscience-protections/index.html> and <https://www.hhs.gov/conscience/religious-freedom/index.html>.

In accordance with the regulatory requirements provided at 45 CFR 75.113 and Appendix XII to 45 CFR Part 75, recipients that have currently active Federal grants, cooperative agreements, and procurement contracts with cumulative total value greater than \$10,000,000 must report and maintain information in the System for Award Management (SAM) about civil, criminal, and administrative proceedings in connection with the award or performance of a Federal award that reached final disposition within the most recent five-year period. The recipient must also make semiannual disclosures regarding such proceedings. Proceedings information will be made publicly available in the designated integrity and performance system (currently the Federal Awardee Performance and Integrity Information System (FAPIIS)). Full reporting requirements and procedures are found in Appendix XII to 45 CFR Part 75. This term does not apply to NIH fellowships.

Treatment of Program Income:

Additional Costs

SECTION IV – CA SPECIFIC AWARD CONDITIONS – 1R01CA277838-01A1

Clinical Trial Indicator: No

This award does not support any NIH-defined Clinical Trials. See the NIH Grants Policy Statement Section 1.2 for NIH definition of Clinical Trial.

INFORMATION: In accordance with the National Cancer Institute's (NCI's) current fiscal year (FY) funding policies, this award has been issued at 83% of the adjusted requested level*.

*Adjusted requested level: The requested level of support with adjustments made in accordance with the budget narrative in the summary statement and applicable grant policies.

INFORMATION: This award, including the budget and the budget period, has been discussed between Rickeya Franklin of the National Cancer Institute and Aaron Parvis on June 1, 2023.

INFORMATION: To help expedite NCI's response to prior approval requests, please submit requests for change of PD/PI, carryover or no cost extensions electronically through the eRA Commons at [Prior Approval Module](#). All other post award requests should be submitted to NCIGrantsPostAward@nih.gov. All electronically submitted requests will be tracked and forwarded to the appropriate Grants Management personnel.

INFORMATION: This award is contingent upon the following: No individual who receives salary support from this project may receive compensation for more than 12 calendar months (i.e., 100%) total effort from all of their sources of support.

SPREADSHEET SUMMARY

AWARD NUMBER: 1R01CA277838-01A1

INSTITUTION: WILLIAM MARSH RICE UNIVERSITY

Budget	Year 1	Year 2	Year 3	Year 4	Year 5
Salaries and Wages	\$119,668	\$119,668	\$119,668	\$119,668	\$119,668
Fringe Benefits	\$14,142	\$14,142	\$14,142	\$14,142	\$14,142
Personnel Costs (Subtotal)	\$133,810	\$133,810	\$133,810	\$133,810	\$133,810
Materials & Supplies	\$42,367	\$42,367	\$42,367	\$42,367	\$42,367
Travel	\$1,660	\$1,660	\$1,660	\$1,660	\$1,660
Other	\$37,964	\$37,964	\$37,964	\$37,964	\$37,964
Subawards/Consortium/Contractual Costs	\$305,440	\$305,438	\$305,438	\$305,438	\$305,438
TOTAL FEDERAL DC	\$521,241	\$521,239	\$521,239	\$521,239	\$521,239
TOTAL FEDERAL F&A	\$144,928	\$130,803	\$130,803	\$130,803	\$130,803
TOTAL COST	\$666,169	\$652,042	\$652,042	\$652,042	\$652,042

Facilities and Administrative Costs	Year 1	Year 2	Year 3	Year 4	Year 5
F&A Cost Rate 1	56.5%	56.5%	56.5%	56.5%	56.5%
F&A Cost Base 1	\$256,510	\$231,510	\$231,510	\$231,510	\$231,510
F&A Costs 1	\$144,928	\$130,803	\$130,803	\$130,803	\$130,803



IACUC - Institutional Animal Care and Use Committee

EFFECTIVE DATE: May 17, 2023
TO: Han Xiao
FROM: Institutional Animal Care and Use Committee

PROJECT TITLE: IACUC-23-061 Evaluation of the therapeutic effects of ALN-EZH2 inhibitor in breast cancer bone metastasis model
SUBMISSION TYPE: IACUC Initial Submission
ACTION: APPROVED

INITIAL APPROVAL DATE: May 17, 2023
EXPIRATION DATE: May 16, 2026

Dear Dr. Han Xiao,

Your protocol IACUC-23-061 - Evaluation of the therapeutic effects of ALN-EZH2 inhibitor in breast cancer bone metastasis model was reviewed at the convened meeting of the Rice IACUC. Full APPROVAL is granted for this study through May 16, 2026.

The Annual Review for this submission is due two months before May 17, 2024. These deadlines can be found on the [IACUC website](https://compliance.rice.edu).

The De Novo /Triennial Review (reviewed as a new protocol) for this submission is due two months before May 16, 2026. These deadlines can be found on the [IACUC website](https://compliance.rice.edu).

Please note the expiration date listed above. If a protocol is allowed to expire, animals will be transferred to a holding protocol and no work may continue until approval is secured.

Any changes to this protocol must be approved by the IACUC before the changes are instituted.

Please let me know if you have any questions.

Sincerely,
Sponsored Projects and Research Compliance (SPARC)
Rice University
713 348 2342
iacuc@rice.edu
<https://research.rice.edu/compliance/iacuc>

From: eraNotifications@mail.nih.gov
To: osrrice@rice.edu
Subject: NoA: 1R01CA277838-01A1 PI: Han Xiao
Date: Sunday, July 2, 2023 11:06:29 PM
Attachments: [1R01CA277838-01A1-Noa.pdf](#)

Grant Number: 1R01CA277838-01A1
Principal Investigator: Xiao, Han(Contact); Zhang, Xiang
Project Title: Modulation of Epigenetic Target in the Bone to Treat Breast Cancer Metastasis
Institution: RICE UNIVERSITY
CFDA: 93.395
OC: 41021
PCC: 0KBP
Award Issue Date: 07/03/2023
Grants Management Officer: Woodward, Shane
Program Official: Kondapaka, Sudhir
Grants Specialist: Franklin, RICKEYA

This is an automated notification - Please do not reply to this message.

Rice University

14110

Internal Processing Form - Application for Grant, Contract, or Cooperative Agreement

Proposal No: 23-0381	Sponsor Deadline: 11/05/2022 4:00 Central
Parent Project: No Project Assigned	Clinical Trials: No
Proposal Type: New	Electronic: Yes (Cayuse 424)
Instrument Type: Grant	Please identify a primary administrative contact who will manage this project should it be awarded: Pedro Prado
Admin Unit: 24220 - Chemistry	If the person you are searching for is not in the system, as a temporary measure, use your name in this field. Affiliated Center/Institute:
Funding Agency: National Institutes of Health (NIH)	ARRA Funded: No Answer
Sponsor Program Name: Research Project Grant (Parent R01)	Contact Person: Alma Farias-Barba
Opportunity/Sponsor Number: PA-20-185	Contact Phone: 713-348-5864
Prime Agency:	Contact Email: af33@rice.edu

Project Title

Modulation of Epigenetic Target in the Bone to Treat Breast Cancer Metastasis

Key Investigators/Research Team

Han Xiao - Lead Principal Investigator

Sponsored Effort: **0%** Cost Shared Effort: **0%** **Person Months:** **1.2** Allocation of Credit: 100%

Dept No: 24220 Dept: Chemistry

Pedro Prado - Proposal Editor

Sponsored Effort: 0% Cost Shared Effort: 0% Person Months: 0 Allocation of Credit: 0%

Dept No: 24220 Dept: Chemistry

Sherrie Kingston - Proposal Editor

Sponsored Effort: 0% Cost Shared Effort: 0% Person Months: 0 Allocation of Credit: 0%

Dept No: C2400 Dept: Wiess School of Natural Sciences

Budget

Lead PI: Han Xiao	Sponsor: National Institutes of Health (NIH)									
Budget Form Type: Summary										
# of Budget Periods: 5										
Comments:										
	<table border="1"> <thead> <tr> <th>Project Dates:</th> <th>Current Period</th> <th>Total Project</th> </tr> </thead> <tbody> <tr> <td>Start:</td> <td>07/01/23</td> <td>07/01/23</td> </tr> <tr> <td>End:</td> <td>06/30/24</td> <td>06/30/28</td> </tr> </tbody> </table>	Project Dates:	Current Period	Total Project	Start:	07/01/23	07/01/23	End:	06/30/24	06/30/28
Project Dates:	Current Period	Total Project								
Start:	07/01/23	07/01/23								
End:	06/30/24	06/30/28								

Cost Sharing

Does this budget period include funds or contributions in the form of [cost sharing](#) or cash matching, whether mandatory or voluntary, from the Dean, Chair, Vice Provost for Research or a third party? Commitments include academic year time (plus fringe benefits and associated F&A), [F&A waiver/reduction](#), equipment, [third party cost share](#), or other requests.

If yes, complete the appropriate cost share form, obtain the required signatures, and upload the document on the attachment tab to this proposal record/R&A form.

No

F&A Rates

	Current Period	Total Project
F&A Rate (1):	56.500 %	56.500 %
F&A Rate (2):	0.000 %	0.000 %
F&A Rate (3):	0.000 %	0.000 %
Effective Rate:	56.500 %	56.500 %

Budget Categories

	Current Period	Total Project
SPONSOR DIRECT COSTS:	\$628,000	\$3,140,000
BASE for F&A Rate (1):	\$256,510	\$1,182,550
BASE for F&A Rate (2):	\$ 0	\$ 0
BASE for F&A Rate (3):	\$ 0	\$ 0
INDIRECT COSTS (F&A):	\$144,928	\$668,141
FEE:	\$ 0	\$ 0
TOTAL SPONSOR PROPOSED COSTS:	\$772,928	\$3,808,141
Internal Cost Sharing:	\$ 0	\$ 0
Third-Party Cost Sharing:	\$ 0	\$ 0
TOTAL PROJECT COSTS:	\$772,928	\$3,808,141

Additional Resources

Do you need additional resources such as personnel, equipment, additional space or the renovation of existing space that are not included on the proposed budget to conduct this project?

No

No Answer Personnel

No Answer New Space/Space Renovation

No Answer Equipment

Please provide a brief explanation for the additional resources:

No Answer

Conflict of Interest

1 Have any of this project's Investigators (see [Policy 216](#) for the definition) reported a financial interest or activity that reasonably appears to be related to the design, conduct, or reporting of the research to be conducted if this proposal is funded?

No

Regulatory Compliance

Does this research involve **HUMAN SUBJECTS**?

No

Does this research involve **ANIMAL SUBJECTS**?

Yes

Has an IACUC application been submitted for this research?

No

Reason for no IACUC application:~ Please indicate your reason for not submitting an IACUC application: ~ **Not required at proposal submission (JIT)**

Research Materials

If this proposal includes the use of hazardous research materials, please check the appropriate category(ies) and contact the Environmental Health & Safety Office for any required approvals: cavender@rice.edu x4444

Does the proposal involve research with any of the following? (please check all that apply)

No Answer Radioactive Materials, Lasers/X-ray, Controlled Substances, Nanomaterials

Yes Biohazards (viruses, etc.)

Yes Chemical Hazards (poisons, explosives, reagents, flammables, carcinogens, etc.)

Yes rDNA

No Answer	Neurotoxin Hazards (botulinum neurotoxins, botulinum neurotoxin-producing species of Clostridium, or preparations or pharmaceuticals containing botulinum neurotoxins, etc...)
No Answer	Materials provided by the sponsor or any other party

Subcontractors
Baylor College of Medicine BCM

Special Budget Items and Keywords

1. Are there administrative and clerical items in your budget?

If Yes, provide justification per OMB A-21 and attach to the Attachments tab.

No

No Answer In State

No Answer Outside State, but within US

No Answer Internationally

2. Does this proposal involve fabricating any equipment, including prototypes, costing \$5,000 or more?

No

No Answer In State

No Answer Outside State, but within US

No Answer Internationally

3. Do you anticipate this project to generate additional revenue?

No

No Answer In State

No Answer Outside State, but within US

No Answer Internationally

4. Will any part of the project be conducted off campus?

No

No Answer In State

No Answer Outside State, but within US

No Answer Internationally

5. Does this proposed project involve education outreach for grades K-12?

No

No Answer In State

No Answer Outside State, but within US

No Answer Internationally

6. Please select the primary key word applicable to this proposal:

Cancer

7. Please select all other key words applicable to this proposal.

No Answer Nano

Yes Bio

No Answer Enviro

No Answer Energy

No Answer Astro&Space

No Answer None of these

CIP Science Code 00.0060

Please select a [Scientific Discipline](#) that describes the type of research contained in this proposal:

Chemistry (00.0060)

Attachments		
Document Name	Document Size	Document Type
Project SummaryAbstract.pdf	70KB	Abstract/Summary
Xiao_R01 BudgetJustification_Nov 2022.pdf	109KB	Budget Justification
Xiao R01 OVERALL Calculating Budget_Nov 2022.xlsm	506KB	Calculating Budget Template
BCM__rate_agreement_2022.pdf	55KB	Other
BCM_Budget Justification_10.26.22.pdf	86KB	Subcontractor Budget Justification
BCM_RR_Subaward Budget_10.26.22_print.pdf	135KB	Subcontractor Budget
BCM_Rice_LOI_Zhang_66301-A1-signed.pdf	171KB	Subcontractor Letter of Intent
BCM_RICE_Subrecipeint_Questionnaire.pdf	246KB	Subcontractor Letter of Intent
BCM_SOW.pdf	87KB	Subcontractor Scope of Work

PA-20-185_ NIH Research Grant (Parent R01).pdf

547KB

Proposal Announcement Guidelines

Submission Notes

There are no notes added to the proposal

Certification

Investigator Approval

Investigator	Authorizing Person	Authorized On
Xiao, Han	Lead Principal Investigator	10/27/2022

Dean/Chair/Director Authorization

Department	Authorizing Person	Authorized On
Chemistry	Prado, Pedro	10/27/2022
Wiess School of Natural Sciences	Kingston, Sherrie	10/27/2022

APPLICATION FOR FEDERAL ASSISTANCE
SF 424 (R&R)

3. DATE RECEIVED BY STATE		State Application Identifier
1. TYPE OF SUBMISSION*		4.a. Federal Identifier CA277838
☐ Pre-application ☒ Application ☐ Changed/Corrected Application		b. Agency Routing Number NOT-CA-21-101
2. DATE SUBMITTED 2022-11-04	Application Identifier	c. Previous Grants.gov Tracking Number
5. APPLICANT INFORMATION UEI*: K51LECU1G8N3		
Legal Name*: William Marsh Rice University Department: Division: Street1*: 6100 Main Street Street2: City*: Houston County: Harris State*: TX: Texas Province: Country*: USA: UNITED STATES ZIP / Postal Code*: 77005-1827		
Person to be contacted on matters involving this application Prefix: Ms. First Name*: Stacy Middle Name: Lynn Last Name*: Ridling Suffix: Position/Title: Grant Specialist III Street1*: 6100 Main Street Street2: MS-16 City*: Houston County: Harris State*: TX: Texas Province: Country*: USA: UNITED STATES ZIP / Postal Code*: 77005-1827 Phone Number*: 713-348-6173 Fax Number: Email: slr5@rice.edu		
6. EMPLOYER IDENTIFICATION NUMBER (EIN) or (TIN)*		1-741109620-A1
7. TYPE OF APPLICANT*		O: Private Institution of Higher Education
Other (Specify): Small Business Organization Type ☐ Women Owned ☐ Socially and Economically Disadvantaged		
8. TYPE OF APPLICATION*		If Revision, mark appropriate box(es).
☐ New ☒ Resubmission ☐ Renewal ☐ Continuation ☐ Revision		☐ A. Increase Award ☐ B. Decrease Award ☐ C. Increase Duration ☐ D. Decrease Duration ☐ E. Other (<i>specify</i>) :
Is this application being submitted to other agencies?* ☐ Yes ☒ No What other Agencies?		
9. NAME OF FEDERAL AGENCY* National Institutes of Health		10. CATALOG OF FEDERAL DOMESTIC ASSISTANCE NUMBER TITLE:
11. DESCRIPTIVE TITLE OF APPLICANT'S PROJECT* Modulation of Epigenetic Target in the Bone to Treat Breast Cancer Metastasis		
12. PROPOSED PROJECT		13. CONGRESSIONAL DISTRICTS OF APPLICANT
Start Date* Ending Date* 07/01/2023 06/30/2028		TX-002

SF 424 (R&R) APPLICATION FOR FEDERAL ASSISTANCE**Page 2****14. PROJECT DIRECTOR/PRINCIPAL INVESTIGATOR CONTACT INFORMATION**

Prefix: First Name*: Han Middle Name: Last Name*: Xiao Suffix: PhD

Position/Title: Assistant Professor

Organization Name*: William Marsh Rice University

Department: Chemistry

Division: Wiess School of Natural Sciences

Street1*: 6100 Main Street

Street2: MS-602

City*: Houston

County: Harris

State*: TX: Texas

Province:

Country*: USA: UNITED STATES

ZIP / Postal Code*: 77005-1827

Phone Number*: 713-348-8239 Fax Number: Email*: hx17@rice.edu

15. ESTIMATED PROJECT FUNDING

a. Total Federal Funds Requested* \$3,808,140.00

b. Total Non-Federal Funds* \$0.00

c. Total Federal & Non-Federal Funds* \$3,808,140.00

d. Estimated Program Income* \$0.00

16. IS APPLICATION SUBJECT TO REVIEW BY STATE EXECUTIVE ORDER 12372 PROCESS?*

- a. YES ☐ THIS PREAPPLICATION/APPLICATION WAS MADE AVAILABLE TO THE STATE EXECUTIVE ORDER 12372 PROCESS FOR REVIEW ON:
- DATE:
- b. NO ☒ PROGRAM IS NOT COVERED BY E.O. 12372; OR
- ☐ PROGRAM HAS NOT BEEN SELECTED BY STATE FOR REVIEW

17. By signing this application, I certify (1) to the statements contained in the list of certifications* and (2) that the statements herein are true, complete and accurate to the best of my knowledge. I also provide the required assurances * and agree to comply with any resulting terms if I accept an award. I am aware that any false, fictitious, or fraudulent statements or claims may subject me to criminal, civil, or administrative penalties. (U.S. Code, Title 18, Section 1001)

☒ I agree*

* The list of certifications and assurances, or an Internet site where you may obtain this list, is contained in the announcement or agency specific instructions.

18. SFLL or OTHER EXPLANATORY DOCUMENTATION

File Name:

19. AUTHORIZED REPRESENTATIVE

Prefix: Mr. First Name*: Aaron Middle Name: David Last Name*: Parvis Suffix:

Position/Title*: Associate Director of Grants and Contracts

Organization Name*: William Marsh Rice University

Department: Sponsored Projects and Research Compliance

Division: Office of Research

Street1*: 6100 Main Street

Street2: MS-16

City*: Houston

County: Harris

State*: TX: Texas

Province:

Country*: USA: UNITED STATES

ZIP / Postal Code*: 77005-1827

Phone Number*: 713-348-5584 Fax Number: Email*: adp5@rice.edu

Signature of Authorized Representative*

Aaron David Parvis

Date Signed*

11/04/2022

20. PRE-APPLICATION File Name:**21. COVER LETTER ATTACHMENT** File Name:

424 R&R and PHS-398 Specific

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Project/Performance Site Location(s)**Project/Performance Site Primary Location**

☐ I am submitting an application as an individual, and not on behalf of a company, state, local or tribal government, academia, or other type of organization.

Organization Name: William Marsh Rice University
UEI: K51LECU1G8N3
Street1*: 6100 Main Street
Street2:
City*: Houston
County: Harris
State*: TX: Texas
Province:
Country*: USA: UNITED STATES
Zip / Postal Code*: 77005-1827
Project/Performance Site Congressional District*: TX-002

Project/Performance Site Location 1

☐ I am submitting an application as an individual, and not on behalf of a company, state, local or tribal government, academia, or other type of organization.

Organization Name: Baylor College of Medicine
UEI: FXKMA43NTV21
Street1*: One Baylor Plaza
Street2:
City*: Houston
County:
State*: TX: Texas
Province:
Country*: USA: UNITED STATES
Zip / Postal Code*: 77030-3144
Project/Performance Site Congressional District*: TX-009

Additional Location(s)

File Name:

RESEARCH & RELATED Other Project Information

1. Are Human Subjects Involved?* ☐ Yes ☒ No	
1.a. If YES to Human Subjects Is the Project Exempt from Federal regulations? ☐ Yes ☐ No If YES, check appropriate exemption number: — 1 — 2 — 3 — 4 — 5 — 6 — 7 — 8 If NO, is the IRB review Pending? ☐ Yes ☐ No IRB Approval Date: Human Subject Assurance Number	
2. Are Vertebrate Animals Used?* ☒ Yes ☐ No	
2.a. If YES to Vertebrate Animals Is the IACUC review Pending? ☒ Yes ☐ No IACUC Approval Date: Animal Welfare Assurance Number D16-00005	
3. Is proprietary/privileged information included in the application?* ☐ Yes ☒ No	
4.a. Does this project have an actual or potential impact - positive or negative - on the environment?* ☐ Yes ☒ No	
4.b. If yes, please explain: 4.c. If this project has an actual or potential impact on the environment, has an exemption been authorized or an environmental assessment (EA) or environmental impact statement (EIS) been performed? ☐ Yes ☐ No 4.d. If yes, please explain:	
5. Is the research performance site designated, or eligible to be designated, as a historic place?* ☐ Yes ☒ No	
5.a. If yes, please explain:	
6. Does this project involve activities outside the United States or partnership with international collaborators?* ☐ Yes ☒ No	
6.a. If yes, identify countries: 6.b. Optional Explanation:	
7. Project Summary/Abstract*	Filename Project_Summary1018160153.pdf
8. Project Narrative*	Project_Narrative1018160154.pdf
9. Bibliography & References Cited	Bibliography1018160155.pdf
10. Facilities & Other Resources	Facilities_Resources1018160156.pdf
11. Equipment	Equipment1018160157.pdf

PROJECT SUMMARY/ABSTRACT

In the clinic, primary breast tumors are usually surgically removed soon after diagnosis, often leaving patients “tumor-free”. However, 20-40% of breast cancer survivors will eventually suffer metastasis to distant organs, sometimes years after surgeries. Bone metastasis is the most frequently occurring metastasis of breast cancer. Our long-term goals are to elucidate the biology underlying the survival and progression of bone metastases, which will inform the design of therapeutic strategies against these latent tumor cells, through a fruitful collaboration between labs at Rice University and Baylor College of Medicine. The overall goal of this proposal is to develop bone tumor-targeting epigenetic inhibitors and demonstrate their efficacy against bone micrometastases as well as further multi-organ metastases seeding from bone lesions. Our preliminary data establish that: (1) epigenetic inhibitors modified with bisphosphonates, have superb binding affinity for bone, and exhibit enhanced bone metastasis sites targeting *in vivo*; and (2) epigenetic inhibitors modified with bisphosphonates exhibited improved activities for inhibiting metastatic seeding from bone lesions to other organs; and (3) an intra-iliac artery (IIA) injection, developed in our lab, can be used to effectively model the bone metastatic niche, providing a powerful platform for evaluating the therapeutic efficacy against bone metastatic cancers and “metastasis-to-metastasis” seeding from bone lesions. Based on these results, the first research direction will focus on engineering current epigenetic inhibitors for breast cancer therapy with bone-homing moiety. Next, we will study their effects on the survival and progression of breast cancer bone metastases using both syngeneic and xenograft nude mouse models. Furthermore, we will dissect the molecular mechanisms underlying the benefits of bone tumor-targeting epigenetic inhibitors. An enhanced therapeutic profile for these bone-specific epigenetic inhibitors on breast cancer will inform the extension of these treatments to other bone cancers and diseases.

PROJECT NARRATIVE

In this proposal, we will develop a general platform to specifically deliver epigenetic inhibitors to the bone and demonstrate their utilities for the treatment of skeletal micrometastases of breast cancer. These active bone-targeted epigenetic inhibitors will afford a promising avenue for the treatment and prevention of bone metastases and multi-organ metastases from the bone lesion site.

FACILITIES AND OTHER RESOURCES

Xiao Lab

Dr. Xiao's laboratories and office are located in the BioScience Research Collaborative (BRC) of Rice University. The BRC is a 10 story, 477,000-square-foot, building where scientists and educators from Rice University and the neighboring Texas Medical Center institutions work collaboratively to perform leading research that benefits human medicine and health. Thoughtfully designed to facilitate and encourage interdisciplinary interactions among interinstitutional researchers, the BRC is equipped for cutting-edge laboratory, theoretical and computational investigations. Research encompasses a wide range of disciplines from chemistry to bioengineering and focuses largely on improving human wellness through science. The BRC is located immediately adjacent to the Texas Medical Center, the largest center of its kind in the world. The TMC encompasses the University of Texas Health Science Center at Houston, the M.D. Anderson Cancer Center, the Baylor College of Medicine, the Texas Children's Hospital, the Houston Methodist Hospital, the Methodist Research Institute, and the Texas A&M University Institute for BioTechnology, among others. More than just a building, the BRC is a catalyst for new and better ways for researchers to collaborate, explore, learn and lead.

Dr. Xiao's laboratory in the Department of Chemistry comprises bench space and fume hoods for 15 researchers and equipment (3000 sq. ft.). The labs are fully equipped for organic synthesis, chemical biology, molecular and cell biology.

- One 2700 square foot lab including four 4-ft and two 6-ft fume hoods, and benchtop space for fifteen people to work simultaneously.
- One 150 square foot cell culture lab (BSL2).
- One 150 square foot instrument lab including Agilent LC-MS, GE Amersham 600 RGB Imager, UV transilluminator, Infinite M PLEX Fluorescence/UV-Vis/Luminescence Plate Reader, AKTA Pure FPLC system, and AKTA Starter FPLC system.
- The PI's office space is 130 sq. ft. (BRC317) and is equipped with a computer. Students and postdocs have desk pods and also a separate computer room (BRC311, 180 sq. ft.) with desks and computers for reading and writing documents. The computers have access to all of Rice University's online journals and electronic literature searching tools (Scifinder, Web of Science, etc.). A scanner, copier and fax machine are also available.

Zhang Lab

Dr. Zhang's research space is located on the 11th floor of the Alkek Research Building at the Baylor College of Medicine. His facilities (approximately 1000 square feet) consist of 6 laboratory modules (with space for 12 personnel) and two tissue culture suites. Dr. Zhang's area is fully equipped for general molecular biology and cell biology experiments. Dr. Zhang has a 120 sq ft office located adjacent to his lab space.

Equipment that Dr. Zhang owns includes two thermal cyclers, hybridization ovens, incubators, DNA and protein electrophoresis units, 1 fume hood, 2 -20° and 1 -80° freezers, 2 4° C refrigerators, two analytical balances, a freeze dryer, CO₂ and O₂ incubators, a self-filling liquid nitrogen tank, 9 personal computers, and 2 laser printers available to project personnel. Dr. Zhang's laboratory is also equipped with centrifuges, electrophoresis equipment, microscopes, and tissue culture hoods and incubators. Available common equipment situated on the same floor (11th) includes a cold room, shaking 37° C incubator, bacterial incubator, a NanoVue spectrophotometer (GE Health Science), sonicator, gel dryers, spectrophotometer, ultracentrifuge and high-speed centrifuges, X-ray developer, liquid nitrogen tanks, scintillation counter, phosphoimaging station, inverted fluorescent microscopes, dissecting-microscopy, and fluorescent confocal microscope. Dr. Zhang also has access to facilities shared within the Breast Center (between 11th and 12th floors), including microarray analysis equipment, and access to two ABI 7500 instruments for quantitative PCR. A newly purchased RareCyte equipment specialized for detection and isolation of CTCs is located in an adjacent room of Dr. Zhang's laboratory and full accessible to Zhang Lab.

Specifically related to this project, Dr. Zhang's laboratory has access to an IVIS Lumina II bioluminescence imager (located within to his mouse room).

The investigators have access to personal computers and internet access to all relevant databases including PubMed, GenBank, TCGA Data Portal, Oncomine, and numerous others. All project computers and printers are networked. There are two computers, a Dell Optiplex990 and a Macbook Pro, in Dr. Zhang's office. In his laboratory, there are seven Dell computers for lab members. Software installed on these computers includes Microsoft Office, Adobe Photoshop and Acrobat 9 Pro, FlowJo, Living Image (for Bioluminescence quantification), Prism GraphPad, and R. Dr. Zhang's laboratory also has access to a super computer cluster and is comprised of hundreds of CPUs (see Biostatistics and Informatics Core).

Research Support from Core Facilities.

Rice University established the Shared Equipment Authority (SEA) in 2001 as a way to provide its faculty superb experimental facilities, research equipment, and support services at an affordable cost. The SEA also provides administrative support for the Electron Microscopy Center, the Biomaterials Laboratory, and the Earth Science Electron Probe Microanalyzer laboratory.

The SEA's management structure is an innovative one. The Authority is governed by an active group of faculty who are committed to the principles of shared infrastructure and fiscal responsibility. These faculty oversee and manage more than 85 pieces of highly sophisticated research equipment, develop administrative policies governing the operation of this equipment, and set schedules and fees for use of it. Rice's research infrastructure includes the following categories of shared equipment:

The Rice SEA supports more than 65 instruments, including:

- **NMR:** Varian 500, 600 and 800 MHz NMRs. All probes are triple resonance (^1H , ^{13}C , ^{15}N) with deuterium decoupling available. The 500 MHz probe is also capable of pulsing ^{31}P . The 600 MHz and 800 MHz probes are Cold Probes with increased sensitivity for looking at dilute samples and running experiments with a weak signal. The carbon channel of the 600 MHz Cold Probe is also cooled to 25K, making it very sensitive for carbon-detected experiments. All instruments have 40 channel shim coils combined with the high quality of our magnets (Oxford Actively Shielded 500 MHz and 600 MHz, Magnex/Varian actively Shielded 800 MHz) give these instruments exceptional field homogeneity. Off-line software for data processing and analysis and for structural calculations is available and maintained by Rice staff.
- **TEM:** JEOL 2100F with Gatan Image Filter, JEOL 2010 TEM with multi-specimen sample holder, JEOL 2010 TEM with low-dose Cryo mode and high tilt angle, cryo-sample holder and Vitrobot for vitreous ice sample preparation.
- **Mass Spectrometry:** Thermo LTQ-Orbitrap, Bruker Micro-ToF LC-MS, Bruker MALDI- TOF BiFlex, Bruker Reflex IV, Bruker AutoFlex II ToF/ToF.
- **Spectroscopy:** Varian Cary 5000 UV-Vis-NIR spectrophotometer, Renishaw inVia Raman Microscope/Spectrometer, Fluorimeter.
- **Microscopy:** Nikon A1-Rsi Confocal Microscope with 5 highly sensitive PMT (4-fluorescences and 1-transmitted light) detectors and a 32-channel PMT spectral detector for 400-750nm bandwidth. The system allows increased light detection efficiency and has a standard high-resolution confocal galvano mode (up to 4096 x 4096 pixels) as well as a high-speed resonant mode capable of scanning 420 fps at 512 x 32 pixels (30fps at 512x512 pixels).
- **Vivarium:** The Rice University vivarium within the BRC utilizes 8,500 sq ft of space for animal housing and research and handling support areas. Professionally certified caretakers see to the physical and emotional well-being of animals within the vivarium. Scientists are available to assist with project design and laboratory animal methodologies. Species that can be housed include mice (320 cages), rats (180 cages), and rabbits (72 cages).

- **Other:** Zeiss LSM 510 Confocal laser scanning microscope, ZetaPALS dynamic light scattering, TA instruments AR 1000 Rheometer, Digital Instruments NanoScope III Atomic force microscope, Beckman Multisizer 3 Coulter Counter, Applied Biosystems ABI Prism 7000 RT-PCR.

These instruments are overseen by dedicated experts, most of whom have earned a Ph.D. These individuals are responsible for training users and maintaining the equipment in excellent running order. They are also available to help plan and carry out experiments that go beyond routine analysis.

Baylor College of Medicine has multiple school-wide cores and facilities that provide state-of-the-art instrumentation and technological support for BCM researchers. These cores receive a school subsidy and are cost-effective for investigators. In addition, there are a number of core facilities specifically developed and supported by the NCI-designated Dan L. Duncan Comprehensive Cancer Center (DLCCCC) to enhance cancer research and reduce costs for BCM faculty participating in the Center.

There are nine Shared Resources within the DLCCCC: Clinical Trials Support Unit, Biostatistics and Informatics Resource, Cell and Vector Production, Genetically Engineered Mouse, Integrated Microscopy, Flow Cytometry, Human Tissue Acquisition and Pathology, Protein Expression and Proteomics, and Molecular Profiling. This Core offers arrays for mouse and human gene expression (to facilitate comparative oncogenomics), cDNA or Affymetrix arrays, custom arrays, Next Generation sequencing, genomic copy number, rearrangements, SNP analysis and methylation arrays. There are several additional Baylor-sponsored cores open to all faculty including: the Frensley Imaging Center, the Genetically Engineered Mouse Core (Transgenic Core), the Cell-Based Assay and Screening Service (C-BASS), and the Mouse Phenotyping Core. In addition to these cores, the Center for Comparative Medicine (CCM) provides the highest quality veterinary and animal care and acts as a resource for the faculty on veterinary and scientific issues relating to the use of laboratory animals. The five core resources directly or potentially relevant to the project are:

- **The Cytometry and Cell Sorting Core (CCSC):** CCSC provides training, instrumentation, technical expertise, and software for flow cytometric analysis and cell sorting. This site provides information about specific equipment, services, fees, training, methods, and resources. CCSC provides full access to the following instruments: BD FACSAria, FACSAriaII, and FACSAriaIII Cell Sorting Flow Cytometer (one of each); BD LSRII Analyzers, BD FACSCanto II Analyzers and BD LSRFortessa Analyzer.
- **Animal Facilities:** The BCM animal facilities are under the control of the Center for Comparative Medicine (CCM), John Sharp, PhD, Director, with a superb staff of veterinarians and technicians. The Center for Comparative Medicine (CCM) provides the highest quality veterinary and animal care and acts as a resource for the faculty on veterinary and scientific issues relating to the use of laboratory animals. CCM also provides a wide array of comparative pathology services to further support BCM's research programs. The BCM animal program is fully accredited by the Association for Assessment and Accreditation of Laboratory Animal Care International and is operated in compliance with the Guide for the Care and Use of Laboratory Animals. The total mouse capacity at BCM (seven animal facilities) is approximately 100,000 cages. All mice in these facilities are housed on ventilated racks. Procedure rooms are adjacent to the animal holding rooms. Sentinels are used to monitor colony health and are evaluated on a quarterly basis.
- **Pathology Resource (Breast Research Program Pathology Core):** The core staffed with specially trained, experienced technologists, and is capable of large volume, highly specialized activity. This Core has already developed many of the histologic assays and tissue processing methodologies necessary for the investigators in the proposal. The Core is experienced in both mouse and human pathology. This Pathology Laboratory space is well equipped to conduct modern pathology-related research. All major equipment necessary for the pathology-related aspects of this proposal are located within the laboratory, including: three microtomes (Microm HM-315) with collimator attachments for ultra-precision alignment (Microm GmbH Type Histo-Collimators); two Dako automated immunostainers; VectraTM system (CRI) for automated biomarker quantification; Nuance multi-spectral deconvolution camera; Leica TCS SP5 confocal microscope; Nikon PCM 2000 confocal microscope with Metamorph software; three clinical-grade light microscopes (Olympus BX 40s); a multi-headed light microscope with digital camera (Olympus BX50 with Nikon DXM 1200 camera); Benchmark Ultra automated stainer (Roche/Ventana);

six manual Sequenza immunostaining racks; six slide incubation chambers; a tissue preparation/grossing center; an automated tissue processor (Tissue-Tek VIP); a tissue embedding station (Tissue-Tek TEC); an automated histochemical stainer (Varistain 24-4); two coverslipping hoods; tissue cryobath; cryostat (Leica CM-1850); laser-capture microdissecting microscope (Arcturus XT™ Microdissection System); It also has the instrument to create high density tissue micro-arrays (Beecher Instruments).

- Integrated Microscopy Core.** IMC provides BCM research labs access to standard brightfield and widefield microscopy, deconvolution microscopy, confocal microscopy, high throughput microscopy, and transmission electron microscopy (TEM). Our instruments include a Nikon Ci-L Brightfield upright microscope with a color camera for imaging histology sections, a Nikon TE-2000 inverted epifluorescence microscope for basic fluorescence microscopy imaging, a GE Healthcare DVLIVE Deconvolution microscope with live cell and high throughput capabilities, a Nikon A1-Rs confocal laser scanning microscope with live cell capability, a spectral detector to allow multiplexing of more than four colors, a resonance scanner for higher speed imaging and new HCA/HCS software/hardware upgrades, and a Vala Sciences IC-200 high throughput microscope (HTM) with live cell capabilities, a temp/gas controlled incubated plate hotel and robotic arm for large, imaging based screens, plus an Hitachi H-7500 transmission electron microscope (TEM). Important accessory equipment includes a Biomek NK span 8 with gripper and a Biotek plate washer for automated compound treatment and multiwell plate processing for immunofluorescence or mRNA FISH, and a Biomek NK 96 pin robot for larger scale experiments; importantly, these robots are identical to systems in the GCC CG for compatibility during larger screens. The AMIAC also has a dedicated cell culture facility for live cell imaging experiments, including automated dispensing of cells into multiwell plates (Titretek). Thick and ultrathin sectioning of blocks for TEM is performed with RMC or Leica ultramicrotomes and several diamond knives. Furthermore, we just received a uniquely-powerful and important new one-of-a-kind commercial platform capable of high throughput imaging using a new low magnification (20x) automated slide or plate scanner (StellarVision™) that produces high resolution images equivalent to standard 100x/1.3 NA microscopes (Biosearch Technologies). For large scale imaging experiments, the AMIAC will use GCC CG's IN Cell 6000, an industry-leading high throughput confocal HTM platform for drug screening that will be essential for some BICA-based studies in Aims 2 and 3. The IN Cell 6000 uses a laser-based light engine combined with a scientific grade large format CMOS camera for capturing low noise images. Increased image quality is achieved via a novel rolling adjustable pinhole system allowing for varying degrees of confocality. These features combine to allow for the 3D imaging of complex samples at resolutions ranging from 4X to 40X magnification.
- Patient-Derived Xenograft and Advanced In Vivo Models Core:** Led by Dr. Michael Lewis and Dr. Andy Sikora, a primary focus of the Core is to develop, and provide to the BCM PDX community, computational and bioinformatics infrastructure to support large scale generation, characterization, and use of PDX and CAM-PDX models from a variety of organ sites, including breast, head and neck, pancreas, brain, etc. as well as other cancer types of interest to BCM investigators including pediatric cancers. The Core will also provide expertise in transplantation and animal handling to those wishing to generate PDX from various cancer types. Finally, the Core will coordinate, and assist with, the evaluation of experimental therapeutics using the PDX and CAM-PDX in vivo platforms in conjunction with those investigators maintaining PDX collections for each organ/disease type. All services will be provided on a chargeback basis. As cancer center member, Dr. Zhang will have full access to the service provided by the Core.

EQUIPMENT

Xiao Lab

Important for the execution of this proposal, the Xiao lab is fully equipped for organic synthesis, chemical biology, molecular and cell biology.

- One 1000 square foot chemistry room, containing Four 4-ft and two 6-ft fume hoods, Buchi Rotavapor R-100, and Biotage Isolera One Flash System enables us to synthesize all the organic chemicals, including noncanonical amino acids, metabolite, small molecule drugs, discussed in the proposal.
- One 780 square foot bio room (BSL1), including two ThermoFisher microbiological incubators, two Thermo MaxQ HP 480 refrigerated incubated floor model shakers, one Sorvall Lynx 4000 refrigerated centrifuge, one Sorvall RC6 refrigerated centrifuge, one benchtop Sorvall ST40R refrigerated centrifuge, two BioRad PCR Thermal Cyclers, AKTA Pure FPLC system, and AKTA Starter FPLC system enables us to perform the proposed biochemistry and molecular biology experiments.
- One 150 square foot cell culture lab (BSL2), including one 4-ft and one 6-ft laminar flow biosafety cabinets, three Forma Steri-Cycle CO₂ incubators, bench top Sorvall ST40 centrifuge, Nikon TMS microscope, and an Olympus BX51 fluorescent microscope enables us to perform the proposed cell biology experiments.
- One 150 square foot instrument lab, containing Agilent LC-MS, GE Amersham 600 RGB Imager, UV transilluminator, Infinite M PLEX Fluorescence/UV-Vis/Luminescence Plate Reader, will facilitate the analysis of compounds and biomolecules prepared.

Other existing major equipment in the Xiao lab includes: Rayonet Photoreactor, ThermoFisher Precision oven, Vacuum Drying box, ThermoScientific VARIOMAG biosystems, STRATAGENE UV Stratalinker 2400, Buy-Rite Sterilizer, Eppendorf refrigerated microfuges, needle puller, Thermo Nanodrop, Gene Pulser Electroporator, Microbalance, SK-O-L180 Digital Orbital shakers, Double Door Refrigerator, FisherScientific Isotemp Refrigerator, FisherScientific Isotemp water bath, TSU 400 ULT -80 freezer, Thermo Revco Ultra Plus -80 freezer, and Locator 4 Plus Cryo Storage System.

All the above equipment are available for the performance of this project.

Zhang Lab

Equipment that Dr. Zhang owns includes two thermal cyclers, hybridization ovens, incubators, DNA and protein electrophoresis units, 1 fume hood, 2 -20o and 1 -80o freezers, 2 4o C refrigerators, two analytical balances, a freeze dryer, CO₂ and O₂ incubators, a self-filling liquid nitrogen tank, 9 personal computers, and 2 laser printers available to project personnel. Dr. Zhang's laboratory is also equipped with centrifuges, electrophoresis equipment, microscopes, and tissue culture hoods and incubators. Available common equipment situated on the same floor (11th) includes a cold room, shaking 37oC incubator, bacterial incubator, a NanoVue spectrophotometer (GE Health Science), sonicator, gel dryers, spectrophotometer, ultracentrifuge and high-speed centrifuges, X-ray developer, liquid nitrogen tanks, scintillation counter, phosphoimaging station, inverted fluorescent microscopes, dissecting-microscopy, and fluorescent confocal microscope. Dr. Zhang also has access to facilities shared within the Breast Center (between 11th and 12th floors), including microarray analysis equipment, and access to two ABI 7500 instruments for quantitative PCR.

Specifically related to this project, Dr. Zhang's laboratory has all surgical equipment required including two surgical microscopes, a bead sterilizer, and other microsurgical instruments. In the animal facility (TMF or BCM), Dr. Zhang's laboratory has access to an IVIS Lumina II bioluminescence imager (located adjacent to his mouse room). Dr. Zhang's laboratory also has access to one microtome and one cryostat with CryoJane add-ons (Leica), which is specifically designed for hard tissues such as bones. The Zhang Lab also has access to several confocal and deconvolution microscopes available from the Integrated Microscopy Core as well as Breast Center.

RESEARCH & RELATED Senior/Key Person Profile (Expanded)

PROFILE - Project Director/Principal Investigator			
Prefix:	First Name*: Han	Middle Name	Last Name*: Xiao
	Suffix: PhD		
Position/Title*:	Assistant Professor		
Organization Name*:	William Marsh Rice University		
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Division:	Wiess School of Natural Sciences		
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County:	Harris		
State*:	TX: Texas		
Province:			
Country*:	USA: UNITED STATES		
Zip / Postal Code*:	77005-1827		
Phone Number*: 713-348-8239	Fax Number:		
E-Mail*: hx17@rice.edu			
Credential, e.g., agency login: XIAO1.HAN			
Project Role*: PD/PI		Other Project Role Category:	
Degree Type: PhD		Degree Year: 2015	
Attach Biographical Sketch*:	File Name:	Biosketch_Xiao1018160166.pdf	
Attach Current & Pending Support:	File Name:		

PROFILE - Senior/Key Person				
Prefix:	First Name*: Xiang	Middle Name	Last Name*: Zhang	Suffix: PhD
Position/Title*:	Professor			
Organization Name*:	Baylor College of Medicine			
Department:	Breast Cancer			
Division:				
Street1*:	One Baylor Plaza			
Street2:				
City*:	Houston			
County:				
State*:	TX: Texas			
Province:				
Country*:	USA: UNITED STATES			
Zip / Postal Code*:	77030-3411			
Phone Number*: 713-798-6239		Fax Number:		
E-Mail*: xiangz@bcm.edu				
Credential, e.g., agency login: ZHANGXS1				
Project Role*: PD/PI		Other Project Role Category:		
Degree Type: PhD		Degree Year: 2006		
Attach Biographical Sketch*:	File Name:	Biosketch_Zhang1018160144.pdf		
Attach Current & Pending Support: File Name:				

PROFILE - Senior/Key Person				
Prefix:	First Name*: Tao	Middle Name	Last Name*: Wu	Suffix: PhD
Position/Title*:	Assistant Professor			
Organization Name*:	Baylor College of Medicine			
Department:	Molecular and Human Genetics			
Division:				
Street1*:	One Baylor Plaza			
Street2:				
City*:	Houston			
County:				
State*:	TX: Texas			
Province:				
Country*:	USA: UNITED STATES			
Zip / Postal Code*:	77030-3411			
Phone Number*: 713-798-2805		Fax Number:		
E-Mail*: Tao.Wu@bcm.edu				
Credential, e.g., agency login: TAOWU01				
Project Role*: Co-Investigator		Other Project Role Category:		
Degree Type: PhD		Degree Year: 2009		
Attach Biographical Sketch*:	File Name:	Biosketch_TaoWu1018160146.pdf		
Attach Current & Pending Support: File Name:				

PROFILE - Senior/Key Person				
Prefix:	First Name*: C	Middle Name Kent	Last Name*: Osborne	Suffix: MD
Position/Title*:	Professor			
Organization Name*:	Baylor College of Medicine			
Department:	Medicine			
Division:				
Street1*:	7200 Cambridge St			
Street2:				
City*:	Houston			
County:	Harris			
State*:	TX: Texas			
Province:				
Country*:	USA: UNITED STATES			
Zip / Postal Code*:	77030-4202			
Phone Number*: 713-798-1641		Fax Number:		
E-Mail*: kosborne@bcm.edu				
Credential, e.g., agency login: CKOSBORNE				
Project Role*: Other (Specify)		Other Project Role Category: Other Significant Contributor		
Degree Type: MD		Degree Year: 1972		
Attach Biographical Sketch*:	File Name:	Biosketch_Osborne1018160148.pdf		
Attach Current & Pending Support: File Name:				

PROFILE - Senior/Key Person				
Prefix:	First Name*: Qian	Middle Name	Last Name*: Zhu	Suffix: PhD
Position/Title*:	Assistant Professor			
Organization Name*:	Baylor College of Medicine			
Department:	Lester & Sue Smith Breast Cent			
Division:				
Street1*:	One Baylor Plaza			
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City*:	Houston			
County:	Harris			
State*:	TX: Texas			
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E-Mail*: qian.zhu2@bcm.edu				
Credential, e.g., agency login:				
Project Role*: Other (Specify)		Other Project Role Category: Other Significant Contributor		
Degree Type: PhD		Degree Year: 2016		
Attach Biographical Sketch*:	File Name:	Biosketch_Qian_Zhu1018160150.pdf		
Attach Current & Pending Support: File Name:				

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Han Xiao

eRA COMMONS USER NAME (credential, e.g., agency login): XIAO1.HAN

POSITION TITLE: Assistant Professor

EDUCATION/TRAINING (*Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.*)

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date MM/YYYY	FIELD OF STUDY
University of Science and Technology of China (USTC), P. R. China	B. S.	06/2010	Organic Chemistry
The Scripps Research Institute, La Jolla, CA	Ph.D.	05/2015	Chemical Biology
Stanford University, Stanford, CA	Postdoctoral Fellow	06/2017	Chemical Biology

A. Personal Statement

The goal of our lab is the development and application of methods to site-specifically label polypeptides, e.g., antibodies, in ways that allow us to use the labeled proteins as tools to precisely probe and manipulate biological systems. Central to precisely labeling proteins is our ability to site-specifically install a bioorthogonal reactive moiety into proteins of interest. Toward this end, since the beginning of my Ph.D. work, I have focused on expanding the repertoire of techniques for genetically incorporating noncanonical amino acids (ncAAs) in both prokaryotic and eukaryotic organisms and applying these techniques for biochemical studies and the development of novel antibody therapeutics. Using an enhanced system to encode multiple, distinct ncAAs in eukaryotic organisms, I developed the first well-defined multi-action antibody conjugate. The utility of this method was demonstrated by generating a full-length antibody, site-specifically conjugated to a drug and a fluorophore, and characterizing its activity *in vitro*. During my postdoctoral work, I used an enzymatic antibody labeling strategy to prepare the first Glyco-Immune Checkpoint inhibitor, an antibody-sialidase conjugate that enables blocking of immunoinhibitory receptors/sialic acid interactions by selective desialylation of the tumor cell glycocalyx. One of the patented drugs (E-602) is currently being evaluated in a phase 1/2 clinical trial.

Despite the clinical success of chemotherapy and biotherapy, reaching targets in the bone microenvironment has proved to be difficult due to the relatively low vascularization of bone tissue and the presence of physical barriers that impair drug penetration. My lab has developed innovative solutions for this purpose, including a pClick antibody labeling, bone-targeting (Bontarg) technology. We have demonstrated that therapeutic proteins modified bone-homing moieties will result in higher concentrations in the bone disease niche, thus providing a targeted therapy against different bone diseases. In this proposal, we aim to develop a series of bone-targeting biological regimens by conjugating bisphosphonates to biologic drugs and demonstrate their efficacy in inhibiting the survival and progression of breast cancer bone metastases. To ensure the success of this proposed project, I have established successful collaborations with labs at Baylor College of Medicine, Rice University, and Texas Children Hospital. Taken together, I feel confident that we have the necessary expertise, preliminary data, and scientific environment to successfully complete the proposed studies.

Citations that I would like to highlight include:

1. Tian, Z.[†], Ling, W.[†], Yu, C., Chen, Y., Xu, Z., Bado, I., Lored, A., Wang, L., Wang, H., Wu, K. L., Zhang, W., Zhang, X. *, **Xiao, H.*** (2021) "Harnessing the Power of the Antibodies to Fight Bone Metastasis." Science Advances, DOI: 10.1126/sciadv.abf2051. ([†]contributed equally) PMC8221630

2. Cao, Y. J.[†], Yu, C.[†], Wu, K. L.[†], Wang, X., Tian, Z., Liu, D., Zhao, L., Li, R., Lored, A., *Chung, A., Xiao, H.** (2021) "Synthesis of Precision Antibody Conjugates using Proximity-Induced Chemistry." *Theranostics*, DOI: 10.7150/thno.62444. PMC8419051
3. Tian, Z.[†], Yu, C.[†], Zhang, W., Wu, K. L., Wang, C., Gupta, R., Zhan, X., Wu, L., Chen, Y., Zhang, X., **Xiao, H.*** (2022) "Bone-Specific Enhancement of Antibody Therapy for Breast Cancer Metastasis to Bone." *ACS Central Science*, DOI: 10.1021/acscentsci.1c01024. PMC8961797
4. **Xiao, H.[†]**, Woods, E.C.[†], Vukojicic, P., and Bertozzi, C.R.* (2016). "Prevision Glycocalyx Editing as a Strategy for Cancer Immunotherapy." *Proceedings of the National Academy of Sciences*, 113(37), 10304-10309. PMCID: PMC5027407 ([†]contributed equally)

B. Positions, Scientific Appointments, and Honors

Positions and Scientific Appointments

2018 – present	Assistant Professor of Bioengineering, Rice University, TX
2017 – present	Assistant Professor of Chemistry, Rice University, TX
2017 – present	Assistant Professor of Biosciences, Rice University, TX
2017 – present	CPRIT Scholar in Cancer Research, Rice University, TX
2015 – 2017	Postdoctoral Research Associate, Department of Chemistry, Stanford University, CA
2010 – 2015	Research Associate, The Scripps Research Institute, CA
2007 – 2010	Research Associate, Department of Chemistry, USTC, P. R. China

Honors:

2021	Breast Cancer Research Program (BCRP) Breakthrough Award – Level 2, Department of Defense
2020	Edward S. and Fofo Lewis Award, Rice University
2019	Maximizing Investigators' Research Award for Early Stage Investigators, National Institute of Health
2018	Hamill Innovation Award, Hamill Foundation
2018	John S. Dunn Foundation Collaborative Research Award, Gulf Coast Consortia
2017	Norman Hackerman - Welch Young Investigator Award, Welch Foundation, TX
2017	CPRIT Faculty Recruitment Award, Rice University, TX
2016	Good Ventures Postdoctoral Fellowship, Life Science Research Foundation
2014	Aldrich Alfred R. Bader Award for Student Innovation, Sigma-Aldrich Co. LLC
2013	Outstanding Self-Financed Students Abroad, Ministry of Education of the People's Republic of China
2010	Honors Degree in Physical Science, University of Science and Technology of China

C. Contributions to Science

1. *Bone-targeting biological drugs.* Dosing the bone microenvironment has proven to be difficult due to relatively low vascularization and the physical barriers of penetration. Because of this reason, protein-based therapy that has established great success in the clinic, only yielded suboptimal outcomes for patients with bone diseases. My lab has developed innovative bone-targeting (Bontarg) technology to engineer biological drugs for bone targeting. The resulting bone-targeting therapeutic regimens yield a better therapeutic profile by delivering a high concentration of therapeutic regimens to the bone and reducing adverse side effects associated with systemic delivery. Using this strategy, we have developed the first bone-targeting antibody and demonstrated its efficacy of preventing breast cancer metastases in the bone as well as inhibiting secondary metastases from bone lesions to multi-organs.
 - a. Tian, Z.[†], Ling, W.[†], Yu, C., Chen, Y., Xu, Z., Bado, I., Lored, A., Wang, L., Wang, H., Wu, K. L., Zhang, W., Zhang, X. *, **Xiao, H.*** (2021) "Harnessing the Power of the Antibodies to Fight Bone Metastasis." *Science Advances*, DOI: 10.1126/sciadv.abf2051. ([†]contributed equally) PMC8221630
 - b. Tian, Z.[†], Yu, C.[†], Zhang, W., Wu, K. L., Wang, C., Gupta, R., Zhan, X., Wu, L., Chen, Y., Zhang, X., **Xiao, H.*** (2022) "Bone-Specific Enhancement of Antibody Therapy for Breast Cancer Metastasis to Bone." *ACS Central Science*, DOI: 10.1021/acscentsci.1c01024. PMC8961797

- c. Xiao, H., Zhang, X. "Bone-Specific Delivery of Antibodies." **U. S. Provisional Patent 63/138,972**, issued February 19, 2021
- d. Xiao, H., Yu, C., Tian, Z. "Engineered Antibodies for Bone-Targeted Therapy." **U. S. Provisional Patent 63/223,875**, issued July 20, 2021.

2. *Next-generation antibody conjugation technology.* Antibody drug conjugates and other bioconjugates are emerging as one of the most important reagents in research, disease diagnostics, and therapy. While revolutionary, first generation antibody labeling strategies result in heterogeneous labeling of antibodies, and thus may diminish binding and stability of the antibodies. As antibody conjugates continue to increase shares in academic research and the therapy pipeline, several strategies, including enzymatic treatment, genetic code expansion, and genetically-encoded tags, have been developed to prepare site-specific homogeneous antibody conjugates. While these methods can be used for preparing homogeneous antibody conjugates, additional chemical treatment or antibody engineering is required, which affects antibody folding, stability, or production yield and cost. My lab developed a novel proximity-induced antibody conjugation technology, named pClick, enabling site-specific covalent bond formation between functional moieties and native antibodies without antibody engineering or additional UV/chemical treatment. Using this method, high conjugation efficiency and specificity could be achieved with IgGs from different species and subclasses. In addition, I also developed the first well-defined multi-action antibody conjugate. Introduction of multiple molecules with diverse chemical and biological activities into a single antibody greatly expands the scope of potential applications of antibody conjugates in research and therapy.

- a. **Xiao, H.[†]**, Chatterjee, A.[†], Choi, S. H., Bajjuri, K. M., Sinha, S. C., and Schultz, P. G.* (2013). "Genetic Incorporation of Multiple Unnatural Amino Acids into Proteins in Mammalian Cells." *Angewandte Chemie International Edition*, 52(52), 14080-14083. ([†] contributed equally)
- b. Yu, C.[†], Tang, J.[†], Lored, A., Chen, Y., Jung, S. Y., Jain, A., Gordon, A., and **Xiao, H.*** (2018). "Proximity-Induced Site-Specific Antibody Conjugation." *Bioconjugate Chemistry*, 29(11), 3522-3526. ([†] contributed equally)
- c. Cao, Y., Rodgers, D. T., Du, J., Ahmad, I., Hampton, E. N., Ma, J. S., Mazagova, M., Choi, S. H., Yun, H. Y., **Xiao, H.**, Yang, P., Luo, X., Lim, R. K., Pugh, H. M., Wang, F., Kazane, S. A., Wright, T. M., Kim, C. H.*, Schultz, P. G.*, Young, T. S.* (2016) "Design of Switchable Chimeric Antigen Receptor T cells Targeting Breast Cancer." *Angewandte Chemie International Edition*, 55, 7520-7524. PMID: PMC5207029
- d. Cao, Y. J.^{†*}, Yu, C.[†], Wu, K. L.[†], Wang, X., Tian, Z., Liu, D., Zhao, L., Li, R., Lored, A., **Chung, A., Xiao, H.*** (2021) "Synthesis of Precision Antibody Conjugates using Proximity-Induced Chemistry." *Theranostics*, DOI: 10.7150/thno.62444. PMC8419051

3. *Glyco-immune Checkpoints in cancer.* Cell surface sialic acids have emerged as an important immune modulation mechanism that is used by tumor cells to evade both innate and adaptive immunity. Recently, tumor-surface sialosides were found to be able to recruit sialic acid-binding Ig-like lectins (Siglecs), a family of immunosuppressive receptors on both innate cells and activated T cells, to evade immunological recognition. I developed the first Glyco-Immune Checkpoint inhibitor, antibody-sialidase conjugates able to block Siglec/sialic acid interactions by selective desialylation of the tumor cell glycocalyx. The antibody-sialidase conjugates were prepared by modern bioconjugation technologies. The superb anti-tumor efficacy was demonstrated in vitro as well as in a syngeneic orthotopic HER2-positive breast cancer model. Unlike current immunotherapies, which only target one immunomodulatory pathway, glycocalyx editing with antibody-enzyme conjugates affects multiple pathways and has the ability to simultaneously stimulate activating pathways and limit inhibitory pathways, across both the adaptive and innate immune compartments. Therefore, precision glycocalyx editing with antibody-enzyme conjugates is considered to provide a promising avenue for cancer immune therapy.

- a. **Xiao, H.[†]**, Woods, E.C.[†], Vukojicic, P., and Bertozzi, C.R.* (2016). "Precision Glycocalyx Editing as a Strategy for Cancer Immunotherapy." *Proceedings of the National Academy of Sciences*, 113(37), 10304-10309. PMID: PMC5027407 (* contributed equally)
- b. Gray, M. A.[†], Stanczak, M. A.[†], Mantuano, N. R., **Xiao, H.**, Pijnenborg, J. F. A., Malaker, S. A., Weidenbacher, P.A., Miller, C. L., Tanzo, J. T., Ahn, G., Woods, E. C., Läubli, H., Bertozzi, C.* (2020)

"Targeted glycan degradation potentiates the anticancer immune response in vivo." *Nature Chemical Biology*, DOI: 10.1038/s41589-020-0622-x. PMCID: PMC7727925

- c. Woods, E. C., **Xiao, H.**, Bertozzi, C. R. "Inhibitory Immune Receptor Inhibition Methods and Compositions." *PCT Int. Appl.* WO 2018/006066 A1, issued June 30, 2017.
- d. Gray, M., Woods, E. C., **Xiao, H.**, Bertozzi, C. R. "Conjugates for Targeted Cell Surface Editing." *PCT Int. Appl.* WO 2018/006034 A1, issued June 20, 2017.

4. *Unnatural Organism with a 21st amino acid genetic code*. Since 1990, more than 100 noncanonical amino acids have been genetically encoded in various organisms using genetic code expansion. Despite the broad application of this technology, the genetic incorporation of these ncAAs requires the exogenous feeding of chemically-synthesized ncAAs and successful uptake of these ncAAs by cells, which significantly restricts the utility of this approach in protein evolution studies and the preparation of novel protein/cell-based drugs. To address these issues, my group is working on the development of prokaryotic and eukaryotic cells with the endogenous ability to biosynthesize different ncAA building blocks and incorporate them into proteins. A collection of cells able to generate and use additional protein building blocks will make the genetic code expansion technology more accessible to scientists in different fields. Furthermore, immune cells containing a 21st amino acid will revolutionize medicine

- a. Chen, Y.[†], Jin, S., Zhang, M., W, K., *Chang, A.*, Wang, S., Tian, Z., Wolynes, P. G., **Xiao, H.*** (2022) "Unleashing the Potential of Noncanonical Amino Acid Biosynthesis for Creation of Cells with Site-Specific Tyrosine Sulfation." *Nature Communications*, DOI: 10.1038/s41467-022-33111-4. PMCID: PMC9481576
- b. Chen, Y., Tang, J., Wang, L., Tian, Z., Cardenas, A., Fang, X., Chatterjee, A., **Xiao, H.*** (2020) "Creation of Bacterial cells with 5-Hydroxytryptophan as a 21st Amino Acid Building Block." *Chem*, 6, 2717-2727. PMCID: PMC7583639
- c. Chen, Y., Lored, A., Gordon, A., Tang, J., Yu, C., Ordonez, J., **Xiao, H.*** (2018) "A noncanonical Amino Acid-based Relay System for Site-specific Protein Labeling." *Chemical Communications*, 54, 7187-7190.
- d. **Xiao, H.[†]**, Nasertorabi, F.[†], Choi, S. H., Han, G. W., Reed, S. A., Stevens, R. C., Schultz, P. G.* (2015). "Exploring the Potential Impact of an Expanded Genetic Code on Protein Function." *Proceedings of the National Academy of Sciences*, 112(22), 6961-6966. PMCID: PMC4460475 (* contributed equally)

5. *Genetically encoding noncanonical amino acids (ncAAs)*. With the exceptions of pyrrolysine and selenocysteine, all organisms are restricted to the same 20 canonical amino acids. The ability to site-specifically incorporate ncAAs into proteins in living cells affords a powerful tool to investigate and manipulate protein structure and function. To this end, I developed a general rapid *in vitro* directed evolution for novel functional proteins with ncAAs. ncAA-containing proteins exhibiting novel functions or enhanced thermostability have been evolved using this system, providing evidence that additional amino acids can provide novel solutions to evolutionary challenges. ncAA-dependent enzyme evolved by this system can be introduced to various organisms, which can provide a promising biocontainment mechanism to prevent the unintended proliferation of genetically modified organisms. Furthermore, replacement of essential enzymes in organisms with evolved ncAA-dependent enzymes will afford a better alternative for autotrophic vaccines.

- a. Wu, K. L., Moore, J. A., Chen, Y., *Lee, C.*, Xu, D., Miller, M. D., Peng, Z., Duan, Q., Philips, G. N., Uribe, R. A., **Xiao, H.*** (2022) "Expanding the Eukaryotic Genetic Code with a Biosynthesized 21st Amino Acid." *Protein Science*, DOI: 10.1002/PRO.4443. PMCID: PMC9601876
- b. Chen, Y.[†], Wu, K. L.[†], Pei, J., Tang, J., Lored, A., Peng, Z., Gupta, R., **Xiao, H.*** (2019) "Addition of Isocyanide-containing Amino Acids to the Genetic Code for Protein Labeling and Activation." *ACS Chemical Biology*, 14, 2793-2799. PMCID: PMC6925311 (* contributed equally)
- c. Chen, Y., Lored, A., Chang, A., Zhang, M., Liu, R., **Xiao, H.*** (2022) "Biosynthesis and Genetic Incorporation of 3,4-Dihydroxy-L-Phenylalanine into Proteins in Escherichia coli." *Journal of Molecular Biology*, 434, 167412. PMCID: PMC9018569

- d. Chatterjee, A.[†], **Xiao, H.[†]**, Bollong, M., Ai, H. W., Schultz, P. G.* (2013). "Efficient Viral Delivery System for Noncanonical Amino Acid Mutagenesis in Mammalian Cells." *Proceedings of the National Academy of Sciences*, 110(29), 11803-11808. PMCID: PMC3718144 (* contributed equally)

Complete List of Published Work in MyBibliography:

http://www.ncbi.nlm.nih.gov/sites/myncbi/1b_KrbpPZpQAn/bibliography/44137323/public/?sort=date&direction=ascending

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Zhang, Xiang

eRA COMMONS USER NAME (credential, e.g., agency login): zhangxs1

POSITION TITLE: Professor

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	Start Date MM/YYYY	Completion Date MM/YYYY	FIELD OF STUDY
Fudan University, Shanghai, China	B.S.	09/01/1996	07/01//2000	Genetics
Columbia University, New York City	Ph.D	08/23/2000	11/25/2005	Molecular Biology

A. Personal Statement

The long-term goal of my laboratory at Baylor College of Medicine (BCM) is to elucidate biological mechanisms and therapeutic strategies of metastasis, the major threat to the lives of solid cancer patients. Specifically, we have been studying two general areas: breast cancer bone metastasis and tumor immunology. In both areas, we are interested in elucidating the interactions between cancer cells and the tumor microenvironment with focus on the spatiotemporal co-evolution of various constituents of the tumor ecosystem and heterogeneity within and between different ecosystems. Toward this end, we have invented a series of unique animal models. We also implement interdisciplinary approaches through collaborations with experts with expertise in microscopy, bioengineering, chemistry, integrated omics, and clinical oncology. Our ultimate goal is to identify effective therapies to conquer metastasis, ideally by leveraging our own immune system. I am extremely enthusiastic to collaborate with Dr. Xiao Han at Rice University to design and test the next generation of bone-targeting EZH2 inhibitors. Our expertise is highly complementary and our ongoing collaboration has been extremely productive (see publication #a below).

Our expertise in related fields can be partially reflected by the following peer-reviewed review articles (not included in **Section C**), as well as publications listed in **Section B4**.

- Tian Z#, Wu L#, Yu C, Chen Y, Xu Z, Bado IL, Loreda A, Wang L, Wang H, Wu K-L, Zhang W, **Xiang H-F Zhang*, Xiao H*** (2021) Harnessing the power of antibodies to fight bone metastasis. *Science Advances*, 7 (26), eabf2051 (# indicates equal first-authorship, * indicates co-senior authors). PMID: PMC8221630
- Gao Y, Bado I, Wang H, Zhang W, Rosen JM, and Xiang H.-F. Zhang (2019). Metastasis Organotropism: Redefining the Congenial Soil. *Dev Cell*. 2019 May 6;49(3):375-391. PMID: PMC6506189
- Muscarella AM, Aguirre S, Hao X, Waldvogel S and Xiang H.-F. Zhang (2021). Exploiting bone niches: Progression of disseminated tumor cells to metastasis. *J Clin Invest*. 131(6):e143764. PMID: PMC7954594.
- Satcher RL, Zhang XH (2021). Evolving cancer-niche interactions and therapeutic targets during bone metastasis. *Nat Rev Cancer*. 2021 Oct 5. doi: 10.1038/s41568-021-00406-5. Online ahead of print.

B. Positions, Scientific Appointments and Honors**Positions and Employment**

- 2022 - Interim Director, Lester and Sue Smith Breast Center, Baylor College of Medicine
- 2020 - William T. Butler Endowed Chair for Distinguished Faculty, Baylor College of Medicine, Houston, TX
- 2020 - Associate Director, Basic Science Research, Lester and Sue Smith Breast, Baylor College of Medicine, Houston, TX

2019-	Professor, Lester and Sue Smith Breast Center and Department of Molecular and Cellular Biology, Baylor College of Medicine, Houston, TX
2017-2019	Associate Professor, Lester and Sue Smith Breast Center and Department of Molecular and Cellular Biology, Baylor College of Medicine, Houston, TX
2011-2016	Assistant Professor, Lester and Sue Smith Breast Center and Department of Molecular and Cellular Biology, Baylor College of Medicine, Houston, TX
2006-2011	Postdoctoral Research Associate, Memorial Sloan Kettering Cancer Center, New York, NY Mentor: Dr. Joan Massagué, Co-Mentor: Dr. Larry Norton.
2000-2006	Graduate Research Assistant, Columbia University, New York, NY Mentor: Dr. Lawrence Chasin

Honors

2020	Laura Ziskin Translational Research Award of Breast Cancer, SU2C-AACR.
2018	Michael E. DeBaKey Excellence in Research Award, Baylor College of Medicine
2018	Sue Eccles Young Investigator Award, Metastasis Research Society
2016	Leadership Award, Teresa Research Foundation
2016	Excellence in Research Award, MCB, Baylor College of Medicine
2015	Era of Hope Scholar Award, Department of Defense, Breast Cancer Research Program
2014	Career Catalyst Research Award, Susan G. Komen Foundation
2011	McNair Scholar, Baylor College of Medicine
2006	Distinction to the Dissertation, Columbia University
2005	James Howard McGregor Award, (Annual departmental prize for the most promising students in the field of vertebrate zoology)
2004-2005	The Peter Sajovic Memorial Prize Fund endowed fellowship (Annual departmental honor for the students doing outstanding work in biology)
2000-2005	Faculty Fellowship, Columbia University
2000	Graduate with Highest Honor, Fudan University
1999	Chinese Academy of Sciences Scholarship, Fudan University
1997	Monsanto Scholarship, Fudan Highest Honors, Fudan University
1996	GE Scholarship, Fudan University
1996	Silver Medal of the 7th International Biology Olympiad Competition (held at Ukraine)

C. Contribution to Science

1. Computational identification of cis-regulatory elements regulating pre-mRNA splicing. My graduate training focused on pre-mRNA splicing. Specifically, my work represented one of the first efforts to apply genomic and bioinformatics approaches to the question of splicing. My early publications utilized genomic annotations of exons and introns to identify cis-regulatory elements that contribute to the efficiency and accuracy of splicing. We also used comparative genomic approaches to address the evolutionary origin of exons and regulatory splicing signals. Our research results have kept been frequently utilized in the literature for prediction of exonic splicing enhancers and silencers.

- a. **Xiang H.-F. Zhang**, Heller K, Helfer I, Leslie C, and Chasin LA (2003). Sequences Information for the splicing of human pre-mRNA identified by support vector machine classification. Genome Research 13(12): 2637-2650. PMID: 14656968 PMCID: PMC403805
- b. **Xiang H.-F. Zhang** and Chasin LA (2004). Computational definition of sequence motifs governing the splicing of constitutive exons. Genes and Development 18(11):1241-1250. PMID: 15145827 PMCID: PMC420350
- c. **Xiang H.-F. Zhang**, Kangsamaksin T, Chao MS, Banerjee JK, and Chasin LA (2005). Exon inclusion is dependent on predictable exonic splicing enhancers. Molecular and Cellular Biology 25(16): 7323-7332. PMID: 16055740 PMCID: PMC1190244
- d. **Xiang H.-F. Zhang** and Chasin LA. (2006). Comparison of multiple vertebrate genomes reveals the birth and evolution of human exons. Proc Natl Acad Sci U S A 103, 13427-13432. PMID: 16938881 PMCID: PMC1569180

2. Organ-specific metastasis mechanisms. With a team of colleagues in Dr. Joan Massague's laboratory, I contributed to a series of studies identifying metastasis-related genes and the underlying mechanisms. Specifically, I combined in silico analyses using clinical data with experimental animal models to decipher and predict signaling pathways underlying metastasis to various organs.

- a. Padua D, **Xiang H.-F. Zhang**, Wang Q, Nadal C, Gerald W, Gomis R, and Massague J (2008). TGF β primes breast tumors for lung metastasis seeding through angiopoietin-like 4. Cell 133(1): 66-77. PMID: PMC2390892
- b. Bos PD, **Xiang H.-F. Zhang**, Nadal C, Shu W, Gomis RR, Nguyen DX, Minn AJ, van de Vijver MJ, Gerald WL, Foekens JA, Massagué J (2009). Genes that mediate breast cancer metastasis to the brain. Nature 459(7249):1005-9. PMID: PMC2698953
- c. Nguyen DX, Chiang AC, **Xiang H.-F. Zhang**, Kim JY, Kris MG, Ladanyi M, Gerald WL, Massagué J (2009). WNT/TCF signaling through LEF1 and HOXB9 mediates lung adenocarcinoma metastasis. Cell 138(1): 51-62. PMID: PMC2742946
- d. Chen Q, **Xiang H.-F. Zhang**, Massagué J (2011). Macrophage binding to receptor VCAM-1 transmits survival signals in breast cancer cells that invade the lungs. Cancer Cell 20 (4):538-49. PMID: PMC3293160

3. Breast cancer bone metastasis. Using a series of unique models, we discovered the initial colonization of disseminated cancer cells requires the microenvironment niche comprised of cells of the osteoblast lineage. The cancer-niche interaction is mediated by heterotypic adherens and gap junctions, which activates the mTOR and calcium pathways in cancer cells to drive progression from single cells to multi-cell micrometastases. The cancer-niche interactions also render cancer cells more stem like and capable for further dissemination to other organs. These studies advanced our understanding of bone metastasis and revealed potential therapeutic targets.

- a. Wang H, Yu C, Gao X, Welte T, Muscarella AM, Zhao H, Zhao Z, Tao J, Lee B, Westbrook TF, Wong STC, Jin X, Rosen JM, Osborne CK, and **Xiang H.-F. Zhang** (2015). The Osteogenic Niche Promotes Early-Stage Bone Colonization of Disseminated Breast Cancer Cells. Cancer Cell. 27(2):193-210. PMID: PMC4326554
- b. Wang H, Lin T, Liu J, Goldstein A, Zhang W, Bado I, Li Z, Zhao H, Wong ST, **Xiang H.-F. Zhang**. (2018) The osteogenic niche is a calcium reservoir of breast cancer bone micrometastases and confers unexpected therapeutic vulnerability. Cancer Cell 12;34(5):823-839.e7. PMID: PMC6239211
- c. Bado IL, Zhang W, Hu J, Xu Z, Wang H, Sarkar, Li L, Wan Y-W, Liu J, Wu W, Lo H-C, Kim IS, Singh S, Janghorban M, Muscarella AM, Goldstein A, Singh P, Jeong H-H, Liu C, Schiff R, Huang S, Ellis MJ, Gaber MW, Gugala Z, Liu Z, **Xiang H.-F. Zhang** (2021) Impact of the bone microenvironment on phenotypic plasticity of ER+ breast cancer cells. Developmental Cell, 56(8):1100-1117. PMID: PMC8062036.
- d. Zhang W, Bado IL, Hu J, Wang Y-W, Wu L, Wang H, Gao Y, Jeong H-H, Xu Z, Hao X, Lege BM, Al-Ouran R, Li L, Li J, Yu L, Singh S, Lo H-C, Niu M, Liu J, Jiang W, Li Y, Wong STC, Cheng C, Liu Z, and **Xiang H.-F. Zhang** (2021). The bone microenvironment invigorates metastatic seeds for further dissemination. Cell, 184(9):2471-2486. PMID: PMC8087656

4. Immuno-regulation of metastasis and diversity of tumor immune microenvironment. The immune system is the major defense against tumorigenesis and tumor progression. However, many tumors acquire the ability to blunt or even invert the functions of the immune system. We are interested in the question of how breast cancers evolve to induce dysfunctions of the immune system both systemically and locally. In particular, we aim to delineate how this is happening in the context of inter- and intra-tumoral heterogeneity. The questions we are trying to attack include: 1) how the systemic and local immune environment varies among tumors of diverse genetic/epigenetic background; 2) how to identify the critical immune cell types/functions in a specific tumor context; and 3) how different tumors evolve dependencies on specific immune cells. Our ultimate goal is to identify therapies that could restore the normal functions of the immune system.

- a. Welte T, Kim IS, Tian L, Gao X, Wang H, Li J, Holdman XB, Herschkowitz JI, Pond A, Xia G, Kurley S, Nguyen T, Liao L, Dobrolecki LE, Pang L, Mo Q, Edwards DP, Huang S, Xin L, Xu J, Li Y, Lewis MT, Wang T, Westbrook TF, Rosen JM*, and **Xiang H.-F. Zhang*** (2016). Oncogenic mTOR signaling

recruits myeloid-derived suppressor cells to promote tumor initiation. Nature Cell Biology 18(6):632-44. PMID: PMC4884142

- b Tian L, Goldstein A, Wang H, Kim IS, Lo HS, Welte T, Sheng K, Dobrolecki LE, Zhang X, Putluri N, Phung T, Mani SA, Stossi F, Sreekumar A, Mancini MA, Zong C, Decker WK, Lewis MT, and **Xiang H.-F. Zhang** (2017). Mutual regulation of tumour vessel normalization and immunostimulatory reprogramming. Nature 544(7649):250-254. PMID: PMC5788037
- c Kim IS, Welte T, Toneff MJ, Wang H, Goldstein A, Sheng K, Niu Y, Liu J, Tan Y, Lo H-C, Arnold J, Gu F, He J, Jebakumar D, Walker K, Mo Q, Westbrook TF, Zong C, Rao A, Sreekumar A, Rosen JM, **Xiang H.-F. Zhang**. (2019), "Immuno-subtyping of breast cancer reveals distinct myeloid cell biology and immunotherapy resistance mechanisms" Nature Cell Biology, 21(9):1113-1126 PMID: PMC6726554.
- d Lo HC, Xu Z, Kim IS, Pingel B, Aguirre S, Kodali S, Liu J, Zhang W, Muscarella AM, Hein SM, Krupnick AS, Neilsen JR, Paust S, Rosen JM, Wang H, and **Xiang H.-F. Zhang** (2020). Resistance to Natural Killer cell immunosurveillance confers a selective advantage to polyclonal metastasis. Nature Cancer, online on June 1, 2020.

5. Novel experimental models and innovative research methods for metastasis research. In order to dissect the metastatic cascade and elucidate underlying mechanisms, we have invented a series of experimental models including novel tumor transplantation approaches and medium throughput ex vivo platforms. In addition, we applied cutting-edge technologies to delineate the spatiotemporal evolution of metastasis. Taken together, the innovative research revealed unappreciated therapeutic vulnerability and biological process related to metastasis.

- a. Wang H, Tian L, Goldstein A, Lo H-C, Liu J, Sheng K, Zbigniew G, Stossi F, Zong C, Li Z, Mancini MA and **Xiang H.-F. Zhang** (2017). Bone-in-culture array (BICA) as a platform to model early-stage bone metastases and discover anti-metastasis therapies. Nature Communications, Apr 21;8:15045 PMID: PMC5413944
- b. Muscarella AM, Dai W, Mitchell PG, Zhang W, Wang H, Jia L, Stossi F, Mancini MA, Chiu W, and Xiang H.-F. Zhang (2020). Unique gap junction- and cofilin-dependent cellular protrusions mediate cancer cell migration by tethering to osteogenic cells. npj Breast Cancer Sep 7;6:42 PMID: PMC7477119
- c. Yu C, Wang H, Muscarella A, Goldstein A, Zeng HC, Bae Y, Lee BH, **Xiang H.-F. Zhang** (2016) Intrailiac Artery Injection for Efficient and Selective Modeling of Microscopic Bone Metastasis. J Vis Exp 2016 26;(115):53982. PMID: PMC5092070
- d. Tian Z#, Wu L#, Yu C, Chen Y, Xu Z, Bado IL, Loreda A, Wang L, Wang H, Wu K-L, Zhang W, **Xiang H-F Zhang***, **Xiao H*** (2021) Harnessing the power of antibodies to fight bone metastasis. Science Advances, 7 (26), eabf2051 (# indicates equal first-authorship, * indicates co-senior authors). PMID: PMC8221630

Published Work in Pubmed:

<https://www.ncbi.nlm.nih.gov/pubmed?term=zhang,%20xiang%20h%20f&cmd=correctspelling>

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Wu, Tao

eRA COMMONS USER NAME (credential, e.g., agency login): TAOWU01

POSITION TITLE: Assistant Professor, Molecular and Human Genetics, Baylor College of Medicine

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date MM/YYYY	FIELD OF STUDY
Nankai University, Tianjin, China	B.S.	07/2003	Biophysics
Institute of Biophysics, Chinese Academy of Sciences, Beijing, China	Ph.D.	11/2009	RNA & Systems Biology
Department of Genetics and Yale Stem Cell Center, Yale University School of Medicine, CT	Postdoctoral Associate	08/2014	Epigenetics & Developmental Biology

A. Personal Statement

The long-term goal of my laboratory is to define the underlying epigenetic regulatory mechanisms of cellular plasticity and cell fates decision in early embryonic development and cancer progression. In the beginning, we focus on pinpointing the functional mechanisms of *non-canonical* DNA/RNA/histone modifications and their crosstalk in embryonic stem cells (ESC) and drug-tolerant persister cancer cell (DTP) models. My academic career started with a book. The Erwin Schrödinger’s “*What Is Life? The Physical Aspect of the Living Cell*” (1944) inspired me to pursue my academic career in deciphering the basic principle of biological systems. During my graduate training with Prof. Runsheng Chen (the founding father of genomics and bioinformatics in China), I developed expertise in bioinformatics, genomics, and RNA biology. Since I was fascinated by one root question in developmental biology how cells can commit to different lineages with the same genome, I joined the Yale Stem Cell Center for my postdoctoral research in epigenetics and stem cell biology. During my early career as a postdoc in Dr. Andrew Xiao’s lab, I played a leading role in defining novel epigenetic markers to distinguish high-quality and low-quality induced pluripotent stem cell (iPSC) clones. This work led to the discovery of a novel functional mechanism of histone variant H2A.X in cell fate decision, which was published in *Cell Stem Cell*. Afterward, I developed a novel method (SMRT-ChIP) to explore the molecular mechanism of H2A.X deposition. Surprisingly, *for the first time*, I discovered a novel non-canonical mammalian DNA methylation, N6-methyladenine (m6dA), in mouse ESC leading to an article in *Nature*. Subsequently, I initiated a collaboration with Dr. Jeremy Rich and found m6dA plays a pivotal role in the oncogenesis of glioblastoma (GBM) stem-like cells. This work opened a new window for identifying novel epigenetic cancer drug targets and was published in *Cell*. At the end of 2018, I started my own lab at Baylor College of Medicine (BCM). Although I am still a “New Investigator” for NIH, I have served as PI on CPRIT Scholar for cancer research and Rivkin Scholar for ovarian cancer research grants. In the last three years, I have built a well-trained team, multidisciplinary collaborations and made significant breakthroughs in several directions (one is in preparation, one is under review, and another is in revision). Based on these, I feel we are well prepared to conduct the proposed R01 research project to answer the fundamental questions in this application.

The ongoing and recently completed projects that I would like to highlight include:
CPRIT (RR180072)
Wu (PI)
08/2018 – 06/2023

Decipher epigenetic mechanisms underlying cancer acquired therapeutic resistance

RIVKIN scientific scholar for ovarian cancer research (2020)

Wu (PI)

03/2020 – 03/2022

Targeting Epigenetic Landscape Reprogramming to Combat Drug-resistance in Ovarian Cancer

Citations:

- a. **Tao Wu**, Yifei Liu, Duancheng Wen, Zito Tseng, Martik Tahmasian, Mei Zhong, Shahin Rafii, Matthias Stadtfeld, Konrad Hochedlinger, and Andrew Xiao. "Histone Variant H2A.X Deposition Pattern Serves as a Functional Epigenetic Mark for Distinguishing the Developmental Potentials of iPSCs." **Cell Stem Cell** 15, no. 3 (2014): 281–94. <https://doi.org/10.1016/j.stem.2014.06.004>. PMID: 25192463.
- b. **Tao Wu**, *et al.*, "Histone Variant H2A.X Deposition Pattern Serves as a Functional Epigenetic Mark for Pluripotency", Invited talk presented at the "*Epigenetics*" (*young scientists*), **Gordon Research Seminar, (International)**, Waltham, MA. (2015).
- c. **Tao P. Wu**, *et al.*, "Novel genomic DNA modification N6-me-dA in mouse ESC and iPSC", Invited talk presented at the "*10-Years of iPSCs*", **Cell Symposium, (International)**, Berkeley, CA. (2016).

B. Positions, Scientific Appointments, and Honors

Positions and Scientific Appointments

2021 – present	Member/Ambassador, European Association for Cancer Research (EACR)
2020 – present	Member, American Association for the Advancement of Science (AAAS)
2020 – present	Member, International Cell Senescence Association (ICSA)
2018 – present	Assistant Professor, Department of Molecular and Human Genetics, Baylor College of Medicine, Houston, TX
2018 – present	Associate Member, Huffington Center on Aging, Baylor College of Medicine, Houston, TX
2018 – present	Associate Member, Dan L Duncan Comprehensive Cancer Center, Baylor College of Medicine, Houston, TX
2018 – present	Associate Member, Therapeutic Innovation Center, Baylor College of Medicine, Houston, TX
2018 – present	Member, American Society of Human Genetics (ASHG)
2017 – present	Member, American Association for Cancer Research (AACR)
2014 – 2018	Associate Research Scientist, Yale Stem Cell Center and Department of Genetics, Yale School of Medicine, New Haven, CT
2013 – present	Member, International Society for Stem Cell Research (ISSCR)

Honors

2020 Rivkin Scientific Scholar Award for Ovarian Cancer Research

2018 CPRIT Scholar "First-Time Tenure-Track Recruitment" Award for Cancer Research

C. Contributions to Science

1. **Define a novel functional mechanism of histone variant H2A.X in iPSC reprogramming.** When I started my post-doctoral work in Andrew Xiao's laboratory at Yale Stem Cell Center in 2010, the induced pluripotent stem cell (iPSC) was just discovered by Shinya Yamanaka, and the mechanisms of epigenetic landscape reprogramming in the iPSC reprogramming process were elusive. A key question in the field is: How do the donor cells reprogram and commit to different lineages and generate heterogeneous iPSC clones? We hypothesized that the histone H2A variant H2A.X might regulate the cell pluripotency and cell fate decision of iPSCs since the histone variants are vital chromosomal players in the mouse's early embryonic development. Based on my genomic analysis, we discovered that H2A.X, which is critical for DNA damage repair and genome integrity, deposited to extraembryonic lineage gene loci and repressed their expression in ESCs. Using native ChIP-seq, we found that H2A.X genomic deposition defects (*epigenomic mutation*) frequently occur in low-quality iPSC lines, whereas the deposition pattern in the

high-quality iPSCs shows the same pattern as in ESCs. H2A.X epigenomic pattern alternation leads to aberrant expression of extraembryonic lineage genes in low-quality iPSCs and predisposes them to extraembryonic lineage differentiation *in vitro* and *in vivo*. This work helps to pave the way for the clinical application of iPSCs in regenerative medicine and cell therapy.

- a. **Tao Wu**, Yifei Liu, Duancheng Wen, Zito Tseng, Martik Tahmasian, Mei Zhong, Shahin Rafii, Matthias Stadtfeld, Konrad Hochedlinger, and Andrew Xiao. "Histone Variant H2A.X Deposition Pattern Serves as a Functional Epigenetic Mark for Distinguishing the Developmental Potentials of iPSCs." *Cell Stem Cell* 15, no. 3 (2014): 281–94. <https://doi.org/10.1016/j.stem.2014.06.004>. PMID: 25192463.
- b. Buganim, Yosef, Styliani Markoulaki, Niek van Wietmarschen, Heather Hoke, **Tao Wu**, Kibibi Ganz, Batool Akhtar-Zaidi, *et al.* "The Developmental Potential of iPSCs Is Greatly Influenced by Reprogramming Factor Selection." *Cell Stem Cell* 15, no. 3 (2014): 295–309. <https://doi.org/10.1016/j.stem.2014.07.003>. PMID: 25192464.
- c. Benchetrit, Hana, Shay Herman, Niek Van Wietmarschen, **Tao Wu**, Kirill Makedonski, Noam Maoz, Nataly Yom Tov, *et al.* "Extensive Nuclear Reprogramming Underlies Lineage Conversion into Functional Trophoblast Stem-like Cells." *Cell Stem Cell* 17, no. 5 (2015): 543–56. <https://doi.org/10.1016/j.stem.2015.08.006>. PMID: 26412562.

2. **Discover the second type of DNA methylation in mammalian genomes.** As my first independent project, I conceived a study to identify the unique property of H2A.X-nucleosome deposition sites for deciphering the deposition mechanism. I developed a novel approach of ChIP-seq, the "SMRT-ChIP", which can simultaneously determine DNA modifications and genomic sequences context of the histone binding sites. This approach leveraged the native chromatin immunoprecipitation (N-ChIP) and the 3rd generation single-molecule real-time sequencing (SMRT) technology. *Unexpectedly, we discovered N6-methyladenine (m6dA, a novel DNA methylation that had never been reported in mammalian species)*. I discovered that *m6dA* was enriched at H2A.X deposition sites and specifically targeted and suppressed younger LINE-1 retrotransposons in mouse ESCs. Moreover, we also identified *ALKBH1* as a demethylase (eraser) of *m6dA*, which means this methylation could be dynamically regulated. For more than forty years, 5-methylcytosine (*5mC*) and its derivatives were the only known functional regulatory DNA methylation in mammals. Thus, this discovery rewrites the epigenetics textbook and has the potential to offer new opportunities for defining novel epigenetic biomarkers and therapeutic targets in human diseases.

- a. **Tao P. Wu**, Tao Wang, Matthew G. Seetin, Yongquan Lai, Shijia Zhu, Kaixuan Lin, Yifei Liu, *et al.* "DNA Methylation on N6-Adenine in Mammalian Embryonic Stem Cells." *Nature* 532, no. 7599 (April 30, 2016): 329–33. <https://doi.org/10.1038/nature17640>. PMID: 27027282.
- b. Zhu, Shijia, John Beaulaurier, Gintaras Deikus, **Tao P. Wu**, Maya Strahl, Ziyang Hao, Guanzheng Luo, *et al.* "Mapping and Characterizing N6-Methyladenine in Eukaryotic Genomes Using Single-Molecule Real-Time Sequencing," *Genome Res* 28, no. 7 (2018): 1–12. <https://doi.org/10.1101/gr.231068.117>. PMID: 29764913.
- c. Pfeifer, G. P. "An elusive DNA base." *Nature* 532, 319–320 (2016). ["News & Views"]

3. **Decipher the functional role of *m6dA* in cancer.** The classical DNA methylation (*5mC*) is one of the best-studied epigenetic modifications and plays an essential role in embryonic development and human diseases. In contrast to *5mC*, house-keeping DNA methylation in mammals, the *m6dA* abundance is ultra-low in mammalian somatic tissues. So, I hypothesized that the non-canonical *m6dA* might be reserved as a stress-responsive regulator and play functional roles in human diseases, such as cancer. To test this hypothesis, I initiated a collaboration with Dr. Jeremy Rich in 2016. I performed a remarkably complex analysis of the genomic data based on the links between *m6dA*, cancer stem cells, and the tumor microenvironment. I have pioneered the application of "SMRT-ChIP" and ultra-sensitive mass spectrometry approaches that I developed in the *Nature* article, with which we confirmed the major discoveries in glioblastoma stem cells (GSC). This work has been published in *Cell*, and I was the driving force in this project. This work sheds new light on cancer epigenetic drug development and will benefit patients with glioblastoma.

- a. Qi Xie*, **Tao P. Wu***, Ryan C. Gimple*, Zheng Li, Briana C. Prager, Qiulian Wu, Yang Yu, *et al.* "N6-Methyladenine DNA Modification in Glioblastoma." **Cell** 175, no. 5 (2018): 1228-1243.e20. <https://doi.org/10.1016/j.cell.2018.10.006>. PMID: 30392959. (*Co-first author)
- b. **Tao P. Wu**, *et al.*, "Novel DNA Methylation N6-Methyladenine-Regulated Retrotransposon LINE-1 in Mammals", Selected talk presented at the "*Mobile Genetic Elements and Genome Plasticity*", **Keystone Symposia**, (International), Santa Fe, NM. (2018).
- c. **Tao P. Wu**, *et al.*, "Novel DNA Methylation in Cancer", Selected talk presented at the "*Cancer Genetics and Epigenetics*", **Gordon Research Conference**, (International), Lucca, Italy. (2019).

4. **Pinpoint an epigenetic mechanism underlying early embryonic cell fate's decision.** After we discovered *m6dA*, we kept exploring its functional roles in different models. With the ESC-TSC (mouse trophoblast stem cells) reprogramming system that mimics the early embryonic cell fates decision, we show that *m6dA* has a vital role in the epigenetic landscape reprogramming during cell fate transition. We found that *m6dA* is elevated in the trans-differentiation to TSC, specifically targeted to genomic sites of stress-induced DNA double helix destabilization (SIDD). Deposition of *m6dA* also antagonizes SATB1 (a nuclear matrix protein) function *in vitro* and *in vivo* by preventing its association to chromatin. I initiated the project and identified that *m6dA* could serve as the boundary insulator between euchromatin and heterochromatin to restrict the spread of euchromatin into condensed heterochromatin. Repression of SIDD–SATB1 interactions mediated by *m6dA* abolished TSC gene regulation and generated trophectoderm phenotype *in vivo*. In this work, we reported an unexpected molecular mechanism for *m6dA* via the anti-reader SATB1 and revealed the crosstalk between DNA modification, DNA topology, and chromatin functional domains in embryonic cell fate's decision.

- a. Zheng Li*, Shuai Zhao*, Raman V. Nelakanti*, Kaixuan Lin*, **Tao P. Wu***, *et al.* "N6-methyladenine in DNA antagonizes SATB1 in early development". **Nature** 583 (7817), 625–630 (2020). DOI: 10.1038/s41586-020-2500-9. PMID: 32669713. (*Co-first author)

5. **Deconvolute the epigenetic regulatory network underlying cellular plasticity.** *At the end of 2018*, I started my lab at BCM and focused on understanding the epigenetic regulatory mechanisms underlying cell senescence, stem cell pluripotency, cancer cell plasticity and therapeutic resistance. Since the prevalent model of cell fate commitment and identity (ID) maintenance is built on gene expression, many works focused on the gene transcriptional regulation network (GRN). Although the interaction between TFs and cis-regulatory elements is critical in GRN, mounting studies indicated that chromatin epigenetic state and remodelers are essential in gene regulation and cell fates transition. *I hypothesized that DNA/RNA/histone modifications could directly crosstalk to each other and be orchestrated by their readers/writers/erasers as an epigenetic network (ERN) to regulate gene expression, define the cellular plasticity and decide the dynamics of cell fate commitment.* To deconvolute this ERN, we will investigate the interaction between DNA and RNA methylation, DNA/RNA methylation and histone modifications, and how the perturbation of the interaction can control the gene expression pattern in cell fate commitment.

- a. Li X, Zhang Z, Luo X, Schrier J, Yang AD, **Wu TP#**. The exploration of N6-deoxyadenosine methylation in mammalian genomes. **Protein Cell**. 2021 Oct;12(10):756-768. doi: 10.1007/s13238-021-00866-3. Epub 2021 Aug 17. PMID: 34405377; PMCID: PMC8464638. (#Corresponding author, invited review article)
- b. Woodcock CB, Yu D, Hajian T, Li J, Huang Y, Dai N, Corrêa IR Jr, **Wu T**, Vedadi M, Zhang X, Cheng X. Human MettL3-MettL14 complex is a sequence-specific DNA adenine methyltransferase active on single-strand and unpaired DNA *in vitro*. **Cell Discov**. 2019 Dec 24;5:63. doi: 10.1038/s41421-019-0136-4. PMID: 31885874; PMCID: PMC6928010.
- c. Zheng Li*, Shuai Zhao*, Raman V. Nelakanti*, Kaixuan Lin*, **Tao P. Wu***, *et al.* "N6-methyladenine in DNA antagonizes SATB1 in early development". **Nature** 583 (7817), 625–630 (2020). DOI: 10.1038/s41586-020-2500-9. PMID: 32669713. (*Co-first author)

Complete List of Published Work in MyBibliography:

<https://www.ncbi.nlm.nih.gov/myncbi/1Z3wEmynn8gUXr/bibliography/public/>

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Osborne, Charles Kent

eRA COMMONS USER NAME (credential, e.g., agency login): ckosborne

POSITION TITLE: Professor of Medicine and Molecular and Cellular Biology

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date MM/YYYY	FIELD OF STUDY
University of Missouri, Columbia, MO	A.B. (Honors)	06/1968	Zoology – Chemistry
University of Missouri, Columbia, MO		05/1972	Medicine
Johns Hopkins Hospital, Baltimore MD	M.D. (Honors)	06/1973	Internal Medicine
	Internship/ Residency Fellow	06/1977	Medical Oncology
National Cancer Institute			

A. Personal Statement

My area of research involves the identification of mechanisms of resistance to targeted therapies in breast cancer particularly treatments targeting the estrogen receptor and the HER2 receptor. We have identified extensive molecular crosstalk between these two pathways whereby ER signaling activates the HER2 pathway and vice versa. We have shown that increased ER expression and signaling can cause resistance to HER targeted therapy and that upregulation of HER receptors and signaling can cause resistance to ER targeted therapy. The clinical implication is that both pathways may need to be blocked in certain patients to circumvent acquired resistance. Another mechanism of resistance to HER-targeted therapy is incomplete blockade of the HER receptor pathway. We have shown that two different combinations of HER-targeted therapy drugs that block the pathway more completely (lapatinib + Herceptin or gefitinib + Herceptin + pertuzumab) are much more effective than either individual drugs or other combinations of these drugs in preclinical models. This data has led to clinical trials of this combination strategy including neoadjuvant trials in our group looking at lapatinib + Herceptin in patients with HER2 amplified tumors. In addition to my laboratory efforts, I have significant administrative duties as Director of the Dan L Duncan Comprehensive Cancer Center for the past 15 years and previously as Director of the Lester and Sue Smith Breast Center at Baylor College of Medicine. I have many years' experience in clinical research having led the Breast Committee of SWOG for more than 10 years. I have served on Advisory Committees for several cancer centers, SPORE and Program Project Grants and was the PI on our Breast Cancer SPORE for 27 years and recently stepped down to give the leadership role to Dr. Matthew Ellis. I have considerable experience mentoring students, post-doctoral fellows, and junior faculty over the past 41 years.

Given my experience as a clinician scientist doing basic, translational, and clinical research I am qualified to serve as a collaborator on this proposal.

Ongoing and recently completed projects that I would like to highlight include:

P30 CA125123
Heslop (PI)
07/01/2007 – 06/30/2025
Baylor College of Medicine Cancer Center

P50 CA186784
Ellis (PI)
08/01/2014 – 07/31/2025
SPORE - Translational Research in Breast Cancer

RP200439
Yang (PI)
Targeting AKT signaling in MAPK4-high Triple Negative Breast Cancer
03/01/2020 – 02/28/2023

W81XWH-20-1-0375
Zhang (PI)
Eliminating Bone Micrometastases by Bone-targeting Antibodies
09/15/2021 – 09/14/2024

W81XWH-17-1-0580
Schiff (PI)
A New Paradigm for De-Escalation of Treatment in HER2-Positive Breast Cancer: Revolutionizing Care with More Effective and Less Toxic Therapy
09/15/2017 – 09/14/2022

B. Positions, Scientific Appointments, and Honors

2020- present	Founding Director, Dan L Duncan Comprehensive Cancer Center, Baylor College of Medicine
2020 -	Director, Dan L Duncan Comprehensive Cancer Center, Baylor College of Medicine
5/99 – 8/14	Director of the Lester and Sue Smith Breast Center, Baylor College of Medicine
5/99-2015	Professor of Medicine and Molecular and Cellular Biology, Tina and Dudley Sharp Chair in Oncology, Baylor College of Medicine, Houston
1995-1999	Division Chief and A.B. Alexander Chair of Medical Oncology, University of Texas Health Science Center at San Antonio
1992-1995	Interim Chief, Division of Medical Oncology, University of Texas Health Science Center at San Antonio
1993-1999	Professor of Cellular and Structural Biology, University of Texas Health Science Center at San Antonio
1985-1999	Professor of Medicine, University of Texas Health Science Center at San Antonio
1981-1985	Associate Professor of Medicine, University of Texas Health Science Center at San Antonio
1977-1981	Assistant Professor of Medicine, University of Texas Health Science Center at San Antonio
1974-1977	Clinical Associate, Medicine Branch, National Cancer Institute, NIH

C. Contributions to Science

I have published more than 400 manuscripts and in the following descriptions I will not comment on the many contributions in the area of prognostic and predictive markers or clinical trials done under my leadership as Chairman of the Breast Cancer Committee for the Southwest Oncology Group.

1. It has been clear for over a century that many breast cancers rely on estrogen for their growth. Early in my career I studied other hormones and growth factors that included insulin, IGF1 and 2, and epidermal growth factor. These studies demonstrated that human breast cancer cells are a target for the action of these polypeptide hormones and that specific receptors are present on the cell surface. We also showed that the inhibition of these pathways could inhibit tumor growth both in *in vivo* and *in vitro* models. These data set the paradigm for identification and targeting of these and other growth factors as a new treatment strategy in breast cancer, a strategy that has proven clinical benefit.
 - a. Osborne, C.K., Hamilton, B., Titus, G. and Livingston, R.B. Epidermal growth factor stimulation of human breast cancer cells in tissue culture. *Cancer Res.* 40: 2361-2366, 1980. PMID: 6966967
 - b. Arteaga, C.L., Coronado, E. and Osborne, C.K. Blockade of the epidermal growth factor receptor inhibits transforming growth factor α -induced but not estrogen-induced growth of hormone-dependent human breast cancer. *Mol. Endocrinol.* 2: 1064-1069, 1988. PMID: 3221874
 - c. Arteaga, C.L., Kitten, L.J., Coronado, E.B., Jacobs, S., Kull, F.C. Jr., Allred, D.C. and Osborne, C.K. Blockade of the type I somatomedin receptor inhibits growth of human breast cancer cells in athymic mice. *J. Clin. Invest* 84: 1418-1423, 1989. PMCID: PMC304004
 - d. Arteaga, C.L. and Osborne, C.K. Growth inhibition of human breast cancer cells in vitro with an antibody against the type I somatomedin receptor. *Cancer Res.* 49: 6237-6241, 1989. PMID: 2553250
2. Also early in my career, I began to study the anti-estrogen tamoxifen to understand its mechanism of action and later mechanisms of resistance. In a series of studies, we showed that tamoxifen's major mechanism of action was to induce a G1 block in cell cycle rather than apoptosis. We then showed that tamoxifen antagonizes certain chemotherapy drugs suggesting that it would be best to give chemotherapy first followed by tamoxifen. This eventually was confirmed by a large clinical trial done in SWOG showing that the sequential use of chemotherapy and tamoxifen was superior than the combined approach and this strategy has persisted as standard of care with all endocrine therapies over the past 20 years.
 - a. Osborne, C.K., Hobbs, K. and Clark, G.M. Effect of estrogens and antiestrogens on growth of human breast cancer cells in athymic nude mice. *Cancer Res.* 45: 584-590, 1985. PMID: 3967234
 - b. Osborne, C.K., Coronado, E.B. and Robinson, J.P. Human breast cancer in the athymic nude mouse: Cytostatic effects of long-term antiestrogen therapy. *Eur. J. Cancer Clin. Oncol.* 23: 1189-1196, 1987. PMID: 3653212
 - c. Osborne, C.K., Kitten, L. and Arteaga, C.L. Antagonism of chemotherapy-induced cytotoxicity for human breast cancer cells by antiestrogens. *J. Clin. Oncol.* 7: 710-717, 1989. PMID: 2715802
 - d. Albain, K.S., Barlow, W.E., Ravdin, P.M., Farrar, W.B., Burton, G.V., Ketchel, S.J., Cobau, C.D., Levine, E.G., Ingle, J.N., Pritchard, K.I., Lichter, A.S., Schneider, D.J., Abeloff, M.D., Henderson, C., Muss, H.B., Green, S.J., Lew, D., Livingston, R.B., Martino, S., Osborne, C.K., for the Breast Cancer Intergroup of North America. adjuvant chemotherapy and timing of tamoxifen in postmenopausal, patients with endocrine-responsive, node-positive breast cancer. *Lancet* 374(9707):2055-2063, 2009. PMCID: PMC3140679
3. Together with my late colleague, Dr. William McGuire, we published much of the early data on the clinical use and value of estrogen and progesterone receptors in breast cancer demonstrating their predictive ability as well as their prognostic significance and defined correlations with other tumor and patient characteristics. We also showed that in patients who had sequential assays, ER is uncommonly lost from the tumor providing an explanation for the sequential use of various endocrine therapies.
 - a. Lippman, M.E., Osborne, C.K., Knazek, R. and Young, N. *In vitro* model systems for the study of hormone dependent human breast cancer. *Beth Israel Seminars in Medicine.* N. Engl. J. Med. 296: 154-159, 1977. PMID: 187935
 - b. Clark, G.M., Osborne, C.K. and McGuire, W.L. Correlations between estrogen receptor, progesterone receptor, and patient characteristics in human breast cancer. *J. Clin. Oncol.* 2: 1102-1109, 1984. PMID: 6491696
 - c. Hull, D.F., Clark, G.M., Osborne, C.K., Knight, W.A. and McGuire, W.L. Multiple estrogen receptor assays in human breast cancer. *Cancer Res.* 43: 413-416, 1983. PMID: 6847780

- d. Osborne, C.K., Fisher, E., Redmond, C., Knight, W.A., Yochmowitz, M.G. and McGuire, W.L. Estrogen receptor, a marker for human breast cancer differentiation and patient prognosis. *Adv. Exp. Biol.* 138: 377-385, 1982. PMID: 7342719
4. More recently, a major effort of our group has been to understand the mechanisms of resistance to endocrine therapy including crosstalk with growth factor receptor pathways. We published many papers describing various mechanisms of resistance in pre-clinical models, most notably upregulation of the HER receptor pathway which occurs upon estrogen receptor targeted treatment. We also showed blockade of these growth factor receptor pathways could reverse resistance to endocrine therapy and in an important series of papers, we showed that the pure anti-estrogen and the ER degrader fulvestrant was a very effective endocrine therapy in *in vivo* pre-clinical models and that it demonstrated activity in tamoxifen-resistant tumors. This led to clinical trials confirming this activity in patients and today fulvestrant is the most effective endocrine therapy available to patients.
 - a. Shou, J., Massarweh, S., Osborne, C.K., Wakeling, A. E, Ali, S., Schiff, R. Mechanisms of tamoxifen resistance: Increased estrogen receptor-HER-2/neu crosstalk in ER/HER2-positive breast cancer. *JNCI* 96(12):926-935, 2004. PMID: 15199112
 - b. Divisova, J., Kuitase, I., Lazard, Z., Weiss, H., Vreeland, F., Hadsell, D.L., Schiff, R., Osborne, C.K., Lee, A.V. The growth hormone receptor antagonist pegvisomant blocks both mammary gland development and MCF-7 breast cancer xenograft growth. *Breast Cancer Res Treat.* 98(3):315-27, 2006. PMID: 16541323
 - c. Creighton, C.J., Li, X., Landis, M., Dixon, M.J., Neumeister, V.M., Sjolund, A., Rimm, D.L., Wong, H., Rodriguez, A., Herschkowitz, J.I., Fan, C., Zhang, X., He, X., Pavlick, A., Gutierrez, M.C., Renshaw, L., Larionov, A.A., Faratian, D., Hilsenbeck, S.G., Perou, C.M., **Lewis, M.T., ** Rosen, J. M., Chang, J.C. (**co-senior authors). Residual breast cancers after conventional therapy display mesenchymal as well as tumor-initiating features. *Proc Natl Acad Sci U S A.* 2009 Aug 18;106(33):13820-5. PMCID: PMC2720409
 - d. Fu, X., Osborne, C.K., Schiff, R. Biology and therapeutic potential of PI3K signaling in ER+/HER2-negative breast cancer. *The Breast Suppl* 2:S12-18, 2013. PMCID: PMC3808116
 5. We have also been interested in developing better therapies for HER2-positive breast cancer and to understand mechanisms of resistance to this therapy. We showed in pre-clinical models and then in a neoadjuvant trial that blocking the HER receptor family with multiple drugs is more effective and can eradicate HER2 amplified breast cancers in *in vivo* models. In the neoadjuvant clinical trial, we showed that about 25-30% of patients with large HER2-positive tumors achieve complete pathologic response with targeted therapy alone suggesting that some patients may not need chemotherapy. In ER-positive, HER2-positive patients, we also showed the need for blocking ER as well as HER2 to prevent ER from becoming a driver and causing a resistance to HER2 therapies.
 - a. Hilsenbeck, S.G., Osborne, C.K. Is There a Role for Adjuvant Tamoxifen in Progesterone Receptor-Positive Breast Cancer? An In silico Clinical Trial. *Clin Cancer Res.* Feb 1;12(3):1049s-55s, 2006. PMID: 16467123
 - b. Hammond ME, Hayes DF, Dowsett M, Allred DC, Hagerty KL, Badve S, Fitzgibbons PL, Francis G, Goldstein NS, Hayes M, Hicks DG, Lester S, Love R, Mangu PB, McShane L, Miller K, Osborne CK, Paik S, Perlmutter J, Rhodes A, Sasano H, Schwartz JN, Sweep FC, Taube S, Torlakovic EE, Valenstein P, Viale G, Visscher D, Wheeler T, Williams RB, Wittliff JL, Wolff AC; American Society of Clinical Oncology; College of American Pathologists. American Society of Clinical Oncology/College of American Pathologists guideline recommendations for immunohistochemical testing of estrogen and progesterone receptors in breast cancer (unabridged version). *Arch Pathol Lab Med.* 2010 Jul;134(7):e48-72. PMID:20586616
 - c. Giuliano, M., Schiff, R., Osborne, C.K., Trivedi, M.V. Biological mechanisms and clinical implications of endocrine resistance in breast cancer. *Breast Suppl* 3:S42-49, 2011. PMID:22015292

Complete list of Published Work in MyBibliography: <http://www.ncbi.nlm.nih.gov/pubmed/?term=osborne+ck>

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Zhu, Qian

eRA COMMONS USER NAME (credential, e.g., agency login): qzhu012

POSITION TITLE: Assistant Professor

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE <i>(if applicable)</i>	Completion Date MM/YYYY	FIELD OF STUDY
University of Ottawa	BSc	09/2008	Biochemistry
University of Ottawa	BSc	09/2009	Computer science
Princeton University	PhD	09/2016	Computer science, computational biology
Dana-Farber Cancer Institute	Postdoctoral	09/2022	Computational biology

A. Personal Statement

I am a computational biologist interested in developing computational and statistical methods, and integrating heterogeneous genomics data sets to tackle questions in cancer and genomics. Currently, my research is focused on single-cell genomics and spatial genomics. In recent years, multiplexed single-molecule RNA FISH (seqFISH) has made it possible to measure the mRNA content of cells while preserving cells' spatial coordinates in tissue. Using this technology, I have provided for the first time a computational framework for decomposing the cell type and spatial contributions to gene expression heterogeneity in normal tissues and in tumors. The methodologies I developed have been further applied to other spatial genomics platforms including the more popular 10X Genomics Visium platform nowadays. My long-term research aim is to develop and apply methods to identify and understand spatial transcriptomic signatures of the tumor ecosystem of triple negative breast cancer (TNBC), an aggressive form of breast cancer for which there are currently limited treatment options. Intra-tumor heterogeneity underlies TNBC treatment responses. I use both single-cell RNA sequencing and spatial sequencing technology (Visium) to identify and analyze the components of TNBC tumor ecosystems through various mouse models of TNBC. The resultant signatures will be integrated with the morphology features of histological images, to elucidate the sources of intra-tumor heterogeneity, glean insights into the treatment response, and find novel predictive biomarkers of the disease.

Ongoing projects that I would like to highlight include:

RR220035
CPRIT
Recruitment of First-Time, Tenure-Track Faculty Members

(Zhu, PI)

08/31/2022 – 08/31/2027

B. Positions, Scientific Appointments, and Honors

2022-
2022-
2016-2022
2009-2016
2015-

Assistant professor, Lester & Sue Smith Breast Center, Baylor College of Medicine, Houston, TX
CPRIT scholar, First-Time Tenure Track Award
Postdoctoral fellow, Dana-Farber Cancer Institute, Boston, MA
Graduate student, Princeton University, Princeton, NJ
Member, International Society of Computational Biology

2010-2011 Teaching assistant, Department of Computer Science, Princeton University, NJ
 2009-2010 Graduate fellowship award, Princeton University, NJ

C. Contributions to Science

1. SeqFISH is a multiplexed RNA-FISH imaging-based technology to reveal the single-cell spatial organization in fixed human tissues. I applied this technology to study the single cell spatial variation in the mouse brain, specifically visual cortex and hippocampus areas. Along with this work is the concurrent development of computational methodologies (hidden Markov random field) and software packages (Giotto) for analyzing and visualizing spatial expression datasets. Recently, we extended Giotto to analyze spatial proteomic datasets related to TNBC.

- a. Zhu, Q., Shah, S., Dries, R. *et al.* Identification of spatially associated subpopulations by combining scRNAseq and sequential fluorescence *in situ* hybridization data. *Nat Biotechnol* **36**, 1183–1190 (2018). <https://doi.org/10.1038/nbt.4260>
- b. Eng, CH.L., Lawson, M., Zhu, Q. *et al.* Transcriptome-scale super-resolved imaging in tissues by RNA seqFISH+. *Nature* **568**, 235–239 (2019). <https://doi.org/10.1038/s41586-019-1049-y>
- c. Zhu, Q. (2019). A Hidden Markov Random Field Model for Detecting Domain Organizations from Spatial Transcriptomic Data. In: Yuan, GC. (eds) Computational Methods for Single-Cell Data Analysis. Methods in Molecular Biology, vol 1935. Humana Press, New York, NY. https://doi.org/10.1007/978-1-4939-9057-3_16
- d. Dries, R*, Zhu, Q*, Dong, R. *et al.* Giotto: a toolbox for integrative analysis and visualization of spatial expression data. *Genome Biol* **22**, 78 (2021). <https://doi.org/10.1186/s13059-021-02286-2> * - co-first authors

2. As a way to dissect transcriptional and epigenetic mechanisms, CUT&RUN is seen by many as an improvement of ChIP-seq. I played a key role in the development of CUT&RUN processing tools. In the lab of Dr. Stuart Orkin, HHMI investigator at Boston Children's Hospital, we gleaned insights into the mechanism of BCL11A repression of globin gene expression. Through my analysis I demonstrated CUT&RUN's improved resolution over ChIP-seq and used the new TF binding site inference to get bp-resolution motif footprints.

- a. Zhu, Q., Liu, N., Orkin, S.H. *et al.* CUT&RUNTools: a flexible pipeline for CUT&RUN processing and footprint analysis. *Genome Biol* **20**, 192 (2019). <https://doi.org/10.1186/s13059-019-1802-4>
- a. Nan Liu*, Victoria V. Hargreaves*, Qian Zhu*, Jesse V. Kurland, Jiyoung Hong, Woojin Kim, Falak Sher, Claudio Macias-Trevino, Julia M. Rogers, Ryo Kurita, Yukio Nakamura, Guo-Cheng Yuan, Daniel E. Bauer, Jian Xu, Martha L. Bulyk, Stuart H. Orkin, Direct Promoter Repression by BCL11A Controls the Fetal to Adult Hemoglobin Switch, *Cell*, Volume 173, Issue 2, 2018, Pages 430-442. e17, ISSN 0092-8674, <https://doi.org/10.1016/j.cell.2018.03.016>. * - co-first authors
- b. Liu, N., Xu, S., Yao, Q. *et al.* Transcription factor competition at the γ -globin promoters controls hemoglobin switching. *Nat Genet* **53**, 511–520 (2021). <https://doi.org/10.1038/s41588-021-00798-y> (2021)

3. Collaborative research in cancer and breast cancer. I have many years of experience collaborating with experimental biologists in AML and breast cancer. These works have produced many interesting insights about the genetic, transcriptomic and epigenetic basis of these cancer types.

- a. Bin Li, Grace Li, Wenqing Cai, Qian Zhu, ..., Christopher Vakoc, and Stuart Orkin. In vivo CRISPR/Cas9 Screening Identifies Pbrm1 as a regulator of myeloid leukemia development. *In review in Cancer Discovery*. (2022)
- b. Aubrey BJ, Cutler JA, Bourgeois W, Donovan KA, Gu S, Hatton C, Perlee S, Perner F, Rahnamoun H, Theall ACP, Henrich JA, Zhu Q, Nowak RP, Kim YJ, Parvin S, Cremer A, Olsen SN, Eleuteri NA, Pikman Y, McGeehan GM, Stegmaier K, Letai A, Fischer ES, Liu XS, Armstrong SA. IKAROS and MENIN coordinate therapeutically actionable leukemogenic gene expression in MLL-r acute myeloid leukemia. *Nat Cancer*. 2022 May;3(5):595-613. doi: 10.1038/s43018-022-00366-1. Epub 2022 May 9. PMID: 35534777; PMCID: PMC9404532.

- c. Christopher J. Nirschl, Mayte Suárez-Fariñas, Benjamin Izar, Sanjay Prakadan, Ruth Dannenfelser, Itay Tirosh, Yong Liu, Qian Zhu, K. Sanjana P. Devi, Shaina L. Carroll, David Chau, Melika Rezaee, Tae-Gyun Kim, Ruiqi Huang, Judilyn Fuentes-Duculan, George X. Song-Zhao, Nicholas Gulati, Michelle A. Lowes, Sandra L. King, Francisco J. Quintana, Young-suk Lee, James G. Krueger, Kavita Y. Sarin, Charles H. Yoon, Levi Garraway, Aviv Regev, Alex K. Shalek, Olga Troyanskaya, Niroshana Anandasabapathy, IFN γ -Dependent Tissue-Immune Homeostasis Is Co-opted in the Tumor Microenvironment, *Cell*. 170(1):127-141. (2017) Pages 127-141.e15, ISSN 0092-8674
- d. Aure, M.R., Leivonen, SK., Fleischer, T. *et al.* Individual and combined effects of DNA methylation and copy number alterations on miRNA expression in breast tumors. *Genome Biol* **14**, R126 (2013). <https://doi.org/10.1186/gb-2013-14-11-r126>

4. An important skill I acquired during graduate school is the ability to conduct large-scale meta-analysis on transcriptomic data. My project aimed to find genes coexpressed to an investigator's query genes of interest using a large body of gene expression data. I developed a system SEEK that can robustly mine coexpressed genes by a query-based weighting of thousands of gene expression datasets. Novel biological insights are discovered by gene-gene correlations extracted from meta-analyses of these datasets. SEEK also retrieves datasets where the investigator's genes are coexpressed.

- a. Zhu, Q., Wong, A., Krishnan, A. *et al.* Targeted exploration and analysis of large cross-platform human transcriptomic compendia. *Nat Methods* **12**, 211–214 (2015). <https://doi.org/10.1038/nmeth.3249>
- b. Young-suk Lee, Arjun Krishnan, Qian Zhu, Olga G. Troyanskaya, Ontology-aware classification of tissue and cell-type signals in gene expression profiles across platforms and technologies, *Bioinformatics*, Volume 29, Issue 23, 1 December 2013, Pages 3036–3044
- c. Qian Zhu, Xavier Tekpli, Olga G Troyanskaya, Vessela N Kristensen, Subtype-specific transcriptional regulators in breast tumors subjected to genetic and epigenetic alterations, *Bioinformatics*, Volume 36, Issue 4, 15 February 2020, Pages 994–999
- d. Christopher Y. Park, Arjun Krishnan, Qian Zhu, Aaron K. Wong, Young-Suk Lee, Olga G. Troyanskaya, Tissue-aware data integration approach for the inference of pathway interactions in metazoan organisms, *Bioinformatics*, Volume 31, Issue 7, 1 April 2015, Pages 1093–1101

Complete List of Published Work in MyNCBI:

<https://www.ncbi.nlm.nih.gov/myncbi/189GvI7KcJ9AG/bibliography/public/>

Contact PD/PI: Xiao, Han

OMB Number: 4040-0001
Expiration Date: 12/31/2022

RESEARCH & RELATED BUDGET - SECTION A & B, Budget Period 1

UEI*: K51LECU1G8N3
Budget Type*: ☒ Project ☐ Subaward/Consortium
Enter name of Organization: William Marsh Rice University

Start Date*: 07-01-2023 End Date*: 06-30-2024 Budget Period: 1

A. Senior/Key Person												
Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary (\$)	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits (\$)*	Funds Requested (\$)*
1 .	Han		Xiao	PhD	PD/PI	128,833.00		1.2		17,178.00	3,470.00	20,648.00
Total Funds Requested for all Senior Key Persons in the attached file												
Additional Senior Key Persons:			File Name:							Total Senior/Key Person		20,648.00

B. Other Personnel							
Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits*	Funds Requested (\$)*
1	Post Doctoral Associates	12			53,000.00	13,568.00	66,568.00
2	Graduate Students	24			74,000.00	0.00	74,000.00
	Undergraduate Students						
	Secretarial/Clerical						
3	Total Number Other Personnel					Total Other Personnel	140,568.00
Total Salary, Wages and Fringe Benefits (A+B)							161,216.00

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, Budget Period 1

UEI*: K51LECU1G8N3
Budget Type*: ☒ Project ☐ Subaward/Consortium
Organization: William Marsh Rice University

Start Date*: 07-01-2023 End Date*: 06-30-2024 Budget Period: 1

C. Equipment Description	
List items and dollar amount for each item exceeding \$5,000	
Equipment Item	Funds Requested (\$)*
Total funds requested for all equipment listed in the attached file	
Total Equipment	
Additional Equipment: File Name:	

D. Travel	Funds Requested (\$)*
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	2,000.00
2. Foreign Travel Costs	
Total Travel Cost	2,000.00

E. Participant/Trainee Support Costs	Funds Requested (\$)*
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other:	
Number of Participants/Trainees	Total Participant Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, Budget Period 1

UEI*: K51LECU1G8N3
Budget Type*: ☒ Project ☐ Subaward/Consortium
Organization: William Marsh Rice University

Start Date*: 07-01-2023 End Date*: 06-30-2024 Budget Period: 1

F. Other Direct Costs		Funds Requested (\$)*
1. Materials and Supplies		51,044.00
2. Publication Costs		
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		368,000.00
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Tuition Remission		28,490.00
9. SEA Instrumentation User Fee		10,000.00
10. Professional Service		2,000.00
11. Animal Housing		5,250.00
Total Other Direct Costs		464,784.00

G. Direct Costs	Funds Requested (\$)*
Total Direct Costs (A thru F)	628,000.00

H. Indirect Costs			
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	Funds Requested (\$)*
1 . Research On_Campus	56.5	256,510.00	144,928.00
Total Indirect Costs			144,928.00
Cognizant Federal Agency		DHHS, Uyen Tran, 214-767-3261	
(Agency Name, POC Name, and POC Phone Number)			

I. Total Direct and Indirect Costs	Funds Requested (\$)*
Total Direct and Indirect Institutional Costs (G + H)	772,928.00

J. Fee	Funds Requested (\$)*

K. Total Costs and Fee	Funds Requested (\$)*
	772,928.00

L. Budget Justification*	File Name: Xiao_R01BudgetJustification_Nov20221018160173.pdf
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RESEARCH & RELATED Budget {F-K} (Funds Requested)

Contact PD/PI: Xiao, Han

OMB Number: 4040-0001
Expiration Date: 12/31/2022

RESEARCH & RELATED BUDGET - SECTION A & B, Budget Period 2

UEI*: K51LECU1G8N3
Budget Type*: ☒ Project ☐ Subaward/Consortium
Enter name of Organization: William Marsh Rice University

Start Date*: 07-01-2024 End Date*: 06-30-2025 Budget Period: 2

A. Senior/Key Person												
Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary (\$)	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits (\$)*	Funds Requested (\$)*
1 .	Han		Xiao	PhD	PD/PI	128,833.00		1.2		17,178.00	3,470.00	20,648.00
Total Funds Requested for all Senior Key Persons in the attached file												
Additional Senior Key Persons:			File Name:							Total Senior/Key Person		20,648.00

B. Other Personnel							
Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits*	Funds Requested (\$)*
1	Post Doctoral Associates	12			53,000.00	13,568.00	66,568.00
2	Graduate Students	24			74,000.00	0.00	74,000.00
	Undergraduate Students						
	Secretarial/Clerical						
3	Total Number Other Personnel				Total Other Personnel		140,568.00
Total Salary, Wages and Fringe Benefits (A+B)							161,216.00

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, Budget Period 2

UEI*: K51LECU1G8N3
Budget Type*: ☒ Project ☐ Subaward/Consortium
Organization: William Marsh Rice University

Start Date*: 07-01-2024 End Date*: 06-30-2025 Budget Period: 2

C. Equipment Description	
List items and dollar amount for each item exceeding \$5,000	
Equipment Item	Funds Requested (\$)*
Total funds requested for all equipment listed in the attached file	
Total Equipment	
Additional Equipment: File Name:	

D. Travel	Funds Requested (\$)*
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	2,000.00
2. Foreign Travel Costs	
Total Travel Cost	2,000.00

E. Participant/Trainee Support Costs	Funds Requested (\$)*
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other:	
Number of Participants/Trainees	Total Participant Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, Budget Period 2

UEI*: K51LECU1G8N3
Budget Type*: ☒ Project ☐ Subaward/Consortium
Organization: William Marsh Rice University

Start Date*: 07-01-2024 End Date*: 06-30-2025 Budget Period: 2

F. Other Direct Costs	Funds Requested (\$)*
1. Materials and Supplies	51,044.00
2. Publication Costs	
3. Consultant Services	
4. ADP/Computer Services	
5. Subawards/Consortium/Contractual Costs	368,000.00
6. Equipment or Facility Rental/User Fees	
7. Alterations and Renovations	
8. Tuition Remission	28,490.00
9. SEA Instrumentation User Fee	10,000.00
10. Professional Service	2,000.00
11. Animal Housing	5,250.00
Total Other Direct Costs	464,784.00

G. Direct Costs	Funds Requested (\$)*
Total Direct Costs (A thru F)	628,000.00

H. Indirect Costs			
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	Funds Requested (\$)*
1 . Research On_Campus	56.5	231,510.00	130,803.00
Total Indirect Costs			130,803.00
Cognizant Federal Agency	DHHS, Uyen Tran, 214-767-3261		
(Agency Name, POC Name, and POC Phone Number)			

I. Total Direct and Indirect Costs	Funds Requested (\$)*
Total Direct and Indirect Institutional Costs (G + H)	758,803.00

J. Fee	Funds Requested (\$)*

K. Total Costs and Fee	Funds Requested (\$)*
	758,803.00

L. Budget Justification*	File Name: Xiao_R01BudgetJustification_Nov20221018160173.pdf
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RESEARCH & RELATED Budget {F-K} (Funds Requested)

Contact PD/PI: Xiao, Han

OMB Number: 4040-0001
Expiration Date: 12/31/2022

RESEARCH & RELATED BUDGET - SECTION A & B, Budget Period 3

UEI*: K51LECU1G8N3
Budget Type*: ☒ Project ☐ Subaward/Consortium
Enter name of Organization: William Marsh Rice University

Start Date*: 07-01-2025 End Date*: 06-30-2026 Budget Period: 3

A. Senior/Key Person												
Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary (\$)	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits (\$)*	Funds Requested (\$)*
1 .	Han		Xiao	PhD	PD/PI	128,833.00		1.2		17,178.00	3,470.00	20,648.00
Total Funds Requested for all Senior Key Persons in the attached file												
Additional Senior Key Persons:			File Name:							Total Senior/Key Person		20,648.00

B. Other Personnel							
Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits*	Funds Requested (\$)*
1	Post Doctoral Associates	12			53,000.00	13,568.00	66,568.00
2	Graduate Students	24			74,000.00	0.00	74,000.00
	Undergraduate Students						
	Secretarial/Clerical						
3	Total Number Other Personnel					Total Other Personnel	140,568.00
Total Salary, Wages and Fringe Benefits (A+B)							161,216.00

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, Budget Period 3

UEI*: K51LECU1G8N3
Budget Type*: ☒ Project ☐ Subaward/Consortium
Organization: William Marsh Rice University

Start Date*: 07-01-2025 End Date*: 06-30-2026 Budget Period: 3

C. Equipment Description	
List items and dollar amount for each item exceeding \$5,000	
Equipment Item	Funds Requested (\$)*
Total funds requested for all equipment listed in the attached file	
Total Equipment	
Additional Equipment: File Name:	

D. Travel	Funds Requested (\$)*
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	2,000.00
2. Foreign Travel Costs	
Total Travel Cost	2,000.00

E. Participant/Trainee Support Costs	Funds Requested (\$)*
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other:	
Number of Participants/Trainees	Total Participant Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, Budget Period 3

UEI*: K51LECU1G8N3

Budget Type*: ☒ Project ☐ Subaward/Consortium

Organization: William Marsh Rice University

Start Date*: 07-01-2025

End Date*: 06-30-2026

Budget Period: 3

F. Other Direct Costs	Funds Requested (\$)*
1. Materials and Supplies	51,044.00
2. Publication Costs	
3. Consultant Services	
4. ADP/Computer Services	
5. Subawards/Consortium/Contractual Costs	368,000.00
6. Equipment or Facility Rental/User Fees	
7. Alterations and Renovations	
8. Tuition Remission	28,490.00
9. SEA Instrumentation User Fee	10,000.00
10. Professional Service	2,000.00
11. Animal Housing	5,250.00
Total Other Direct Costs	464,784.00

G. Direct Costs	Funds Requested (\$)*
Total Direct Costs (A thru F)	628,000.00

H. Indirect Costs			
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	Funds Requested (\$)*
1 . Research On_Campus	56.5	231,510.00	130,803.00
Total Indirect Costs			130,803.00
Cognizant Federal Agency	DHHS, Uyen Tran, 214-767-3261		
(Agency Name, POC Name, and POC Phone Number)			

I. Total Direct and Indirect Costs	Funds Requested (\$)*
Total Direct and Indirect Institutional Costs (G + H)	758,803.00

J. Fee	Funds Requested (\$)*

K. Total Costs and Fee	Funds Requested (\$)*
	758,803.00

L. Budget Justification*	File Name:
	Xiao_R01BudgetJustification_Nov20221018160173.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

Contact PD/PI: Xiao, Han

OMB Number: 4040-0001
Expiration Date: 12/31/2022

RESEARCH & RELATED BUDGET - SECTION A & B, Budget Period 4

UEI*: K51LECU1G8N3
Budget Type*: ☒ Project ☐ Subaward/Consortium
Enter name of Organization: William Marsh Rice University

Start Date*: 07-01-2026 End Date*: 06-30-2027 Budget Period: 4

A. Senior/Key Person												
Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary (\$)	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits (\$)*	Funds Requested (\$)*
1 .	Han		Xiao	PhD	PD/PI	128,833.00		1.2		17,178.00	3,470.00	20,648.00
Total Funds Requested for all Senior Key Persons in the attached file												
Additional Senior Key Persons:			File Name:							Total Senior/Key Person		20,648.00

B. Other Personnel							
Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits*	Funds Requested (\$)*
1	Post Doctoral Associates	12			53,000.00	13,568.00	66,568.00
2	Graduate Students	24			74,000.00	0.00	74,000.00
	Undergraduate Students						
	Secretarial/Clerical						
3	Total Number Other Personnel					Total Other Personnel	140,568.00
Total Salary, Wages and Fringe Benefits (A+B)							161,216.00

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, Budget Period 4

UEI*: K51LECU1G8N3
Budget Type*: ☒ Project ☐ Subaward/Consortium
Organization: William Marsh Rice University

Start Date*: 07-01-2026 End Date*: 06-30-2027 Budget Period: 4

C. Equipment Description	
List items and dollar amount for each item exceeding \$5,000	
Equipment Item	Funds Requested (\$)*
Total funds requested for all equipment listed in the attached file	
Total Equipment	
Additional Equipment: File Name:	

D. Travel	Funds Requested (\$)*
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	2,000.00
2. Foreign Travel Costs	
Total Travel Cost	2,000.00

E. Participant/Trainee Support Costs	Funds Requested (\$)*
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other:	
Number of Participants/Trainees	Total Participant Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, Budget Period 4

UEI*: K51LECU1G8N3

Budget Type*: ☒ Project ☐ Subaward/Consortium

Organization: William Marsh Rice University

Start Date*: 07-01-2026

End Date*: 06-30-2027

Budget Period: 4

F. Other Direct Costs	Funds Requested (\$)*
1. Materials and Supplies	51,044.00
2. Publication Costs	
3. Consultant Services	
4. ADP/Computer Services	
5. Subawards/Consortium/Contractual Costs	368,000.00
6. Equipment or Facility Rental/User Fees	
7. Alterations and Renovations	
8. Tuition Remission	28,490.00
9. SEA Instrumentation Fee	10,000.00
10. Professional Service	2,000.00
11. Animal Housing	5,250.00
Total Other Direct Costs	464,784.00

G. Direct Costs	Funds Requested (\$)*
Total Direct Costs (A thru F)	628,000.00

H. Indirect Costs			
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	Funds Requested (\$)*
1 . Research On_Campus	56.5	231,510.00	130,803.00
Total Indirect Costs			130,803.00
Cognizant Federal Agency		DHHS, Uyen Tran, 214-767-3261	
(Agency Name, POC Name, and POC Phone Number)			

I. Total Direct and Indirect Costs	Funds Requested (\$)*
Total Direct and Indirect Institutional Costs (G + H)	758,803.00

J. Fee	Funds Requested (\$)*

K. Total Costs and Fee	Funds Requested (\$)*
	758,803.00

L. Budget Justification*	File Name:
	Xiao_R01BudgetJustification_Nov20221018160173.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

Contact PD/PI: Xiao, Han

OMB Number: 4040-0001
Expiration Date: 12/31/2022

RESEARCH & RELATED BUDGET - SECTION A & B, Budget Period 5

UEI*: K51LECU1G8N3
Budget Type*: ☒ Project ☐ Subaward/Consortium
Enter name of Organization: William Marsh Rice University

Start Date*: 07-01-2027 End Date*: 06-30-2028 Budget Period: 5

A. Senior/Key Person												
Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary (\$)	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits (\$)*	Funds Requested (\$)*
1 .	Han		Xiao	PhD	PD/PI	128,833.00		1.2		17,178.00	3,470.00	20,648.00
Total Funds Requested for all Senior Key Persons in the attached file												
Additional Senior Key Persons:			File Name:							Total Senior/Key Person		20,648.00

B. Other Personnel							
Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits*	Funds Requested (\$)*
1	Post Doctoral Associates	12			53,000.00	13,568.00	66,568.00
2	Graduate Students	24			74,000.00	0.00	74,000.00
	Undergraduate Students						
	Secretarial/Clerical						
3	Total Number Other Personnel					Total Other Personnel	140,568.00
Total Salary, Wages and Fringe Benefits (A+B)							161,216.00

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, Budget Period 5

UEI*: K51LECU1G8N3
Budget Type*: ☒ Project ☐ Subaward/Consortium
Organization: William Marsh Rice University

Start Date*: 07-01-2027 End Date*: 06-30-2028 Budget Period: 5

C. Equipment Description	
List items and dollar amount for each item exceeding \$5,000	
Equipment Item	Funds Requested (\$)*
Total funds requested for all equipment listed in the attached file	
Total Equipment	
Additional Equipment: File Name:	

D. Travel	Funds Requested (\$)*
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	2,000.00
2. Foreign Travel Costs	
Total Travel Cost	2,000.00

E. Participant/Trainee Support Costs	Funds Requested (\$)*
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other:	
Number of Participants/Trainees	Total Participant Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, Budget Period 5

UEI*: K51LECU1G8N3

Budget Type*: ☒ Project ☐ Subaward/Consortium

Organization: William Marsh Rice University

Start Date*: 07-01-2027

End Date*: 06-30-2028

Budget Period: 5

F. Other Direct Costs	Funds Requested (\$)*
1. Materials and Supplies	51,044.00
2. Publication Costs	
3. Consultant Services	
4. ADP/Computer Services	
5. Subawards/Consortium/Contractual Costs	368,000.00
6. Equipment or Facility Rental/User Fees	
7. Alterations and Renovations	
8. Tuition Remission	28,490.00
9. SEA Instrumentation User Fee	10,000.00
10. Professional Service	2,000.00
11. Animal Housing	5,250.00
Total Other Direct Costs	464,784.00

G. Direct Costs	Funds Requested (\$)*
Total Direct Costs (A thru F)	628,000.00

H. Indirect Costs			
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	Funds Requested (\$)*
1 . Research On_Campus	56.5	231,510.00	130,803.00
Total Indirect Costs			130,803.00
Cognizant Federal Agency	DHHS, Uyen Tran, 214-767-3261		
(Agency Name, POC Name, and POC Phone Number)			

I. Total Direct and Indirect Costs	Funds Requested (\$)*
Total Direct and Indirect Institutional Costs (G + H)	758,803.00

J. Fee	Funds Requested (\$)*

K. Total Costs and Fee	Funds Requested (\$)*
	758,803.00

L. Budget Justification*	File Name:
	Xiao_R01BudgetJustification_Nov20221018160173.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

BUDGET JUSTIFICATION

Rice University

Senior Personnel

Han Xiao, PhD, Principal Investigator (Contact PI), (effort = 1.2 academic months per year). Dr. Xiao is Assistant Professor in the Department of Chemistry and his research interests focus on the development of various chemical tools that allow us to precisely probe and manipulate biological systems. He will serve as the Contact PI and will be responsible for the overall coordination of the project. Dr. Xiao will be responsible for planning and conducting experiments, supervision of the postdoctoral researcher, graduate student, and communicating research through conference presentations and publications.

Other Personnel

Han Sun, PhD, Postdoctoral Researcher, (effort = 12 calendar months per year). This individual will coordinate the day-to-day management of the laboratory, assist with assessment and analysis of experiments, and be working on the evaluation of efficacy of bone-targeting EZH2 inhibitors.

Yixian Wang, Graduate Research Assistant, (effort = 12 calendar months per year). The graduate student will be responsible for the synthesis of EZH2 inhibitors. The amount requested of \$37,000 reflects the anticipated rate for graduate students in the Department of Chemistry at Rice University.

Chenhang Wang, Graduate Research Assistant, (effort = 12 calendar months per year). The graduate student will be responsible for the synthesis and evaluation of bone-targeting EZH2 inhibitors in vitro. The amount requested of \$37,000 reflects the anticipated rate for graduate students in the Department of Chemistry at Rice University.

Fringe Benefits

Fringe benefits are charged at the currently approved rates of 20.2% for faculty and 25.6% for staff.

Travel

Funds are requested to allow the PI and the postdoc/graduate student to attend scientific conferences and present the findings from this research. Possible conferences might include: Bones and Teeth Conference GRC, ASBMR annual meeting, Bioorganic Chemistry Conference GRC. An estimated cost for travel is provided in the table below.

Domestic (\$2,000 per year)

Description	Lodging	Per Diem	No. of Days	Subtotal	Registration Fee	Airfare	Total
Domestic Scientific Conference	\$190	\$60	4	\$1000	\$450	\$550	\$2000

Other Direct Costs

Materials and Supplies:

Laboratory supplies are necessary to conduct the proposed project. This includes the cost for chemicals, antibodies, molecular biology reagents, glassware, general consumables such as syringes, needles, pipettes, vials, filter paper, cleaning solutions, safety lab coats, goggles, face masks, and gloves. The amount of \$40,452 per year is allocated for laboratory supplies.

In addition, the research study will require the purchase of 1324 over the 5 year project period (see Research Strategy) at Rice University, including athymic nude, C57BL/6 and BALB/c mice. The average cost of one mouse is about \$40 based on the actual expenditure in the past year. Therefore, the amount of \$10,592 per year is allocated for this purpose.

Subaward/Consortium Costs:

The proposed research study will establish a subaward agreement with the following domestic institution:

Baylor College of Medicine

Dr. Xiang Zhang will serve as Principal Investigator on this multi-PI project.

The total amount of \$368,000 (\$230,000 directs + \$138,000 indirects) is requested per year.

Total cost for the 5 year period is \$1,840,000 (\$1,150,000 directs + \$690,000 indirects).

Other – Animal Housing:

Animal Maintenance cost at Rice is 1.05/cage/day. Based on the actual records of the past year and the expected total number of mice of 1324, we expect to maintain 50 cages for 100 days each year. Based on these numbers, the cost of animal per diem maintenance will be \$5,250 each year.

Other – Tuition Remission:

Tuition remission is requested for the two graduate students. The amount is calculated at the current rate of 38.5% of the compensation for each graduate student.

$\$37,000 \times 38.5\% = \$14,245/\text{year} \times 2 \text{ graduate students} = \$28,490$

Other – Professional Services

Editing and proofreading service is requested for manuscripts and reports that directly result from this research project. The editing service is necessary to help communicate information clearly and concisely and make research findings more accessible to the public. The amount of \$2,000 is requested per year.

Other – Shared Facility Core Services

The Rice Shared Equipment Authority (SEA) sets user fees for on-site analytical instruments.

\$10,000 is requested for core services including mass spectrometer, imaging, and flow cytometry facilities to analyze the samples generated in the project.

Indirect Costs:

Indirect costs are charged at Rice University's federally negotiated DHHS research rate of 56.5% of modified total direct costs (MTDC). MTDC excludes tuition remission costs in years 1 – 5 of the project, and the portion of the subaward costs in excess of \$25,000.

RESEARCH & RELATED BUDGET - Cumulative Budget

	Totals (\$)	
Section A, Senior/Key Person		103,240.00
Section B, Other Personnel		702,840.00
Total Number Other Personnel	15	
Total Salary, Wages and Fringe Benefits (A+B)		806,080.00
Section C, Equipment		
Section D, Travel		10,000.00
1. Domestic	10,000.00	
2. Foreign		
Section E, Participant/Trainee Support Costs		
1. Tuition/Fees/Health Insurance		
2. Stipends		
3. Travel		
4. Subsistence		
5. Other		
6. Number of Participants/Trainees		
Section F, Other Direct Costs		2,323,920.00
1. Materials and Supplies	255,220.00	
2. Publication Costs		
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs	1,840,000.00	
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Other 1	142,450.00	
9. Other 2	50,000.00	
10. Other 3	10,000.00	
11. Other 4	26,250.00	
12. Other 5		
13. Other 6		
14. Other 7		
15. Other 8		
16. Other 9		
17. Other 10		
Section G, Direct Costs (A thru F)		3,140,000.00
Section H, Indirect Costs		668,140.00

Section I, Total Direct and Indirect Costs (G + H)	3,808,140.00
Section J, Fee	
Section K, Total Costs and Fee (I + J)	3,808,140.00

Contact PD/PI: Xiao, Han

OMB Number: 4040-0001
Expiration Date: 12/31/2022

RESEARCH & RELATED BUDGET - SECTION A & B, Budget Period 1

UEI*: FXKMA43NTV21
Budget Type*: ☐ Project ☒ Subaward/Consortium
Enter name of Organization: Baylor College of Medicine

Start Date*: 07-01-2023 End Date*: 06-30-2024 Budget Period: 1

A. Senior/Key Person													
Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary (\$)	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits (\$)*	Funds Requested (\$)*	
1 .	Xiang		Zhang	PhD	PD/PI	203,700.00	1.2			20,370.00	5,307.00	25,677.00	
2 .	Tao		Wu	PhD	Co-Investigator	128,750.00	0.6			6,438.00	1,635.00	8,073.00	
Total Funds Requested for all Senior Key Persons in the attached file													
Additional Senior Key Persons:			File Name:								Total Senior/Key Person		33,750.00

B. Other Personnel							
Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits*	Funds Requested (\$)*
2	Post Doctoral Associates	18			87,831.00	24,486.00	112,317.00
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
2	Total Number Other Personnel					Total Other Personnel	112,317.00
Total Salary, Wages and Fringe Benefits (A+B)							146,067.00

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, Budget Period 1

UEI*: FXKMA43NTV21
Budget Type*: ☐ Project ☒ Subaward/Consortium
Organization: Baylor College of Medicine

Start Date*: 07-01-2023 End Date*: 06-30-2024 Budget Period: 1

C. Equipment Description	
List items and dollar amount for each item exceeding \$5,000	
Equipment Item	Funds Requested (\$)*
Total funds requested for all equipment listed in the attached file	
Total Equipment	
Additional Equipment: File Name:	

D. Travel	Funds Requested (\$)*
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	3,000.00
2. Foreign Travel Costs	
Total Travel Cost	3,000.00

E. Participant/Trainee Support Costs	Funds Requested (\$)*
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other:	
Number of Participants/Trainees	Total Participant Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, Budget Period 1

UEI*: FXKMA43NTV21
Budget Type*: ☐ Project ☒ Subaward/Consortium
Organization: Baylor College of Medicine

Start Date*: 07-01-2023 End Date*: 06-30-2024 Budget Period: 1

F. Other Direct Costs			Funds Requested (\$)*
1. Materials and Supplies			54,233.00
2. Publication Costs			
3. Consultant Services			
4. ADP/Computer Services			
5. Subawards/Consortium/Contractual Costs			
6. Equipment or Facility Rental/User Fees			
7. Alterations and Renovations			
8. Animal Maintenance			7,700.00
9. Laboratory Analysis/Core Charges			19,000.00
Total Other Direct Costs			80,933.00

G. Direct Costs		Funds Requested (\$)*
Total Direct Costs (A thru F)		230,000.00

H. Indirect Costs			
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	Funds Requested (\$)*
1 . MTDC	60	230,000.00	138,000.00
Total Indirect Costs			138,000.00
Cognizant Federal Agency	DHHS, Arif Karim, 214-767-3261		
(Agency Name, POC Name, and POC Phone Number)			

I. Total Direct and Indirect Costs		Funds Requested (\$)*
Total Direct and Indirect Institutional Costs (G + H)		368,000.00

J. Fee	Funds Requested (\$)*

K. Total Costs and Fee	Funds Requested (\$)*
	368,000.00

L. Budget Justification*	File Name: BCM_Budget_Justification_10_26_221018160151.pdf
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RESEARCH & RELATED Budget {F-K} (Funds Requested)

Contact PD/PI: Xiao, Han

OMB Number: 4040-0001
Expiration Date: 12/31/2022

RESEARCH & RELATED BUDGET - SECTION A & B, Budget Period 2

UEI*: FXKMA43NTV21
Budget Type*: ☐ Project ☒ Subaward/Consortium
Enter name of Organization: Baylor College of Medicine

Start Date*: 07-01-2024 End Date*: 06-30-2025 Budget Period: 2

A. Senior/Key Person													
Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary (\$)	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits (\$)*	Funds Requested (\$)*	
1 .	Xiang		Zhang	PhD	PD/PI	203,700.00	1.2			20,370.00	5,307.00	25,677.00	
2 .	Tao		Wu	PhD	Co-Investigator	128,750.00	0.6			6,438.00	1,635.00	8,073.00	
Total Funds Requested for all Senior Key Persons in the attached file													
Additional Senior Key Persons:			File Name:								Total Senior/Key Person		33,750.00

B. Other Personnel							
Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits*	Funds Requested (\$)*
2	Post Doctoral Associates	18			87,831.00	24,486.00	112,317.00
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
2	Total Number Other Personnel					Total Other Personnel	112,317.00
Total Salary, Wages and Fringe Benefits (A+B)							146,067.00

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, Budget Period 2

UEI*: FXKMA43NTV21
Budget Type*: ☐ Project ☒ Subaward/Consortium
Organization: Baylor College of Medicine

Start Date*: 07-01-2024 End Date*: 06-30-2025 Budget Period: 2

C. Equipment Description	
List items and dollar amount for each item exceeding \$5,000	
Equipment Item	Funds Requested (\$)*
Total funds requested for all equipment listed in the attached file	
Total Equipment	
Additional Equipment: File Name:	

D. Travel	Funds Requested (\$)*
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	3,000.00
2. Foreign Travel Costs	
Total Travel Cost	3,000.00

E. Participant/Trainee Support Costs	Funds Requested (\$)*
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other:	
Number of Participants/Trainees	Total Participant Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, Budget Period 2

UEI*: FXKMA43NTV21
Budget Type*: ☐ Project ☒ Subaward/Consortium
Organization: Baylor College of Medicine

Start Date*: 07-01-2024 End Date*: 06-30-2025 Budget Period: 2

F. Other Direct Costs			Funds Requested (\$)*
1. Materials and Supplies			54,233.00
2. Publication Costs			
3. Consultant Services			
4. ADP/Computer Services			
5. Subawards/Consortium/Contractual Costs			
6. Equipment or Facility Rental/User Fees			
7. Alterations and Renovations			
8. Animal Maintenance			7,700.00
9. Laboratory Analysis/Core Charges			19,000.00
Total Other Direct Costs			80,933.00

G. Direct Costs		Funds Requested (\$)*
Total Direct Costs (A thru F)		230,000.00

H. Indirect Costs			
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	Funds Requested (\$)*
1 . MTDC	60	230,000.00	138,000.00
Total Indirect Costs			138,000.00
Cognizant Federal Agency	DHHS, Arif Karim, 214-767-3261		
(Agency Name, POC Name, and POC Phone Number)			

I. Total Direct and Indirect Costs		Funds Requested (\$)*
Total Direct and Indirect Institutional Costs (G + H)		368,000.00

J. Fee	Funds Requested (\$)*

K. Total Costs and Fee	Funds Requested (\$)*
	368,000.00

L. Budget Justification*	File Name: BCM_Budget_Justification_10_26_221018160151.pdf
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RESEARCH & RELATED Budget {F-K} (Funds Requested)

Contact PD/PI: Xiao, Han

OMB Number: 4040-0001
Expiration Date: 12/31/2022

RESEARCH & RELATED BUDGET - SECTION A & B, Budget Period 3

UEI*: FXKMA43NTV21
Budget Type*: ☐ Project ☒ Subaward/Consortium
Enter name of Organization: Baylor College of Medicine

Start Date*: 07-01-2025 End Date*: 06-30-2026 Budget Period: 3

A. Senior/Key Person													
Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary (\$)	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits (\$)*	Funds Requested (\$)*	
1 .	Xiang		Zhang	PhD	PD/PI	203,700.00	1.2			20,370.00	5,307.00	25,677.00	
2 .	Tao		Wu	PhD	Co-Investigator	128,750.00	0.6			6,438.00	1,635.00	8,073.00	
Total Funds Requested for all Senior Key Persons in the attached file													
Additional Senior Key Persons:			File Name:								Total Senior/Key Person		33,750.00

B. Other Personnel							
Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits*	Funds Requested (\$)*
2	Post Doctoral Associates	18			87,831.00	24,486.00	112,317.00
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
2	Total Number Other Personnel					Total Other Personnel	112,317.00
Total Salary, Wages and Fringe Benefits (A+B)							146,067.00

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, Budget Period 3

UEI*: FXKMA43NTV21
Budget Type*: ☐ Project ☒ Subaward/Consortium
Organization: Baylor College of Medicine

Start Date*: 07-01-2025 End Date*: 06-30-2026 Budget Period: 3

C. Equipment Description	
List items and dollar amount for each item exceeding \$5,000	
Equipment Item	Funds Requested (\$)*
Total funds requested for all equipment listed in the attached file	
Total Equipment	
Additional Equipment: File Name:	

D. Travel	Funds Requested (\$)*
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	3,000.00
2. Foreign Travel Costs	
Total Travel Cost	3,000.00

E. Participant/Trainee Support Costs	Funds Requested (\$)*
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other:	
Number of Participants/Trainees	Total Participant Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, Budget Period 3

UEI*: FXKMA43NTV21
Budget Type*: ☐ Project ☒ Subaward/Consortium
Organization: Baylor College of Medicine

Start Date*: 07-01-2025 End Date*: 06-30-2026 Budget Period: 3

F. Other Direct Costs			Funds Requested (\$)*
1. Materials and Supplies			54,233.00
2. Publication Costs			
3. Consultant Services			
4. ADP/Computer Services			
5. Subawards/Consortium/Contractual Costs			
6. Equipment or Facility Rental/User Fees			
7. Alterations and Renovations			
8. Animal Maintenance			7,700.00
9. Laboratory Analysis/Core Charges			19,000.00
Total Other Direct Costs			80,933.00

G. Direct Costs		Funds Requested (\$)*
Total Direct Costs (A thru F)		230,000.00

H. Indirect Costs			
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	Funds Requested (\$)*
1 . MTDC	60	230,000.00	138,000.00
Total Indirect Costs			138,000.00
Cognizant Federal Agency	DHHS, Arif Karim, 214-767-3261		
(Agency Name, POC Name, and POC Phone Number)			

I. Total Direct and Indirect Costs		Funds Requested (\$)*
Total Direct and Indirect Institutional Costs (G + H)		368,000.00

J. Fee	Funds Requested (\$)*

K. Total Costs and Fee	Funds Requested (\$)*
	368,000.00

L. Budget Justification*	File Name: BCM_Budget_Justification_10_26_221018160151.pdf
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RESEARCH & RELATED Budget {F-K} (Funds Requested)

Contact PD/PI: Xiao, Han

OMB Number: 4040-0001
Expiration Date: 12/31/2022

RESEARCH & RELATED BUDGET - SECTION A & B, Budget Period 4

UEI*: FXKMA43NTV21
Budget Type*: ☐ Project ☒ Subaward/Consortium
Enter name of Organization: Baylor College of Medicine

Start Date*: 07-01-2026 End Date*: 06-30-2027 Budget Period: 4

A. Senior/Key Person													
Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary (\$)	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits (\$)*	Funds Requested (\$)*	
1 .	Xiang		Zhang	PhD	PD/PI	203,700.00	1.2			20,370.00	5,307.00	25,677.00	
2 .	Tao		Wu	PhD	Co-Investigator	128,750.00	0.6			6,438.00	1,635.00	8,073.00	
Total Funds Requested for all Senior Key Persons in the attached file													
Additional Senior Key Persons:			File Name:								Total Senior/Key Person		33,750.00

B. Other Personnel							
Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits*	Funds Requested (\$)*
2	Post Doctoral Associates	18			87,831.00	24,486.00	112,317.00
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
2	Total Number Other Personnel					Total Other Personnel	112,317.00
Total Salary, Wages and Fringe Benefits (A+B)							146,067.00

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, Budget Period 4

UEI*: FXKMA43NTV21
Budget Type*: ☐ Project ☒ Subaward/Consortium
Organization: Baylor College of Medicine

Start Date*: 07-01-2026 End Date*: 06-30-2027 Budget Period: 4

C. Equipment Description	
List items and dollar amount for each item exceeding \$5,000	
Equipment Item	Funds Requested (\$)*
Total funds requested for all equipment listed in the attached file	
Total Equipment	
Additional Equipment: File Name:	

D. Travel	Funds Requested (\$)*
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	3,000.00
2. Foreign Travel Costs	
Total Travel Cost	3,000.00

E. Participant/Trainee Support Costs	Funds Requested (\$)*
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other:	
Number of Participants/Trainees	Total Participant Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, Budget Period 4

UEI*: FXKMA43NTV21

Budget Type*: ☐ Project ☒ Subaward/Consortium

Organization: Baylor College of Medicine

Start Date*: 07-01-2026

End Date*: 06-30-2027

Budget Period: 4

F. Other Direct Costs			Funds Requested (\$)*
1. Materials and Supplies			54,233.00
2. Publication Costs			
3. Consultant Services			
4. ADP/Computer Services			
5. Subawards/Consortium/Contractual Costs			
6. Equipment or Facility Rental/User Fees			
7. Alterations and Renovations			
8. Animal Maintenance			7,700.00
9. Laboratory Analysis/Core Charges			19,000.00
Total Other Direct Costs			80,933.00

G. Direct Costs	Funds Requested (\$)*
Total Direct Costs (A thru F)	230,000.00

H. Indirect Costs			
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	Funds Requested (\$)*
1 . MTDC	60	230,000.00	138,000.00
Total Indirect Costs			138,000.00
Cognizant Federal Agency		DHHS, Arif Karim, 214-767-3261	
(Agency Name, POC Name, and POC Phone Number)			

I. Total Direct and Indirect Costs	Funds Requested (\$)*
Total Direct and Indirect Institutional Costs (G + H)	368,000.00

J. Fee	Funds Requested (\$)*

K. Total Costs and Fee	Funds Requested (\$)*
	368,000.00

L. Budget Justification*	File Name:
	BCM_Budget_Justification_10_26_221018160151.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

Contact PD/PI: Xiao, Han

OMB Number: 4040-0001
Expiration Date: 12/31/2022

RESEARCH & RELATED BUDGET - SECTION A & B, Budget Period 5

UEI*: FXKMA43NTV21
Budget Type*: ☐ Project ☒ Subaward/Consortium
Enter name of Organization: Baylor College of Medicine

Start Date*: 07-01-2027 End Date*: 06-30-2028 Budget Period: 5

A. Senior/Key Person													
Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary (\$)	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits (\$)*	Funds Requested (\$)*	
1 .	Xiang		Zhang	PhD	PD/PI	203,700.00	1.2			20,370.00	5,307.00	25,677.00	
2 .	Tao		Wu	PhD	Co-Investigator	128,750.00	0.6			6,438.00	1,635.00	8,073.00	
Total Funds Requested for all Senior Key Persons in the attached file													
Additional Senior Key Persons:			File Name:								Total Senior/Key Person		33,750.00

B. Other Personnel							
Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary (\$)*	Fringe Benefits*	Funds Requested (\$)*
2	Post Doctoral Associates	18			87,831.00	24,486.00	112,317.00
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
2	Total Number Other Personnel					Total Other Personnel	112,317.00
Total Salary, Wages and Fringe Benefits (A+B)							146,067.00

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, Budget Period 5

UEI*: FXKMA43NTV21
Budget Type*: ☐ Project ☒ Subaward/Consortium
Organization: Baylor College of Medicine

Start Date*: 07-01-2027 End Date*: 06-30-2028 Budget Period: 5

C. Equipment Description	
List items and dollar amount for each item exceeding \$5,000	
Equipment Item	Funds Requested (\$)*
Total funds requested for all equipment listed in the attached file	
Total Equipment	
Additional Equipment: File Name:	

D. Travel	Funds Requested (\$)*
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	3,000.00
2. Foreign Travel Costs	
Total Travel Cost	3,000.00

E. Participant/Trainee Support Costs	Funds Requested (\$)*
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other:	
Number of Participants/Trainees	Total Participant Trainee Support Costs

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, Budget Period 5

UEI*: FXKMA43NTV21

Budget Type*: ☐ Project ☒ Subaward/Consortium

Organization: Baylor College of Medicine

Start Date*: 07-01-2027

End Date*: 06-30-2028

Budget Period: 5

F. Other Direct Costs			Funds Requested (\$)*
1. Materials and Supplies			54,233.00
2. Publication Costs			
3. Consultant Services			
4. ADP/Computer Services			
5. Subawards/Consortium/Contractual Costs			
6. Equipment or Facility Rental/User Fees			
7. Alterations and Renovations			
8. Animal Maintenance			7,700.00
9. Laboratory Analysis/Core Charges			19,000.00
Total Other Direct Costs			80,933.00

G. Direct Costs		Funds Requested (\$)*
Total Direct Costs (A thru F)		230,000.00

H. Indirect Costs			
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	Funds Requested (\$)*
1 . MTDC	60	230,000.00	138,000.00
Total Indirect Costs			138,000.00
Cognizant Federal Agency		DHHS, Arif Karim, 214-767-3261	
(Agency Name, POC Name, and POC Phone Number)			

I. Total Direct and Indirect Costs		Funds Requested (\$)*
Total Direct and Indirect Institutional Costs (G + H)		368,000.00

J. Fee	Funds Requested (\$)*

K. Total Costs and Fee	Funds Requested (\$)*
	368,000.00

L. Budget Justification*	File Name:
	BCM_Budget_Justification_10_26_221018160151.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

Budget Justification

Baylor College of Medicine

Key Personnel

Xiang Zhang, Ph.D., 10% (1.2 cal mos) is the MPI of this project. He will oversee this project and provide scientific leadership to the research team. Dr. Zhang has been working in the field of cancer metastasis for over 13 years, and published over 50 papers in first-rate journals. He will work with graduate student and research staff on experimental design, analysis and manuscript preparation. He will also teach the incoming postdoctoral fellow experimental skill to fulfill the proposed experiments.

Tao Wu, Ph.D., 5% (0.6 cal mos) will be a Co-Investigator. Dr. Wu's research focuses on how the epigenetic factors dynamically regulate the cell fate decision and the response to external stimulation, specifically underlying cancer therapeutic resistance, which well suits this project. Dr. Wu is currently a tenure-track assistant professor at BCM. His lab has established a full pipeline for epigenomic profiling and analysis.

Other Personnel

Xiaoxin Hao, Ph.D., 100% (12 cal mos), is a Postdoctoral Associate in Zhang Lab. His previous Ph.D work focuses on hematopoiesis. During his training in Zhang Lab, he has acquired strong skills for bone metastasis research. He also has superb skills to conduct most molecular and cellular biology.

Zhan Xu, Ph.D., 50% (6 cal mos) is a Postdoctoral Associate in Zhang Lab. He received his Ph.D training in the field of bone biology in this project. He has all the required skills for implementation of bone-related experiments (e.g., Aim 3.2). In addition, he is also well trained for bone metastasis assays.

Other Significant Contributor

C. Kent Osborne, Ph.D., 0% (0 cal mos) will serve as Other Significant Contributor on the project. Dr. Osborn is the funding director of Dan L. Duncan Comprehensive Cancer Center, Dudley and Tina Sharp Chair for Cancer Research, and the Professor of Medicine at Baylor College of Medicine. He is a clinical oncologist with over 40 years of experience and a research focus on breast cancer. Dr. Osborne will provide advice for future translation of the proposed studies as needed and there is no measurable effort for his contribution.

Qian Zhu, Ph.D., 0% (0 cal mos) will serve as an Other Significant Contributor on the project. Dr. Zhu is a leading computational biologist specialized in integrated single cell omics and spatial transcriptomics. His previous works established new methodologies for discovery of spatial features of gene/pathway expression. He will help perform analysis described in Aim 3.1. Since the dataset will be obtained as a part of another project and the proposed analyses can be done using the same pipeline, he is not requesting salary coverage from this application.

Fringe Benefits

The fringe benefits are specifically identified to each employee and are charged individually as direct cost.

Other Direct Costs

Materials and Supplies: The amount requested in each of the following categories has been derived by cost accounting using actual expenditures (from purchase orders) over the past year on a project involving similar quantity of experiments and personnel.

Lab supplies (glassware, rubber items, plastic, paper supplies) for all routine molecular and cellular experiments: \$5,000.

Antibodies for immunofluorescence staining, western blot, and flow cytometry: \$8,000.

Chemicals including buffers, ingredients of buffers, culture mediums, fluorescence dyes, reagents for molecular experiments: \$4,233.

Kits for molecular experiments including but not limited to DNA/RNA/Protein purification, RT-PCR, and DNA transduction: \$5,000.

Animal Purchase: We request funds to cover the purchase cost of 400 animals per year (see Research Strategy), including NSG, athymic nude and C57/B6 mice. The average cost of one mouse is about \$80

based on the actual expenditure in the past year). Therefore, the total cost for each year is expected to be \$32,000.

Travel

Funds are requested for travel of both the PI and postdoctoral associate to attend one national meeting per year (e.g., San Antonio Breast Cancer Symposium): \$3,000 each year.

Animal Maintenance cost at BCM is 0.77/cage/day. For the 400 purchased mice each year, based on the actual records of the past year and the expected total number of mice, we expect to maintain 100 cages with each cage for 100 days for each year. Based on these numbers, the cost of animal per diem maintenance will be \$7,700 each year.

Laboratory Analysis/Core Charges: Based on the actual expenditure of the past year, funds will cover the cost of service from the following:

Pathology Core at BCM for regular paraffin-embedding and microtome sectioning: \$1,000

Integrated Microscopy Core: \$1,000

Flow Cytometry Core: \$5,000

Next generation sequencing Core: \$12,000

Subtotal: \$19,000 each year.

Indirect Costs

BCM's facilities and administrative cost rates are established in accordance with the cost principles outlined in OMB PART 200—UNIFORM ADMINISTRATIVE REQUIREMENTS, COST PRINCIPLES, AND AUDIT REQUIREMENTS FOR FEDERAL AWARDS and agreed upon in negotiation with the federal government. The Indirect Cost Rate at Baylor College of Medicine is 60% from July 1, 2022 to June 30, 2026 for On-Campus Research.

RESEARCH & RELATED BUDGET - Cumulative Budget

	Totals (\$)	
Section A, Senior/Key Person		168,750.00
Section B, Other Personnel		561,585.00
Total Number Other Personnel	10	
Total Salary, Wages and Fringe Benefits (A+B)		730,335.00
Section C, Equipment		
Section D, Travel		15,000.00
1. Domestic	15,000.00	
2. Foreign		
Section E, Participant/Trainee Support Costs		
1. Tuition/Fees/Health Insurance		
2. Stipends		
3. Travel		
4. Subsistence		
5. Other		
6. Number of Participants/Trainees		
Section F, Other Direct Costs		404,665.00
1. Materials and Supplies	271,165.00	
2. Publication Costs		
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Other 1	38,500.00	
9. Other 2	95,000.00	
10. Other 3		
11. Other 4		
12. Other 5		
13. Other 6		
14. Other 7		
15. Other 8		
16. Other 9		
17. Other 10		
Section G, Direct Costs (A thru F)		1,150,000.00
Section H, Indirect Costs		690,000.00

Section I, Total Direct and Indirect Costs (G + H)	1,840,000.00
Section J, Fee	
Section K, Total Costs and Fee (I + J)	1,840,000.00

Contact PD/PI: Xiao, Han

Total Direct Costs less Consortium F&A

NIH policy (NOT-OD-05-004) allows applicants to exclude consortium/contractual F&A costs when determining if an application falls at or beneath any applicable direct cost limit. When a direct cost limit is specified in an FOA, the following table can be used to determine if your application falls within that limit.

Categories	Budget Period 1	Budget Period 2	Budget Period 3	Budget Period 4	Budget Period 5	TOTALS
Total Direct Costs less Consortium F&A	490,000	490,000	490,000	490,000	490,000	2,450,000

PHS 398 Cover Page Supplement

OMB Number: 0925-0001

Expiration Date: 09/30/2024

1. Vertebrate Animals Section

Are vertebrate animals euthanized? ☒ Yes ☐ No

If "Yes" to euthanasia

Is the method consistent with American Veterinary Medical Association (AVMA) guidelines?

☒ Yes ☐ No

If "No" to AVMA guidelines, describe method and provide scientific justification

.....

2. *Program Income Section

*Is program income anticipated during the periods for which the grant support is requested?

☐ Yes ☒ No

If you checked "yes" above (indicating that program income is anticipated), then use the format below to reflect the amount and source(s). Otherwise, leave this section blank.

*Budget Period	*Anticipated Amount (\$)	*Source(s)
----------------	--------------------------	------------

PHS 398 Cover Page Supplement

3. Human Embryonic Stem Cells Section

*Does the proposed project involve human embryonic stem cells? ☐ Yes ☒ No

If the proposed project involves human embryonic stem cells, list below the registration number of the specific cell line(s) from the following list: http://grants.nih.gov/stem_cells/registry/current.htm. Or, if a specific stem cell line cannot be referenced at this time, check the box indicating that one from the registry will be used:

☐ Specific stem cell line cannot be referenced at this time. One from the registry will be used.

Cell Line(s) (Example: 0004):

4. Human Fetal Tissue Section

*Does the proposed project involve human fetal tissue obtained from elective abortions? ☐ Yes ☒ No

If "yes" then provide the HFT Compliance Assurance

If "yes" then provide the HFT Sample IRB Consent Form

5. Inventions and Patents Section (Renewal applications)

*Inventions and Patents: ☐ Yes ☐ No

If the answer is "Yes" then please answer the following:

*Previously Reported: ☐ Yes ☐ No

6. Change of Investigator/Change of Institution Section

☐ Change of Project Director/Principal Investigator

Name of former Project Director/Principal Investigator

Prefix:

*First Name:

Middle Name:

*Last Name:

Suffix:

☐ Change of Grantee Institution

*Name of former institution:

PHS 398 Research Plan

OMB Number: 0925-0001

Expiration Date: 09/30/2024

Introduction	
1. Introduction to Application (for Resubmission and Revision applications)	Introduction1018160158.pdf
Research Plan Section	
2. Specific Aims	Specific_Aims1018160159.pdf
3. Research Strategy*	Research_Strategy1018160171.pdf
4. Progress Report Publication List	
Other Research Plan Section	
5. Vertebrate Animals	Vertebrate_Animals1018160161.pdf
6. Select Agent Research	
7. Multiple PD/PI Leadership Plan	MultiPI_Leadership_Plan_202211021018160168.pdf
8. Consortium/Contractual Arrangements	ConsortiumContractualArrangement1018160169.pdf
9. Letters of Support	LettersSupport_Xiao1018160174.pdf
10. Resource Sharing Plan(s)	Resource_Sharing_Plan1018160163.pdf
11. Authentication of Key Biological and/or Chemical Resources	AuthenticationKeyBiological1018160164.pdf
Appendix	
12. Appendix	

This is a resubmission for R01 CA277838. This application was submitted in response to PA20-185 and received a percentile of 24th percentile. We are happy to find that three reviewers and panel members seemed to be very enthusiastic about the significance of the project and its potential to formulate novel antibody therapy against bone metastasis of breast cancer. The panel suggested that the scientific premise is strong and there are considerable technical and conceptual novelties. We made the following revisions:

1) Why not considering to modify FDA-approved EZH2 inhibitor tazemetostat. Tazemetostat as an FDA-approved drug would be a better choice. *Response: We appreciate the reviewer's comment. We have included both GSK126 and tazemetostat with established efficacy and toxicity as model drugs. We successfully prepared the bone-targeting tazemetostat (Fig. 9C). To our delight, consistent with GSK126, the resulting bone-targeting tazemetostat doesn't alter its inhibitory activity against EZH2 (Fig. 9E), but exhibits significantly higher uptake in bone tumor than wildtype tazemetostat (Fig. 10A and D). We will next evaluate the efficacy of bone-targeting tazemetostat for the treatment of BCa metastases in the bone as well as the inhibition of secondary metastases from bone lesions to multi-organs as demonstrated in Fig. 12I.*

2) Some preliminary data demonstrating GSK126's anti-metastatic breast cancer activity in vivo would strengthen the application. *Response: We have successfully established the superiority of bone-targeting GSK126 to prevent or impede secondary metastases from bone lesions in a xenograft MDA-MB-231 model as well as an immunocompetent model (Fig. 12I and J). 2×10^5 MDA-MB-231 cells were introduced to the bone by an intra-iliac artery (IIA) injection, followed by the treatment of wildtype or bone-targeting GSK126 inhibitors. As shown in Figure 12I, treatment with bone-targeting GSK126 significantly reduces the frequency of secondary metastases to the hindlimb, vertebrae, forelimbs, ribs, sternum, cranium, liver, spleen, and brain. In our preliminary study, bone-targeting GSK126 also showed encouraging efficacy to inhibit secondary metastases in C57BL mice (Fig. 12J). Next, we will increase the mouse number to get the statistical data.*

3) Mouse models in this proposal use immunodeficient mice and the type and usage of immunocompetent mice is not well described. *Response: Since 57% of patients with bone metastasis are hormone receptor (HR)-positive,¹ we have included several ER+/HER2- xenograft and PDX models, including MCF-7, ZR-75-1, and HCl011/WHIM9 in this study. Furthermore, we have included several syngeneic triple negative breast cancer (TNBC) models, including 4T1 and EO771 models. In our preliminary study with LLC tumor-bearing immunocompetent mice, the bone-targeting GSK126 established an enhanced efficacy to inhibit the metastasis-to-metastasis from bone lesions (Fig. 12J). We will not be able to use ER+ syngeneic models due to the lack of syngeneic mouse models for this subtype of breast cancer. This has been a long-standing, field-wide challenge. We have been delightedly establishing new collaborations to overcome this challenge. In this case, we will employ a series of newly developed Rat-based models that develop ER+ mammary tumors that spontaneously metastasize. This series of models are developed by Dr. Yi Li's group, and are currently being characterized collaboratively by both of our laboratories (please see Dr. Yi Li's support letter).*

4) Some concerns of modified EZH2i compromise the significance of proposal; It is unclear whether bone-targeting EZH2i could keep the same activity as EZH2i. *Response: To explore the effect of bisphosphonate modification, we have evaluated the potencies of wildtype (Taz) and bone-targeting tazemetostat (Taz-ALN) against EZH2 using a reported EZH2 activity assay.² In our preliminary data, we didn't observe a significant activity difference between wildtype and bone-targeting tazemetostat (Fig. 9E). The piperazine ring of GSK126 and the morpholine ring of tazemetostat are solvent-exposed, thus modification of GSK126 and tazemetostat at these sites will have minimal disruption on its activity.^{3,4} In fact, several chemical groups (e.g., 4-Hydroxythalidomide, thienotriazolodiazepine, Olaparib, and others) have been conjugated to the morpholine ring of tazemetostat.⁴⁻⁷ The modifications of tazemetostat at the morpholine ring don't alter its activity to inhibit EZH2, thus we didn't expect our bone-targeting EZH2i has reduced activity. Furthermore, in case that we observed any activity reduction after bisphosphonate conjugation, we have designed a series of cleavable bone-targeting EZH2i (Fig. 9C, 9D, and 11). After the cleavage by Cathepsin K, wild type EZH2i will be generated.*

5) Other concerns of the reviewers are addressed here: Reviewer 2. There is a lack of knowledge of EZH expression in primary human samples, which could be very different to mouse models or cell line data. *Response: Through a collaboration with Dr. Arun Sreekumar and Dr. Ruth Asirvatham, we have obtained a number of samples of paired primary tumors and bone metastases. We will examine the EZH2 expression in these samples to complement our findings in pre-clinical models. Figure 5b is a bit confusing. It suggests the EZH2 signature is less in bone, contrary to what is written in the text; Response: EZH2 is a repressor, so lower gene expression suggests higher EZH2 activity; Reviewer 3. The use of bisphosphonate conjugate as bone-targeting strategy for anticancer drugs is an established technology. Response: As Reviewer 1 said "The concept that delivery of epigenetic inhibitors to bone to overcome therapeutic resistance, eradicate these metastatic seeds, and prevent secondary metastases from bone niches is innovative" and Reviewer 2 said "The proposal is also very innovative in that it is the first to target bone metastases epigenetically", this study will, for the first time, reveal how the specific delivery of epigenetic inhibitors to the bone cancer niche can affect bone tumor progression and multi-organ metastases development from the bone lesions.*

Metastasis to distant organs is the major cause of solid cancer-related death. Bone is the most frequent destination of metastasis. In breast cancer, over 47% of first-site metastases occurs to the bone, as compared to 19% to lung, 5% to liver and 2% to brain⁸. Patients with the skeletal system as the only site of metastasis usually have a better prognosis than those with visceral organs involved.^{9,10} In more than two-thirds of cases, bone metastases will not be confined to the skeleton, but rather are associated with subsequent metastases to other organs and eventually deaths of patients.^{9,10} Recent genomic analyses suggested that the majority of metastases result from seeding from other metastases, rather than primary tumors.^{11–13} Some metastases first found in non-bone organs can be seeded from sub-clinical bone micrometastases (BMMs) as well, as cancer cells arrived in the bone can acquire more aggressive phenotypes even before establishing overt metastases.¹⁴ Taking advantage of a recently developed approach that selectively delivers cancer cells to hind limb bones,^{15–17} we also uncovered frequent metastatic seeding from established bone lesions to multiple other organs. *Thus, strategies to inhibit BMMs can prevent metastasis to the bone as well as secondary metastases to multi-organs from the bone.*

Epigenetic manipulation can reshape the tumor microenvironment and restore effective antitumor response, thus providing a promising avenue for cancer therapy. Enhancer of zeste homolog 2 (EZH2), a classic histone methyl transferase, has been recognized as a key modulate in cancer progression that associates with the poor overall survival in patients with metastatic cancers.^{18–21} Studies indicate that inhibition of EZH2 by small molecular inhibitors significantly reduces cancer cell growth and tumor formation.^{22–24} Despite enthusiasm shared by the community for EZH2 inhibitor (EZH2i) as a cancer therapy, studies have raised concerns over immune suppression associated with EZH2 inhibition, which neutralizes EZH2is' antitumor activities. Indeed, knockdown of human EZH2 in T cells dampens their antitumor immunity *in vivo*.²⁵ Systemic EZH2 blockade also results in an increased number of myeloid-derived suppressor cells, which suppresses T cell functions.²⁶ Furthermore, studies have raised concerns over potentially toxic or even adverse effects associated with EZH2 inhibition in normal tissues.^{27–29} EZH2 is reported to be a crucial factor in maintaining the self-renewal capacity of different tissue-specific stem and progenitor cells.^{27,28} *Thus, systemic EZH2 inhibition counteracts the antitumor activities of EZH2i by inhibiting immune cell functions and may cause greater toxicity by impairing normal functions of EZH2.*

In this proposal, we propose to develop a series of bone tumor-targeting EZH2is and study their effects on treating and preventing cancer metastases to the bone and other organs via a fruitful collaboration between labs at Rice University and Baylor College of Medicine (**Fig. 1**). *Our preliminary data* establish that: 1) Interactions between cancer cells and the osteogenic bone microenvironment elevate EZH2 level, rendering EZH2 as a promising therapeutic target for bone cancers; 2) Inhibition of EZH2 restores the sensitivity of bone metastases to endocrine therapies; 3) EZH2i modified with bone-targeting moiety, bisphosphonate, has superb binding affinity for bone *in vitro* and *in vivo* and exhibits enhanced activities for inhibiting metastatic seeding from bone lesions to other organs *in vivo*; 4) using an intra-iliac artery (IIA) injection developed in Zhang lab (BCM), can be used to effectively model the bone metastatic niche, providing a powerful platform for evaluating the therapeutic efficacy against bone metastatic cancer and "metastasis-to-metastasis" seeding from bone lesions. Based on these results, our *central hypothesis* is that bisphosphonate modification of EZH2i will result in higher EZH2i concentrations in the bone metastatic niche, relative to other tissues, thus providing a targeted therapy against bone metastases as well as further multi-organ metastases seeding from bone lesions.

Aim 1. Prepare and characterize bone tumor-targeting EZH2 inhibitors for the treatment of bone metastasis. To date, no strategy has been reported for a specific delivery of epigenetic modulators to the bone. In this aim, we will develop both bone tumor-targeting EZH2 inhibitor and a prodrug that can inhibit EZH2 in osteolytic metastases and as well explore their efficacies in the treatment of bone metastases.

Aim 2. Evaluate therapeutic benefits of bone tumor-targeting EZH2 inhibitors *in vivo*.

EZH2 is up-regulated in the bone microenvironment and drives ER+ bone metastases toward a basal and stem-like state. We will, for the first time, evaluate bone tumor-specific epigenetic therapy as a means of treating BCa metastases in bone as well as inhibiting secondary metastases from bone lesions to multi-organs.

Aim 3. Dissect the molecular mechanisms underlying the benefits of bone tumor-targeting EZH2 inhibitors. Inhibition of EZH2 is emerging as a new therapeutic approach for the treatment of cancers. However, previous reports suggest that systemic EZH2 inhibition suppresses T cell response, thus dampening antitumor immunity. Our preliminary data indicates EZH2i modified with bone-homing moiety exhibited improved antitumor activities. In this aim, we will dissect how bone-targeting EZH2is diminish the stemness of metastatic cells and reprogram its tumor microenvironment, while potentially circumventing the undesirable immunosuppression effects observed during systemic EZH2 inhibition.

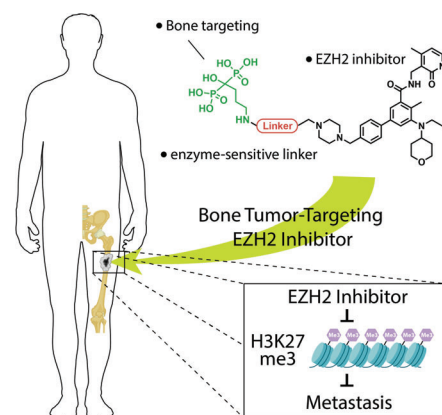


Figure 1. Epigenetic modulators can be specifically delivered to the bone by conjugation of bisphosphonate molecules using enzyme-sensitive linkers.

BACKGROUND AND SIGNIFICANCE

There is an urgent need to improve outcomes of patients with bone metastases. Despite an over 90% 5-year survival rate, 20-40% of breast cancer (BCa) survivors will eventually suffer metastases to distant organs, sometimes years after surgeries.³⁰ Over 47% first-site metastases occurs to the bone, as compared to 19% to the lung, 5% to the liver and 2% to the brain.⁹ BCa patients with bone metastases result in symptoms of pain and immobility, and skeletal-related events (SREs) such as fractures, bone pain, spinal cord compression, and frequently hypercalcemia. SREs reduce the quality of life and increase mortality. The one-year survival rate of BCa patients with bone metastases is 51%, but the five-year survival rate drops to 13%.^{31,32} *Thus, therapies with improved outcomes for BCa patients with bone metastases are highly desired.*

“Metastasis to Metastasis” from the bone lesion is prevalent. Patients with skeletons as the only site of metastasis usually have a better prognosis than those with visceral organs involved.^{9,10} Unfortunately, in more than two-thirds of cases, bone metastases will not be confined to the skeleton, but rather are associated with subsequent metastases to other organs.^{9,10} Recent genomic analyses suggested that the majority of metastases result from seeding from other metastases, rather than primary tumors.¹¹⁻¹³ Some metastases first found in non-bone organs can be seeded from sub-clinical bone micrometastases (BMMs), as cancer cells arrived in the bone can acquire more aggressive phenotypes even before establishing overt metastases.¹⁴ Taking advantage of new bone metastasis models and approaches,¹⁵⁻¹⁷ we also uncovered frequent metastatic seeding from established bone lesions to multiple other organs (**Fig. 4**). *Thus, strategies to inhibit BMMs can prevent BCa metastasis to the bone as well as secondary metastases to multi-organs from the bone.*

Bone metastases is more frequent to occur in estrogen receptor-positive (ER+) breast cancer. The ER+ subtype accounts for over 70% of all breast cancer, and after recurring, causes over 24,000 deaths per year in the US. Adjuvant endocrine therapies target ER and significantly reduce metastatic recurrences. However, 20-40% of patients still develop metastases, often after a prolonged latency.^{33,34} Endocrine resistance is a major problem encountered in the clinic.³⁵ Bone is the most frequently affected organ by ER+ breast cancer, which is usually luminal-like.^{36,37} Compared to the basal-like subtype, luminal A/B breast cancer exhibits a 2.5-fold increased frequency of bone metastasis, but a 2.5-fold decreased frequency of lung metastasis³⁶. We recently found that the direct and paracrine interactions between ER+ breast cancer cells and the osteogenic bone microenvironment elicit epigenetic adaptation, a process independent of clonal selection and leading to persistent endocrine resistance (**Fig. 5 and 6**). *Thus, it is imperative to develop novel therapy to overcome therapeutic resistance, eradicate these metastatic seeds, and prevent secondary metastases from bone niches.*

Epigenetic modulation is emerging as an important cancer treatment, but tumor-targeted delivery is needed. Epigenetic changes are increasingly being appreciated as key events in cancer progress and metastasis. Thus, strategies to modulate the epigenetic state of the cancer cells provide a promising therapeutic intervention for the precision treatment of a broad spectrum of cancers with similar epigenetic characteristics. Enhancer of zeste homolog 2 (EZH2), a classic histone methyl transferase, has been recognized as a key modulate in breast cancer progression that associates with the poor overall survival in patients with metastatic breast cancers.¹⁸⁻²¹ Studies indicate that inhibition of EZH2 by small molecular inhibitors significantly reduces cancer cell growth and tumor formation.²²⁻²⁴ Our previous studies indicate that blockade of EZH2 may reverse the microenvironment-induced epigenomic reprogramming, alleviate acquired endocrine resistance of BMMs, and reduce subsequent metastatic recurrences (**Fig. 6 and 7**).³⁸ Our preliminary data also indicate that EZH2 is a master regulator of secondary metastases from bone lesions (**Fig. 7**). We showed that the inducible knockdown of EZH2 in cancer cells dramatically decreased secondary metastases (**Fig. 7E, F, and G**). These data add support to the potential clinical significance of EZH2 as a therapeutic target. *Despite enthusiasm shared by the community for EZH2 inhibitor (EZH2i) as a cancer therapy, studies have raised concerns over the immune suppression and potentially toxicity associated with systemic EZH2 inhibition.*^{25-29,39} In a preclinical mouse model, knockdown of human EZH2 in T cells has been shown to dampen the antitumor immunity *in vivo* and significantly increase lung metastasis.²⁵ An analysis of 135 ovarian patients' tissue samples indicates EZH2 mediates potent antitumor immunity of T cells in human cancer and is associated with long-term survival.²⁵ Furthermore, the systemic EZH2 inhibition also results in an increased number of myeloid-derived suppressor cells and lower number of both CD4+ and CD8+ T cells, which weakens T cell-mediated antitumor immunity.²⁶ Because of this reason, EZH2i that has established great success in the preclinical studies using immunodeficient hosts, only yielded suboptimal outcomes for cancer patients in the clinical trials.⁴⁰ For example, GSK126 has been identified as an effective drug in Karpas-422 xenograft tumors.⁴¹ But only a modest anti-tumor activity was observed at the tolerable doses of GSK126 in the phase I clinical trial.⁴² In our study of secondary metastasis, the impact of EZH2 was evident only when the inhibition was targeted to cancer cells by pre-treatment or inducible genetic knockdown (**Fig. 7**). In sharp contrast, systemic administration of EZH2i could not reduce multi-organ secondary metastases, suggesting a more complex effect when EZH2i is applied to multiple cell types (**Fig. 12**). The safety profile of systemic delivery of EZH2i is another concern. EZH2 is reported to be a crucial factor in maintaining the self-renewal capacity of different tissue-specific stem and progenitor cells.^{27,28} Systemic EZH2 inhibition may cause greater toxicity by impairing EZH2's activity in other cell types as well as its emerging non-enzymatic

functions.^{43–45} *Thus, a tumor-specific delivery approach using the minimal effective dosage is likely to be a wise strategy for epigenetic therapy.*

New concepts and multidisciplinary methods to address unaddressed problems: Bone tumor-targeted epigenetic prodrug is a promising therapeutic intervention. EZH2 is a reliable marker for cancer stemness and plays a central role in loss of ER and gain of plasticity. Our recent studies show that cancer-cell-intrinsic EZH2 expression level was specifically increased in the bone microenvironment (BME) compared with other organs, which renders EZH2 inhibition as an attractive approach to treat bone cancer cells.^{38,46} Dosing the bone microenvironment has proven to be difficult due to relatively low vascularization and the physical barriers of penetration. Attempts to ensure effective concentrations of a therapeutic in bone drive high concentrations in other tissues, resulting in adverse systematic effects or side effects that may limit or exclude the use of a therapeutic in the bone. Thus, the potential benefits of passive targeting are lost.^{47,48} Taken the advantage of the high mineral concentration unique to the bone hydroxyapatite matrix, bisphosphonate (BP) has been used to selectively deliver small molecule drugs, including bis-(2-chloroethyl)-amine derivatives, cisplatin analogues, doxorubicin, methotrexate, gemcitabine, camptothecin, and others, to the bone for the treatment of osteoporosis, primary and metastatic bone neoplasms, and other bone disorders.^{49–65} *An important feature of bisphosphonates is that the uptake of the bisphosphonate drugs in bone tumor sites is 10-20 fold higher than in healthy bone tissue*, since they have been shown to favor binding to osteolytic bone metastasis niches with higher levels of hydroxyapatite crystallinity.^{66–68} *Our preliminary data* establish that: (1) EZH2i modified with bisphosphonates, have superb binding affinity for the bone, and exhibit enhanced bone metastasis sites targeting *in vivo*; and (2) EZH2i modified with bisphosphonates exhibited improved activities for inhibiting metastatic seeding from bone lesions to other organs; and (3) an intra-iliac artery (IIA) injection, developed in our co-investigator, Dr. Zhang's lab (BCM), can be used to effectively model the bone metastatic niche, providing a powerful platform for evaluating the therapeutic efficacy against bone metastatic cancers and “metastasis-to-metastasis” seeding from bone lesions. Based on these results, our *central hypothesis* is that bisphosphonate modification of EZH2 inhibitor will result in higher EZH2 inhibitor concentrations in the bone metastatic niche, relative to other tissues, thus providing a targeted therapy against bone metastases as well as further multi-organ metastases seeding from bone lesions.

In this proposal, we will develop a bone tumor-targeting EZH2i (bEZH2i) and demonstrate its efficacy against bone micrometastases as well as further multi-organ metastases seeding from bone lesions. (Fig. 1). To ensure the specific release of bEZH2i in bone metastases, we will design a series of bone tumor-targeting EZH2 prodrugs that are effectively cleaved in a cathepsin K-dependent manner. Cathepsin K is essential for osteoclast-mediated bone degradation in which it functions in proteolytic pathways to promote the breast cancer bone metastasis.^{69–72} Furthermore, the cathepsin K expression level was specifically increased in cancer cells in BME compared with cells residing in other organs.^{73,74} Thus, the new bone tumor-targeting EZH2 prodrugs exploit both the bone tumor-targeting capability of bisphosphonates and the unique microenvironment of bone metastases to ensure a selective release of epigenetic modulators in the bone metastatic niches.

Long-term intent to translate and other potential users: *If successful*, our study will generate the first bone tumor-targeting epigenetic therapy for the precision treatment of breast cancer bone metastases, which will impact at least 9% of BCa patients carrying bone-only metastases. This study may also benefit BCa patients with other site metastases, since frequent metastasis-to-metastasis seeding from bone to other organs has been identified.^{11–13} Although we will focus on EZH2 inhibitors, this platform can also be applied for the design of bone tumor-targeting EZH2 proteolysis targeting chimeras (PROTACs). EZH2 inhibitors only showed limited efficacy in some cancers, e.g., SWItch/sucrose nonfermentable mutant tumors, castration-resistant prostate tumors, since these tumors rely on both catalytic and noncatalytic activities of EZH2.^{75–77} Thus, EZH2 PROTACs can completely block the oncogenic functions of EZH2 may provide a new strategy for cancer therapy.^{4,7} Our goal is to provide a rationale and establish feasibility for phase I clinical trials. *Most importantly*, our study may also benefit bone cancers and bone metastasis patients with other cancer types, since bone is also the preferential metastatic target site for prostate, lung, kidney, and thyroid primary cancers.

A multidisciplinary team: The proposed study is led by two interdisciplinary investigators who are supported by a team of senior collaborators with extensive clinical, and biostatistical expertise (**Fig. 2**). **Han Xiao, Ph.D.**—his research focuses on the development of chemical biology tools as novel therapeutic strategies. He was the first to prepare antibody-bisphosphonate conjugates (**Fig. 8**) and tested their efficacy for the treatment of bone metastases *in vivo*.^{78,79} **Xiang Zhang, Ph.D.**—his laboratory has over 12 years experience to investigate early-stage bone colonization of disseminated breast cancer cells and its therapeutic strategies. His lab was the first to discover that EZH2 is up-regulated in the bone microenvironment and drives ER+ BMMs toward a basal and stem-like state (**Fig. 5**).^{38,46} Moreover, EZH2 inhibition reverses endocrine resistant and reduces the stemness and metastasis capacity of cancer cells (**Figs. 6 and 7**). Drs. Zhang and Xiao have previously established successful collaboration based on the common interest in targeting the bone microenvironment associated with metastatic cancer.^{78,79} Drs. Xiao and Zhang will be supported by: **C. Kent Osborne, M.D.**—his research involves the identification of mechanisms of resistance to targeted therapies in breast cancer particularly treatments

targeting the estrogen receptor and the HER2 receptor. **Tao Wu, Ph.D.**—his research involves the deciphering the drivers of epigenetic mutations in cancer and development of new epigenetic therapies for cancer. The team will communicate via biweekly conference calls and in-person or Zoom meetings monthly.

INNOVATION

Conceptually, the proposed research will, for the first time, reveal how the specific delivery of epigenetic modulators to the bone metastatic niches can affect bone tumor progression and multi-organ metastases development from the bone lesions. Epigenetic therapy has proved to be effective for the treatment of haematological tumors and holds great potential for the treatment of other solid tumors.^{80–82} While most of these epigenetic targets are upregulated in the neoplastic tissue, it is well accepted that the activities of these targets remain at a low level in normal tissues.²⁴ Accordingly, data suggest that systemic epigenetic therapy may dampen the antitumor activities of the immune system and lead to excessive toxicity.^{25–29,39} Because of this reason, epigenetic therapy that has established great success in the treatment of tumors using immunodeficient hosts in preclinical trials, only yielded suboptimal outcomes for cancer patients in clinical trials.²⁵ Thus, a tumor-targeted delivery of epigenetic modulators will be a wise option in the clinical exploration of epigenetic therapy. In this study, the combined expertise from Xiao and Zhang Laboratories will allow elucidation of how bone tumor-targeted epigenetic therapy may alter the stemness of metastatic cells and reprogram its tumor microenvironment, while avoiding the potential side effects of systemic epigenetic therapy. The concept adopted in this proposal is innovative, because we currently have no strategy to specifically deliver epigenetic modulators to the metastatic bone tumors and our preliminary data suggest that specific delivery of EZH2 inhibitors to the bone tumor can be a powerful strategy for the treatment of breast cancer micrometastasis and the prevention of secondary metastatic dissemination from the initial bone lesions.

Methodologically, the proposed research is made feasible by the molecular engineering techniques in the Xiao lab at Rice University and a set of unique techniques and animal models to inspect, extract and monitor microscopic metastases in bone, developed in the Zhang lab at Baylor College of Medicine.^{15–17,83} Xiao lab developed the first bone-targeting antibodies and prepared several small molecule drugs, including monomethyl auristatin F, photosensitizers, for cancer therapy.^{78,79,84–88} IIA injection developed in the Zhang lab is a novel injection technology for micrometastases models, allowing for the selective delivery of cancer cells into hind limb bones without causing tissue damage.^{15–17} This technology enables sufficient time for some indolent cells to eventually colonize the bone as well as a large number of cancer cells to specifically colonize the bone, thereby enriching micrometastases in early stages, allowing swift detection and robust quantification of micrometastases. Furthermore, this injection technology provides a unique approach to studying metastasis-to-metastasis seeding from bone, since the timing and development of metastatic colonization can be strictly controlled and it prevents mice from succumbing to metastases in other vital organs before bone colonization.

Mechanistically, the application of bEZH2i will allow us to dissect the differential roles of EZH2 in various cell types during metastasis. BMMs closely interact with the microenvironment niches comprised of osteoblasts, osteoclasts, endothelial cells, and other cell types, which will likely be affected by bEZH2i as well. Furthermore, other bone marrow cells, especially the hematopoietic stem and progenitor cells, may also require EZH2 for normal functions. Therefore, although we expect that bEZH2i can be delivered in a more bone tumor-targeted manner, it remains critical to evaluate the “off-target” effects and to understand the mechanism of action of bEZH2i. Toward this end, we will characterize the impact of traditional EZH2i, bEZH2i, and inducible shRNA-mediated EZH2 knockdown (in cancer cells), and determine the consequent tumor-intrinsic, microenvironmental,

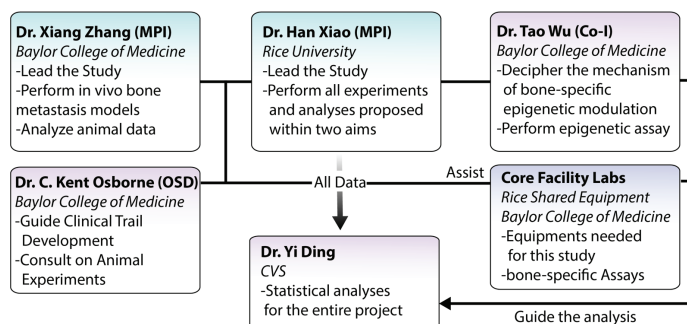


Figure 2. Organizational Structure.

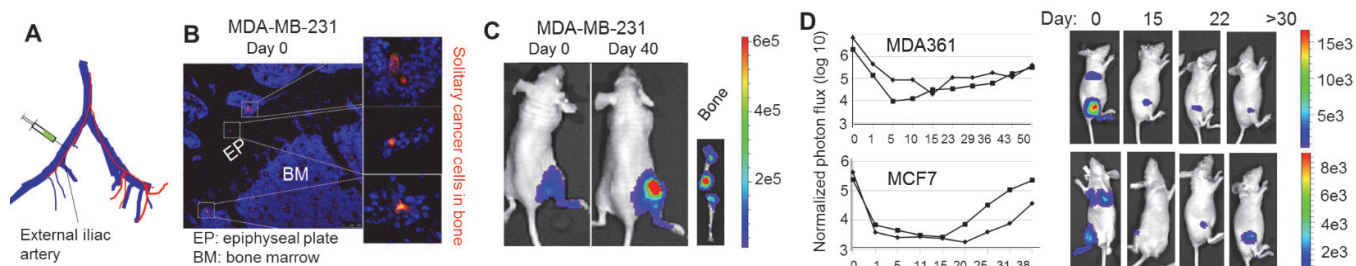


Figure 3. Indolent bone metastasis models based on intra-iliac artery (IIA) injection. (A) A schematic of IIA injection. (B) Representative pictures showing bioluminescence signals from cancer cells colonized in bone extracted from a mouse after IIA injection with 5×10^5 MDA-MB-231 cells. (C) Single cancer cells (MDA-MB-231) delivered to the bone and bone marrow by IIA injection. (D) Bone metastatic growth/dormancy traced by bioluminescence imaging. Two indolent cell lines, MDA-MB-361 and MCF-7, were injected via IIA. Left panel: The curves of bioluminescence signals (normalized to signals on Day 0) from two representative mice are shown for each cell line. Right panels: representative bioluminescence images at different time points post IIA injection of cancer cells.

and systemic effects. The comparisons among these treatments will help delineate cell compartment-specific roles of EZH2 in bone metastasis, which will be instrumental for further improvement of EZH2-targeting therapies and for generalization of bEZH2i to other cancer types that involve bone and bone marrow (e.g., osteosarcoma and leukemia).

Finally, this proposal will yield the first bone tumor-targeted epigenetic therapy. Bone cancer is driven by paracrine crosstalk among cancer cells, osteoblasts, and osteoclasts, which together constitute an osteolytic vicious niches. Bisphosphonates can specifically target these osteolytic bone niches over healthy bones, since they have been shown to favor binding to hydroxyapatite surfaces with higher levels of crystallinity.^{66–68} Our preliminary data indicate that bisphosphonate-modified epigenetic modulator has superb binding affinity for bone *in vitro*, and exhibits enhanced bone targeting and anti-tumor activity *in vivo* compared to the unmodified inhibitor.

To achieve a more specific delivery to bone tumor sites, we will engineer this bisphosphonate-modified epigenetic modulator caged with enzyme-cleavage linkers. The resulting caged drug will be only active in the bone cancer niches, due to the fact that the bone tumor sites are generally characterized by low pH and high level of cathepsin K.^{73,74} The resulting bone tumor-targeting epigenetic therapy regimens are likely to yield a better therapeutic profile by delivering a high concentration of therapeutic regimens to the bone and reducing adverse side effects associated with systemic delivery. The successful implementation of this strategy will also allow for the systematic assessment of repurposing epigenetic therapies currently in the preclinical and clinical study. Furthermore, this strategy can be potentially applied for the design of bone tumor-targeting EZH2 degrader. In preclinical studies, selective epigenetic enzyme degradation is emerging as a powerful approach for the treatment of breast cancer.^{4,7}

PRELIMINARY STUDIES

IIA – a novel technique to enrich microscopic bone metastases. We adopted intra-iliac artery (IIA) injection to selectively deliver a determined number of cancer cells into hind limb bones (Fig. 3A).^{15,17} We inoculated 5×10^5 MDA-MB-231 cells labeled with firefly luciferase and red fluorescent protein into the IIA of nude mice. A few hours after injection, single cancer cells were detected in the bone marrow close to the epiphyseal plate (Fig. 3B). Continuous monitoring through bioluminescence imaging revealed that MDA-MB-231 cells can readily colonize bone post-injection and form macrometastases (Fig. 3C). We also performed IIA injection using several ER+ cell lines including MDA-MB-361 (ER+HER2+) and MCF-7 (ER+HER2-). MDA-MB-361 cells maintained detectable signals for over fifty days, whereas the signals of MCF-7 cells initially decreased but spontaneously outgrew two weeks after injection (Fig. 3D). This approach provides three major advantages over traditional approaches. First, IIA injection concentrates tumor cells to hind limbs, prevents mice from succumbing to metastases in other vital organs, and allows sufficient time for some indolent cells to eventually colonize. Second, IIA injection delivers a large number of cells specifically to bone, thereby enriching micrometastases in early stages and allowing swift detection and robust quantification of micrometastases. Finally, IIA injection avoids physical damage to the bone and allows cancer cells to home to bone via circulation, thereby better mimicking the natural metastatic process compared to intratibial or intra-femoral injections.

Temporally lagged multi-organ metastases in mice carrying IIA-introduced bone lesion. IIA has been employed to investigate early-stage bone colonization. Both aggressive (e.g., MDA-MB-231) and relatively indolent (e.g., MCF-7) cells can colonize bone albeit following different kinetics. In both cases, cancer cell distribution is highly bone-specific at early time points, allowing us to dissect cancer-bone interactions without the confounding effects of tumor burden

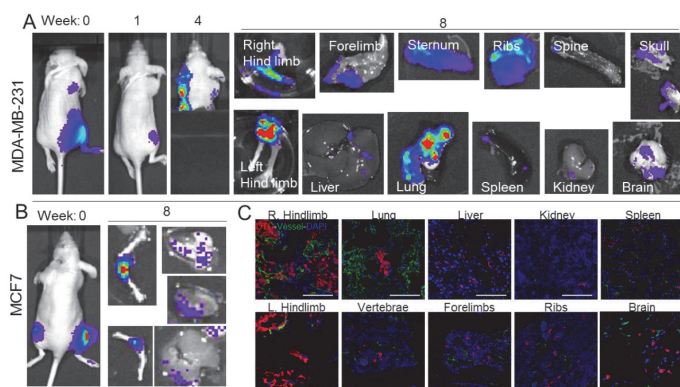


Figure 4. Temporally lagged multi-organ metastases following establishment of IIA-introduced bone lesions. (A) Representative bioluminescence images of temporal course of multi-organ metastasis arisen from IIA-introduced bone lesions of MDA-MB-231 cells. (B) The same as (A) except for MCF7 cells. (C) The confocal imaging confirms the presence of fluorescent protein-tagged MDA-MB-231 tumor cells across various tissues. Red, tumor cells; Green, CD31/V-Ecadherin+ endothelium; Blue, nucleus staining. Scale bar, 100 um.

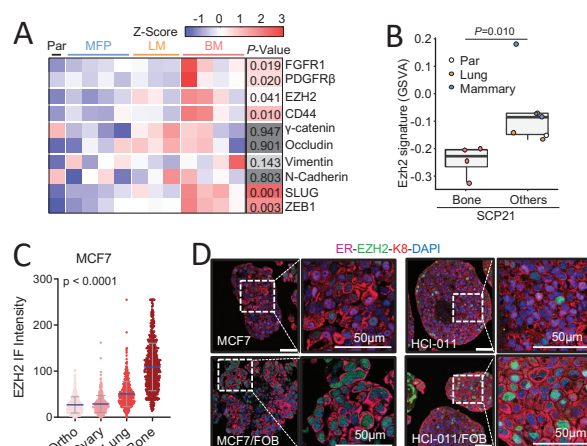


Figure 5. Upregulation of EZH2 in the bone cancer microenvironment. (A) Expression levels of proteins in the parent (Par), lung metastasis (LM), bone metastasis (BM)-entrained SCP21s. (B) Levels of EZH2 signature genes (GSVA) in bone entrained and other SCP21 cells. (C) IF EZH2 expression levels of MCF7 tumors at lung, ovary, bone, and mammary gland. (D) IF showing co-expression of ER and EZH2 in 3D monocultures and co-cultures (+FOB) models of MCF7 and PDX HCl011. Scale bars: 50 mm.

in other organs.^{93,94} However, as bone lesions progress, metastases begin to appear in other organs, including additional bones, lungs, liver, kidney, and brain, usually 8-12 weeks after IIA injection (**Fig. 4A**). This phenomenon is not specific for the highly invasive MDA-MB-231 cells, but was also observed in more indolent MCF-7 cells (**Fig. 4B**).

Bone microenvironment increases EZH2 activities in cancer cells. EZH2 is a reliable marker for cancer stemness.²⁴ Our preliminary studies strongly indicate EZH2 activation specifically in cancer cells colonizing bone.^{38,46} After examining stemness markers of SCP21 cells in different microenvironments, bone-entrained cells appeared to express a high level of EZH2 (**Fig. 5A and B**). This result was also validated in ER+ MCF-7 and PDX HCI011 cells. The intrinsic EZH2 expressions of both MCF-7 and PDX HCI011 cells were specifically increased in BME compared with other organs (**Fig. 5C and D**).

Short-term inhibition of EZH2 restores sensitivity of bone micrometastases to endocrine therapies. Since EZH2 mediates bone microenvironment-induced endocrine resistance, we demonstrated that inhibition of EZH2 should reverse this resistance and synergize with endocrine therapies.³⁸ The synergy is especially strong on bone-entrained MCF-7 cells (**Fig. 6A**). We also conducted *in vivo* experiments to test if combinatory treatment of tazemetostat and fulvestrant at the microscopic metastasis stage (to mimic adjuvant therapy) could lead to decreased bone colonization. At overt metastasis phase (4 weeks after “adjuvant” treatment), fulvestrant was again used for treatment to control tumor progression. In this setting, tazemetostat and fulvestrant had little or modest effects respectively as single agents. However, the combined treatment strongly inhibited bone colonization and rendered 50% of animals tumor-free by the end of experiment (**Fig. 6B-D**).³⁸ This data serves as proof-of-principle for combining anti-EZH2 treatment with SOC endocrine therapy to prevent metastatic recurrences.

EZH2 mediates the secondary metastases of some breast cancer cells. My lab has demonstrated that EZH2 mediates secondary metastasis from bone lesions (**Fig. 7A**) and transient inhibition of EZH2 completely abolished the enhanced metastasis of bone-entrained SCP21 (**Fig. 7B, C, and D**) and MCF7-SCP2 cells (REF).⁴⁶ To confirm the intrinsic role of EZH2 during the “Metastasis to Metastasis” from the bone lesion, we generated an inducible knockdown EZH2 cell line. Interestingly, whereas EZH2 knockdown did not alter primary bone lesion development, but dramatically reduced secondary metastasis to other organs (**Fig. 7E, F, and G**). These results indicate that EZH2 is a major regulator of secondary metastases from bone lesions.

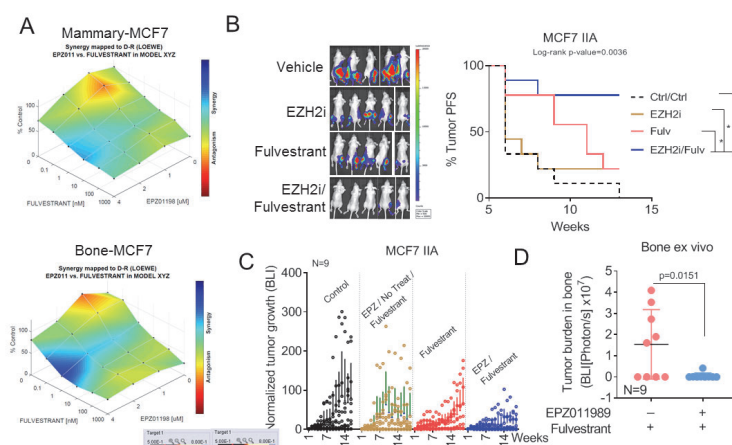


Figure 6. EZH2i enhances endocrine response in breast cancer bone metastasis. **(A)** LOEWE analysis of fulvestrant and tazemetostat combination in 3D co-culture of mammary (top) and bone-entrained (bottom) MCF7 cells. **(B)** Representative images of fulvestrant and EPZ011989 combination treatment on IIA-induced bone metastases. * p-values <0.05. **(C)** Normalized BLI of IIA-induced bone metastases normalized to day 1 of treatment. **(D)** Ex vivo quantification of tumor bioluminescence intensity in bone.

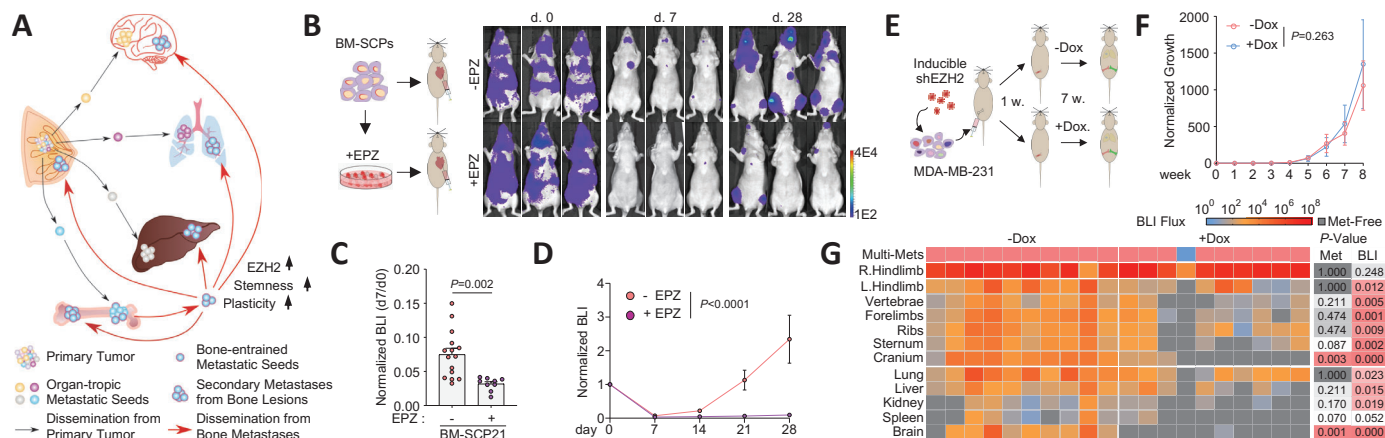


Figure 7. **(A)** Models for secondary metastatic spreads from established bone metastases. **(B-D)** The schematic diagram and representative BLI images **(B)**, normalized BLI intensity at day 7 **(C)**, and the colonization kinetics **(D)** of BM-SCP21 cells with 5-day's EZH2 inhibitor treatment by intra-cardiac injection. **(E)** Experimental design assessing the multi-site metastases from bone lesions with inducible depletion of EZH2 protein. Mice were injected with MDA-MB-231 cells stably transduced with inducible shRNAs targeting EZH2 at right hindlimb via IIA injection, followed by induction with or without 1mg/ml doxycycline. **(F)** Growth kinetics of the primary bone lesions in mice receiving doxycycline or control water, assessed by *in vivo* bioluminescent imaging. **(G)** Heat map of ex vivo BLI intensity and status of metastatic involvement in tissues from animals with EZH2 depleted or control bone metastases.

Establishing the First Platform to Deliver Biologics to the Bone for the treatment of Bone Metastatic Breast Cancers.

My lab has recently established an innovative bone targeting (BonTarg) technology that enables the specific delivery of therapeutic antibodies to bone tissue by conjugating bone-targeting moieties (Fig. 8).^{78,95} To demonstrate the possibility of specifically delivering therapeutic antibodies to the bone, trastuzumab (Tras) targeting the human epidermal growth factor receptor 2 (HER2) and a BP drug, Alendronate (ALN), were used as a model. ALN is a second-generation BP drug used in the clinic to treat osteoporosis and bone metastases.⁹⁶ We chemically linked the monoclonal antibody cancer drug trastuzumab to ALN using a combination of bioorthogonal chemistry methods (Fig. 8A).⁸³ We demonstrated for the first time that site-specific conjugation of bisphosphonates to large therapeutic proteins can enhance their concentration in the bone metastatic niche, relative to healthy bone and other tissues, thus providing a targeted therapy against bone metastases (Fig. 8B, C, and D).⁷⁸ In two bone metastasis xenograft models, the bone-targeting trastuzumab (Tras-ALN) exhibited improved activities of delaying bone colonization, as well as inhibition of further metastatic seeding from bone to other organs.⁷⁸ As shown in Fig. 8E, F, G, and H, treatment with Tras-ALN resulted in more significant tumor growth inhibition, compared with Tras-treated mice, demonstrating that Tras-ALN is more efficacious against HER2+ cells in vivo. Overall, modification of therapeutic agent like an antibody with bisphosphonates significantly inhibits cancer-directed to, or developing in the bone.

APPROACH

Aim 1. Prepare and characterize bone tumor-targeting EZH2 inhibitors for the treatment of bone metastasis.

Introduction. EZH2 is aberrantly over-expressed in a number of cancers. Most important, our recent studies indicate that EZH2 is upregulated in the bone cancer microenvironment (Fig. 5).^{38,46} EZH2 inhibitors have successfully realized their clinical potential and have become very useful in treating a variety of pathologies, but immune suppression and potentially toxicity associated with systemic EZH2 inhibition limited its use.²⁷⁻²⁹ Targeted delivery of drugs to the bone microenvironment is of great interest due to problems associated with surgical resection of skeletal lesions, increased reports of chemoresistant bone tumors, as well as the adverse systematic effects of a high dose of a therapeutic to ensure effective concentrations in bone. In this aim, we will prepare the bone tumor-targeting EZH2i (bEZH2i) and EZH2 prodrug by introducing bone-targeting moiety. We will use two potent, highly selective EZH2is, tazemetostat (Taz) and GSK126 as model drugs.

Aim 1.1. Prepare bone tumor-targeting EZH2 small molecule inhibitors.

Methods: (1) Preparation of bEZH2is. Bisphosphonates are widely used to control bone metastases.^{47,97,98} These negatively charged chemicals prefer to adhere to high crystallized hydroxyapatite in the bone metastatic niches, thus have been used to deliver a wide range of drugs to the bone metastases.⁴⁹⁻⁶⁵ To investigate if epigenetic modulator can be delivered to bone metastatic niches by the modification of BP, we have conjugated ALN to the piperazine ring of GSK126, and Taz (Fig. 9A, B and C). The co-crystal structure of human polycomb repressive complex 2 (PRC2) and GSK126 indicate that the piperazine ring of GSK126 is solvent-exposed, thus modification of the piperazine ring of GSK126 has minimal disruption on its activity.^{3,4}

(2) Preparation of bone tumor-targeting EZH2 prodrug. Next, we will prepare bone tumor-targeting GSK126 and Taz prodrug (GSK126-va-ALN and Taz-va-ALN, Fig. 9C). This bone tumor-targeting prodrug has little activity in its native state and gains high potency after the GSK126 is released in the presence of cathepsin K, a major bone-resorbing protease in the bone metastatic niches.⁹⁹ The pyridone moiety of GSK126 and Taz deeply buried in the hPCR2 structure is essential for their activities and pyridone alkylation has been shown to efficiently cage the inhibitor activity.^{3,100}

Aim 1.2. Evaluate bone tumor-targeting EZH2 inhibitors in vitro.

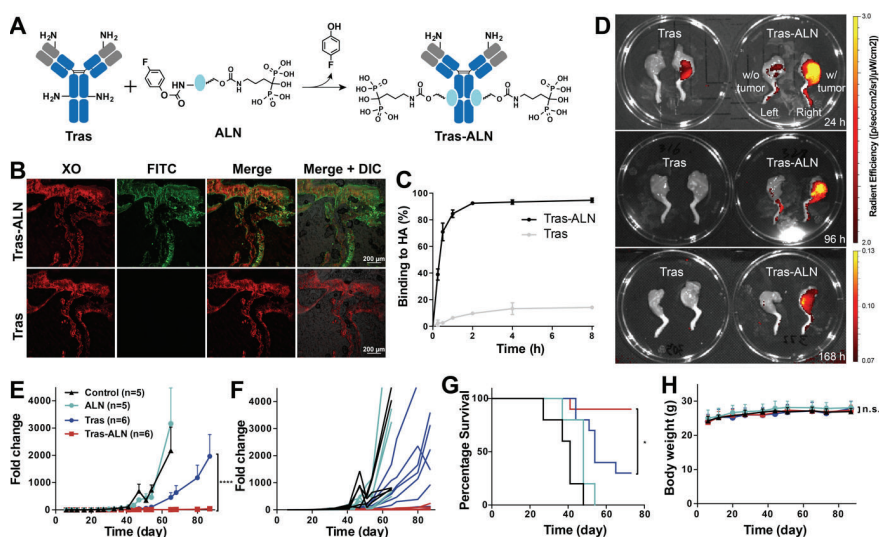


Figure 8. (A) Preparation of site-specifically labeled trastuzumab-ALN conjugates (Tras-ALN) using pClick. **(B)** Differential bone targeting ability of wild type Tras and Tras-ALN conjugate. Scale bars, 200 μ m. **(C)** Binding kinetics of Tras and Tras-ALN on hydroxyapatite and native bone. **(D)** Ex vivo fluorescence images of lower limbs of athymic nude mice bearing MDA-MB-361 tumor 24 h, 96 h, or 168 h after the retro-orbital injection Cy7.5-labeled Tras and Tras-ALN. **(E)** mean luminescent intensity of MDA-MB-361 tumors in mice treated with Tras or Tras-ALN. **(F)** Fold change in Individual luminescent intensity of HER2-positive MDA-MB-361 tumors in mice treated with Tras or Tras-ALN. **(G)** Kaplan-Meier plot of the time-to-euthanasia of mice treated as described in (E). **(H)** Body weight change of mice.

Methods: (1) Quantification of the bone binding affinity bEZH2is. To quantify the affinity difference among the bEZH2is, we will incubate GSK126, bone-targeting EZH2is with hydroxyapatite or native bones and characterize their bone binding affinities as shown in Fig. 8B and 8C. In our preliminary study, GSK126 and GSK126-ALN were used to stain nondecalfied bone sections from BALB/c mice. We observed the blue fluorescent signal for the section stained with the GSK126-ALN or Taz-ALN, but not wild type GSK126 or Taz (Fig. 10A). Our preliminary data indicate that the wild type GSK126 exhibited slight binding affinity with HA, while more than 60% of GSK126-ALN bound to HA within 5 h (Fig. 10B).

(2) In vitro EZH2i release from bEZH2i prodrug. Prodrug GSK126-va-ALN or Taz-va-ALN (40 μ M) will be incubated with cathepsin K (30 nM) for 0, 10, 20, 30, 60, and 120 min. The reaction product will be analyzed by LCMS.

(3) Measurement of the potency of bone targeting EZH2is. We will measure the half-maximum inhibitory concentrations (IC₅₀s) of EZH2is using the EZH2 methyltransferase inhibition assay published previously.^{101,102} Briefly, H3 peptide will be coated on the plate, followed by the incubation with S-adenosylmethionine (SAM), PRC2 complex, and different concentrations of EZH2is. The methylation level of H3K27 will be determined by the HRP-labeled anti-H3K27 antibody. Our preliminary data indicate that bisphosphonate modification of Taz doesn't alter its activity to inhibit EZH2 (Fig. 9E).

(4) In vitro cytotoxicity of bone targeting EZH2is. We next determine the cytotoxicity of GSK126, GSK126-ALN, GSK126-va-ALN, Taz, Taz-ALN, and Taz-va-ALN using MCF-7, ZR-75-1, MDA-MB-231, MDA-MB-468, E0771, and 4T1 cells using both 2D and 3D cultures. Briefly, breast cancer cells will be placed in 96 well plates at 1000 cells per well and followed by the incubation with different concentrations of EZH2i and bEZH2is. After 3d of incubation, the cell viability will be identified using the MTT assay or fluorescence. Our preliminary data indicate that both GSK126 and GSK126-ALN effectively inhibited cell proliferation (Fig. 9F, G, and H).

Aim 1.3. In vivo disposition study of bone tumor-targeting EZH2 inhibitors.

Methods: (1) In vivo pharmacokinetic studies of bEZH2is. BALB/c mice (4 weeks) will be randomly divided into 4 groups for each study. Wild type GSK126 or Taz (10 mg/kg intraperitoneally in sterile PBS),^{103,104} GSK126-ALN or Taz-ALN (same regimen as GSK126), and GSK126-va-ALN or Taz-va-ALN (same regimen as GSK126), or PBS buffer will be administered to the mice. At the time points (2, 5, 10, 15, 30, 45, 60, 90, 120, 150, 180, 210, and 240 post injection), the blood samples will be collected from the tail vein and analyzed as described previously.¹⁰³ Each group will include 6 animals, and a total of 48 animals will be required for this study.

(2) In vivo biodistribution studies of bEZH2is. To demonstrate if bisphosphonate modification can enhance the bone binding affinity for EZH2is, we have used a syngeneic mouse model. We inoculated 5×10^5 4T1 cells labeled with firefly luciferase and red fluorescent protein into the right leg of BALB/c mice via intra-iliac artery (IIA) injection. Wild type GSK126 or Taz (10 mg/kg i.p.),^{103,104} GSK126-ALN or Taz-ALN (same regimen as GSK126), or GSK126-va-ALN or Taz-va-ALN (same regimen as GSK126), will be admitted. Mice will be euthanized 1, 2, 4, 8, 16, and 32 hr post injection, and the major organs, including kidney, liver, brain, healthy bone, and bone metastatic lesions will be removed, homogenized, and analyzed using LCMS as reported previously.¹⁰³ In our preliminary data, GSK126-ALN and Taz-ALN showed more than 10-fold higher uptake in bone tumor than wildtype drugs (Fig. 10C and D). Each group will include 6 animals, and a total of 216 animals

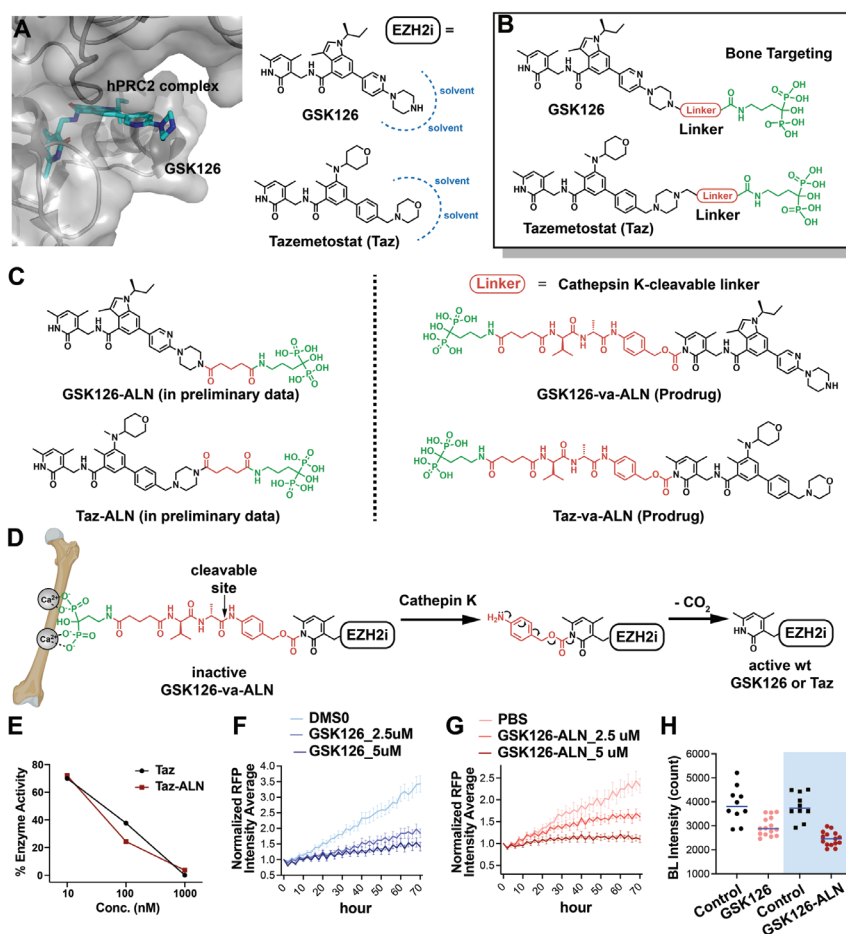


Figure 9. (A) Structure of hPRC2 bound to GSK126 inhibitor. **(B)** GSK126 can be specifically delivered to the bone by conjugation of ALN using cathepsin K-sensitive linkers. **(C)** Structures of GSK126-ALN and GSK126-va-ALN. **(D)** Proteolytic pathway of GSK126-va-ALN to release GSK126. **(E)** Potency of Taz and Taz-ALN against EZH2. **(F-H)** MCF7 cells were cultured in 3D in regular media for 24h before starting a 72-hour treatment. The growth curves were obtained based on the integrated red fluorescence intensity of the cells (F and G). BLI imaging was performed 3 days post treatment (H). Each dot represents a technical replicate.

Our preliminary data indicate that bisphosphonate modification of Taz doesn't alter its activity to inhibit EZH2 (Fig. 9E).

will be required for this study.

Potential Caveats and Alternative Strategy. (1) Cathepsin K cannot efficiently cleave the designed linker within GSK126-va-ALN to release GSK126. We will test the other cathepsin K-cleavage linkers, including Val-Cit, Val-Lys, and Val-Arg dipeptide linkers.⁹⁹ (2) In case that we observed any activity reduction after bisphosphonate conjugation, we will make bone-targeting EZH2is with cathepsin K or pH-cleavable linkers (Fig. 11).

Aim 2. Evaluate therapeutic benefits of bone-targeting EZH2is in vivo.

Introduction. 20-40% of all BCa patients will develop bone metastases.³⁰ Moreover, minimal residual diseases (MRD) in the bone marrow, including disseminated tumor cells and bone micrometastases, are associated with poor overall survival and are possible precursors of all BCa metastases.^{105,106} Thus, developing treatments that target bone metastases as well as MRD is a key focus for curing bone metastatic BCa. We expect that EZH2is with bone-targeting moiety to be a superior treatment for MRD, as compared to un-modified EZH2i. The affinity of ALN to bones may help overcome physical and biological barriers of the bone microenvironment against therapeutic drugs, thereby enriching and retaining epigenetic modulators in the bone microenvironment. Thus, EZH2is with ALN modification may achieve a higher

concentration in the bone without systemic toxicity. In this aim, we plan to first establish the superiority of bEZH2is to prevent BCa metastases in the bone as well as secondary metastases to other organs from bone lesions. We will not only examine anti-metastasis efficacies but also signs of additional toxicity.

Aim 2.1. Demonstrate bone tumor-targeting EZH2 inhibitor as an adjuvant endocrine therapy to treat bone metastases in vivo.

Rationale: Bone metastases more frequently occur to estrogen ER+ breast cancer.^{107,108} The direct and paracrine interactions between ER+ breast cancer cells and the osteogenic bone microenvironment elicit epigenetic adaptation, a process independent of clonal selection and leading to persistent endocrine resistance. Our recent studies suggest EZH2 is a master regulator of this reprogramming process. EZH2 integrates signals received from the osteogenic niche and in turn activates multiple downstream pathways related to endocrine resistance (Fig. 5 and 6). Therapeutic blockade of EZH2 also reverses the microenvironment-induced epigenomic reprogramming, alleviates acquired endocrine resistance of BMs, and reduces subsequent metastatic recurrences (Fig. 6 and 7). In this aim, we plan to establish the superiority of bone tumor-targeting EZH2 inhibitors to improve endocrine therapy.

Methods: (1) *In vivo efficacy of bone-targeting EZH2 inhibitors.* We will be primarily using 4 ER+ models: two cell lines, MCF-7 and ZR75-1, which are amenable to genetic manipulations, and two PDXs, HCI011, and WHIM9, which were derived from patients with bone metastases and can spontaneously metastasize to the bone in animals. For all cell lines, 2×10^5 cells will be introduced to the bone by IIA injection. Nine groups will be used: untreated control, fulvestrant (25 mg/kg PO daily)¹⁰⁹, GSK126 (10 mg/kg IP daily),⁴¹ GSK126-ALN (same regimen as GSK126), GSK126-va-ALN (same regimen as GSK126), fulvestrant+GSK126 (25 mg/kg PO + 10 mg/kg IP daily), fulvestrant+GSK126+ALN (25 mg/kg PO + 10 mg/kg IP daily + 5.15 mg/kg IP daily), fulvestrant+GSK126-ALN (same regimen as fulvestrant+GSK126), and fulvestrant+GSK126-va-ALN (same regimen as fulvestrant+GSK126). We will repeat the study with Taz and use 15 mice per group. Five mice in each group will be euthanized after 1-2 weeks to examine the therapeutic effects on MRD including single cell DTCs and multiple cell BMs. The remaining 10 animals will be followed by bioluminescence imaging or PET imaging till the end points described in previous aims. After euthanization, half of the bone specimens will be used for H&E confirmation of lesions and TRAP assays of osteoclast activities, and the other half will be used for micro-CT to assess alteration of bone structures and bone mass. A total of 540 athymic nude mice and 540 SCID/Beige mice will be required (4 models).

(2) *Comparison of maximal tolerable dosage and minimum effective dosage between wild type EZH2i and bEZH2i analogs.* Using the same experimental procedures as described above and the MCF-7 model, we will vary the dosage of GSK126, GSK126-ALN, GSK126-va-ALN, Taz, Taz-ALN, and Taz-va-ALN at 0, 10, 20, 40,

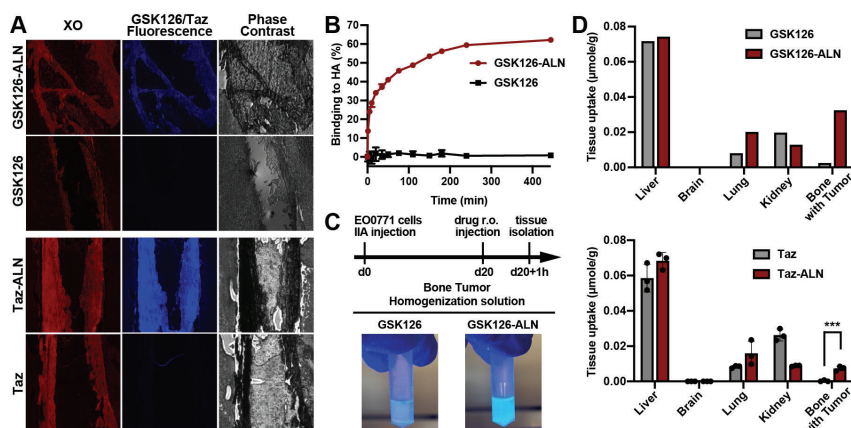


Figure 10. (A) Differential bone targeting ability of GSK126 and GSK126-ALN. Nondecalfied bone sections from BALB/c mice were incubated with 1 mg/mL GSK126 (blue) or GSK126-ALN overnight, followed by staining with 4 μg/mL xylenol orange (XO, red). Scale bars, 100 μm. (B) Binding kinetics of GSK126 and GSK126-ALN to hydroxyapatite (HA). (C) In vivo biodistribution schedule and fluorescence of bone tumor homogenization solutions from mice injected with either GSK126 or GSK126-ALN. (D) Comparison of the biodistribution profiles of GSK126 and GSK126-ALN in 4T1-bearing mice 1 h after the injection.

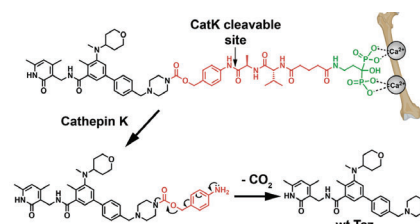


Figure 11. Synthesis of bone-targeting tazemetostat with a cathepsin K-cleavable linker.

80, 160 mg/kg with fulvestrant (25 mg/kg PO daily).¹⁰⁹ Metastatic progression will be monitored and quantitated using bioluminescence imaging. Potential toxicity will be evaluated by measurement of mouse weight and food consumption twice a week. Signs of pain and distress, such as difficulty to eat, walk, or groom normally, or other signs of systemic toxicity will be recorded. If significant signs of toxicity are observed, the corresponding dosage group will be euthanized, and histological comparison of heart, liver, kidney, and spleen will be conducted using tumor-free, healthy animals as reference.. Metastasis burden will be quantitated and compared between remaining experimental groups. 15 animals each group and 540 animals in total will be required.

Aim 2.2. Evaluate benefits of bone-targeting EZH2 inhibitor against secondary metastases from the bone.

Rationale: MRD in the bone marrow (single-cell DTCs and multi-cell BMMs) is not only the likely precursor of overt bone metastases but may also fuel further metastases to other organs. This is supported by correlation between MRD and metastasis of all sites,^{105,106}. IIA-injected animals exhibited metastases to almost every organ (Fig. 4). EZH2 is a master regulator of secondary metastases from bone lesions and knockdown of EZH2 in cancer cells dramatically decreased secondary metastases (Fig. 12A, B, and C).⁴⁶

However, systematic administration of EZH2i, including EPZ011989 and GSK126, could not inhibit multi-organ secondary metastases, suggesting more complex effects of EZH2i on multiple cell types (Fig. 12D, E, F, G, and I). Herein, we plan to establish the superiority of bEZH2is to prevent or impede secondary metastases from bone lesions. Five groups will be used: untreated control, ALN (5.15 mg/kg IP daily), GSK126 (10 mg/kg IP daily),⁴¹ a mixture of GSK126 and ALN (10 mg/kg IP daily+ 5.15 mg/kg IP daily), and GSK126-ALN (same regimen as GSK126). Our preliminary data indicate that only the GSK126-ALN conjugate, but not the mixture of GSK126 and ALN, significantly inhibits osteolytic micrometastases (Fig. 12G, H, and I). Furthermore, we also tested the efficacy of GSK126-ALN in the immunocompetent model. Lewis lung cancer (LLC) cells were introduced to the bone by IIA injection in right hind limbs, followed by the treatment of a mixture of GSK126 and ALN (10 mg/kg IP daily+ 5.15 mg/kg IP daily), and GSK126-ALN (16.58 mg/kg IP daily). To our delight, GSK126-ALN established an enhanced efficacy in preventing multi-organ metastases (Fig. 12 J). In this aim, the effect of bEZH2is to reduce the metastatic capacity of cancer cells will be further evaluated in MDA-MB-231 and MCF-7 xenograft cancer models as well as EO771 and 4T1 syngeneic mouse models.

Methods: Wild-type syngeneic BALB/c mice and athymic nude mice will be used as host animals for EO771/4T1 and MDA-MB-361/MCF-7 cells, respectively. 2×10^5 cancer cells will be introduced to the bone by IIA injection in right hind limb. Five groups will be used: untreated control, GSK126 or Taz (10 mg/kg IP daily),^{103,104} GSK126-ALN or Taz-ALN (10 mg/kg IP daily), GSK126-va-ALN or Taz-va-ALN (10 mg/kg IP daily), and a mixture of GSK126/Taz and ALN (10 mg/kg IP daily+ 5.15 mg/kg IP daily). After 12 weeks, various organs will be isolated and cancer cells will be subjected to histological validation using anti-GFP antibody. 15 per group and a total of

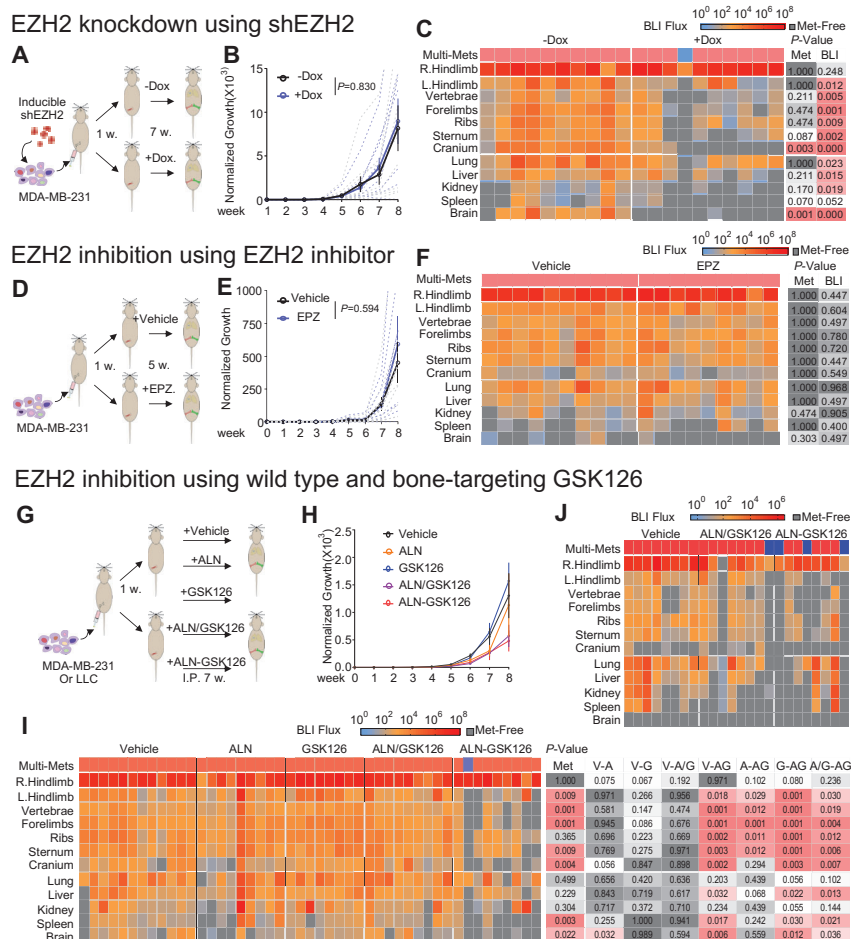


Figure 12. (A) Experimental design assessing the multi-site metastases from bone lesions with inducible depletion of EZH2. (B) Growth kinetics of the primary bone lesions in mice receiving doxycycline or control water. (C) Heatmap of ex vivo BLI intensity and status of metastatic involvement in tissues from animals with inducible depletion of EZH2. (D) Experimental design assessing the multi-site metastases from mice treated with EPZ011989. (E) Growth kinetics of the primary bone lesions in mice receiving EPZ011989 or control water. (F) Heatmap of ex vivo BLI intensity and status of metastatic involvement in tissues from animals treated with EPZ011989. (G) Experimental design assessing the multi-site metastases from MDA-MB-231-bearing mice treated with GSK126 and its analogs. (H) Growth kinetics of the primary bone lesions in mice receiving GSK126 and its analogs. (I) Heatmap of ex vivo BLI intensity and status of metastatic involvement in tissues from MDA-MB-231-bearing mice treated with GSK126 and its analogs. (J) Heatmap of ex vivo BLI intensity and status of metastatic involvement in tissues from LLC-bearing mice treated with GSK126 and its analogs.

600 animals will be required.

Potential Caveats and Alternative Strategy.

(1) We will use IIA injection to introduce bone metastases. Although this approach skips early steps of the metastasis cascade, two major advantages are provided in addition to what has been described in Innovation. First, seeding of cancer cells can be synchronized so that the relative timing of therapeutic intervention can be strictly controlled. Second, IIA injection concentrates tumor cells to hind limbs, prevents mice from succumbing to metastases in other vital organs, and allows us to focus on efficacies on bone metastases only. This approach was developed in my laboratory.^{16,17} Therefore, we do not anticipate any technical problems. If moribund toxicity is observed, we will terminate the experiments and reduce the dosage by 2-5 times. (2) In some situations, GFP, RFP, and luciferase expression may trigger adaptive immune responses.¹¹⁰ If antigenicity is observed in future experiments, we will use “glowing head” mice in which luciferase and GFP are selectively expressed in pituitary glands, and therefore, are immunologically tolerated.¹¹⁰ We have obtained and backcrossed these mice to FVB backgrounds.

Aim 3. Dissect the molecular mechanisms underlying the efficacy of bone tumor-targeting EZH2 inhibitors.

Introduction: EZH2 Inhibition is emerging as a new therapeutic approach for cancer treatment. Targeting EZH2 in bone metastases reduces their phenotypic plasticity and abolishes further dissemination. Herein, we will dissect the impact of EZH2 on epigenomic landscape and elucidate downstream pathways. EZH2 also regulates development and homeostasis of multiple normal tissues. Thus, it is important to determine the toxicity of EZH2is. The ALN moiety enriches EZH2is in the bone, which is predominantly occupied by osteoblasts, osteoclasts, and hematopoietic cells. Therefore, we will characterize the undesirable side effects due to EZH2 inhibition in these cells. This aim will provide a more comprehensive understanding of EZH2's roles in the bone metastasis ecosystem and lay foundation for further optimization of bone-targeting EZH2is.

Aim 3.1 Elucidate how bone-targeting EZH2 inhibitors reduce further dissemination.

Rationale: Two lines of evidence suggest a specific role of EZH2 in further dissemination, rather than local progression, of bone metastasis. First, inducible knockdown of EZH2 in cancer cells abolished further metastases but did not slow down growth of original bone metastasis¹. Second, bEZH2i achieved effects similar to the inducible knockdown as shown in **Fig. 12**. Curiously, non-targeted EZH2i did not impact secondary metastasis, suggesting that side effects of EZH2 inhibition on other cells may offset the therapeutic effect on cancer cells. In an ER+ model, EZH2i alone did not appear to suppress bone metastasis but can reverse endocrine resistance.² Thus, EZH2 in bone metastatic cells mediates processes clearly distinct from mere proliferation. In this aim, we will dissect the molecular nature of this phenomenon.

Methods: (1) To examine global epigenomic impact of bEZH2is on bone metastatic cells. EZH2 is a methyl transferase and catalyzes tri-methylation of H3K27, thereby silencing gene expression. Using the approaches shown in **Fig. 13A**, we will harvest cancer cells that are extracted from bone metastasis and perform ATAC-seq to profile chromatin conformation, Chip-seq of H3K27me3 to examine enzymatic activities of EZH2, and RNAseq to measure the transcriptional readout. Untreated, bEZH2is-treated bone metastases will be compared at multi-omic levels. We will also include metastases with inducible knockdown of EZH2 to provide a genetic reference for distinguishing ontarget and offtarget effects of bEZH2is. Of note, reduction of EZH2 protein in the knockdown cells will also help delineate enzymatic and non-enzymatic functions³. We will determine if compared to EZH2i, bEZH2i more effectively alter the epigenomic landscape of metastatic cells. Moreover, we will identify most profoundly changed genes in metastatic cells that may lead to the observed decrease of metastasis-to-metastasis seeding. 15 animals each group and a total of 240 animals will be required. These analyses will be done in collaboration with Dr. Tao Wu, an epigenomics expert at BCM.

(2) To identify downstream pathways mediating the phenotypic plasticity. EZH2 has a profound impact on many other pathways, including NOTCH⁴, WNT⁵, STAT3⁶ and NFkB⁷, all of which have been implicated in cancer cell stemness and metastasis. In fact, NOTCH, STAT3 and WNT pathways appear to be elevated in cancer cells in co-cultures with MSCs (**Fig 13**), and will be further examined in the multiomic data described in the previous

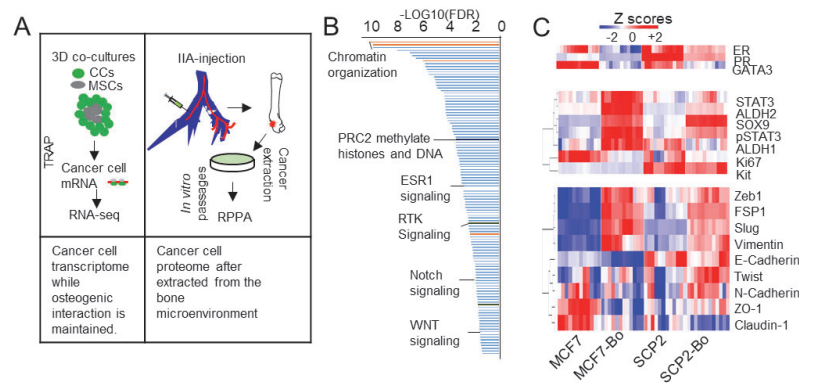


Figure 13. Global reprogramming effects of the osteogenic cells on ER+ breast cancer cells (A) Schematics show experimental strategies to profile transcriptome of cancer cells while interacting with MSCs via TRAP-seq of 3D co-cultures of MCF-7 cells and MSCs (left) and partial proteome of MCF-7 parental or SCP2 cells extracted from IIA-derived bone lesions and expanded in vitro by several passages (right). **(B)** Output of gene ontology enrichment analysis using differentially expressed genes between MCF-7 cells with and without co-cultures of MSCs in 3D. **(C)** Selected groups of (phosphor)-proteins in parental MCF-7, bone-entrained MCF-7 (MCF7-Bo), original SCP2, and bone-entrained SCP2 (SCP2-Bo) assessed by RPPA.

paragraph. We hypothesize that these pathways may act downstream of EZH2 and/or in cooperation with EZH2 in mediating the enhanced stemness and further metastasis. To test their functional relevance, we will use the 3D co-culture system and PDX/PDO models to examine if corresponding inhibitors may reverse the expression of stem cell markers such as ALDH, CD44 and SOX2, which are up-regulated in bone metastatic cells as we previously observed². Specifically, we will use anti-Jag1 (clone 15D1142, 100 pM) and DAPT43 (10 μ M) to interrupt the Notch pathway, JW6744 (1-5 μ M) and NSC66803645 (10 μ M) to inhibit WNT signaling, and Napabucasin⁸ (2 μ M) and Stattic⁹ (50 μ M) to inhibit the STAT3 pathway. Of note, it would be unrealistic to test these inhibitors *in vivo*. We expect to identify one of a few of these pathways as key downstream effectors, which will then be moved on to *in vivo* metastasis-to-metastasis experiments as shown in **Fig. 12** by combining bEZH2i with inhibitors of the corresponding pathways.

(3) *To evaluate the EZH expression in primary human bone metastatic samples.*

Through ongoing collaborations with Dr. Arun Sreekumar, we have collected a unique sample cohort consisting of matched paraffin embedded specimens from primary breast tumors and bone metastasis placed on glass slides, that were pathologically curated from eleven breast cancer patients (**Fig 14**). On consecutive slides, we will perform IHC of EZH2 and multiplexed immunofluorescence staining of EZH2 together with osteogenic cell markers (e.g., Col-I or ALP).⁹⁴ The former will allow us to semiquantitatively compare expression level and subcellular localization of EZH2 between matched primary and metastatic tumors. The latter will reveal to what extent EZH2 upregulation is dependent upon interactions with osteogenic cells, as we previously showed.³⁸ In the meantime, we are obtaining spatial transcriptomic profiles from these samples as a part of another project. The resultant data will also be analyzed in the context of this project to evaluate how EZH2 activities may spatially correlate with enhanced stemness features (e.g., expression of ALDH, SOX2, CD44 and the pathways mentioned in the previous paragraph). These data will be evaluated by Dr. Qian Zhu (see support letter), a leading computational biologist specialized in integrated single cell omics and spatial transcriptomics. The high-resolution spatial omics will provide an opportunity to investigate associations across different locations within each sample, thereby offering much stronger power.

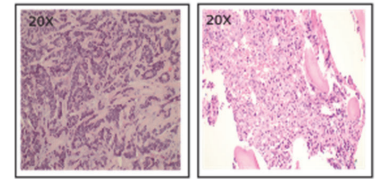


Figure 14. Representative IHC images of a pair of primary tumor and bone metastasis.

Aim 3.2 Examine the impact of bone-targeting EZH2 inhibitors on osteogenic niche cells and other bone TME cells.

Rationale: Bone resident cells, especially osteoblasts and osteoclasts, play crucial roles in metastatic colonization. The paracrine vicious cycle between osteoclasts and cancer cells is the driving force of advanced stage of metastasis¹⁰⁻¹², whereas the direct cell-cell interactions with osteoblasts and their precursors (designated as the osteogenic niche) support early phase of colonization¹³⁻¹⁵. Interestingly, EZH2 regulates both cell types. Inhibition of EZH2 accelerates bone formation and fracture healing¹⁶. Therefore, it is important to assess how bEZH2is may indirectly alter tertiary metastasis via impacting bone cells. Equally imperative is to determine how bEZH2is may affect normal bone biology and lead to undesirable side effects.

Methods: (1) *Determine how EZH2 inhibition affects the tumor-promoting effects of osteogenic cells.* We will use MSCs pre-treated with EZH2i or siRNAs against EZH2 for the 3D co-culture assay shown in **Fig. 13**. After 2-3 days co-culture, gene expression of cancer cells will be profiled using TRAP-seq following the same procedures in our previous study. We will determine whether the key changes induced by MSCs (**Fig. 13C**) are reversed upon EZH2 inhibition in MSCs. In addition, we will also perform *in vitro* osteogenesis assay to evaluate whether EZH2i treatment affects differentiation potential of MSCs.

(2) *Determine the potential side effects of bEZH2is on bone homeostasis and fracture healing.* We will perform histomorphometry analysis on long bones from 8-week old animals undergone treatments of control, GSK126 or Taz, GSK126-ALN or Taz-ALN, and GSK126-va-ALN or Taz-va-ALN for at least two weeks. We will focus on effects on frequency and activities of osteoblasts and osteoclasts. We will also assess the bone formation rate by injecting Calcein and Alizarin Red S 14 and 19 days after the onset of treatment. The outcome of these experiments will reveal any side effects of bEZH2is on bone homeostasis. We will next examine the fracture healing process. To this end, bone drilling will be performed on adult animals undergone control, GSK126 or Taz, GSK126-ALN or Taz-ALN, GSK126-va-ALN or Taz-va-ALN treatment for 2 weeks. Micro-CT will be used to determine healing progress after 2.5 weeks while the same treatments are maintained. 15 animals each group and a total of 120 animals will be required. All of these assays have been established in Zhang Lab at BCM as shown in **Fig. 15**.

Aim 3.3 Determine if bone-targeting EZH2 inhibitors cause immunosuppression.

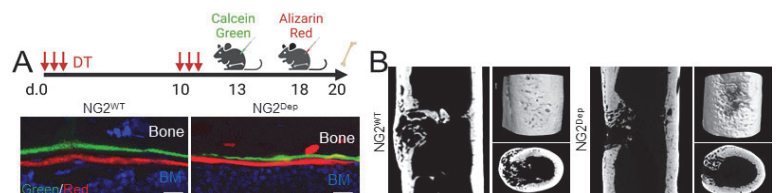


Figure 15. Bone remodeling assays. A. Bone formation assay using two different dyes. The distance between the two color borders indicates bone formation rate. B. Micro-CT of needle-drilled bones for measurement of fracture healing. These data are taken from a study focusing on NG2+ cells in Zhang lab. NG2^{Dep}: Depletion of NG2+ cells.

Rationale: Non-targeted EZH2i did not achieve expected efficacy in many clinical settings. One potential reason is that systemic EZH2 inhibition suppresses T cell response,^{17,18}. The bone-targeting moiety retains most EZH2i in the bone and the bone marrow is major site of hematopoiesis. Thus, it is even more crucial to examine the impact of b EZH2is on immunity—in both tumor-naïve and tumor-carrying hosts.

Methods: (1) *Characterize the shift of immune landscape of animals undergoing bEZH2i treatment with or without tumor burden.* We will randomize 8-week old tumor-free mice into the following treatment groups with N=15 in each group. 1) untreated; 2) GSK126 or Taz (10 mg/kg IP daily)^{103,104}; 3) GSK126-ALN or Taz-ALN (same as GSK126); and 4) GSK126-va-ALN or Taz-va-ALN (same as GSK126). In parallel, we will transplant 4T1 tumors to mammary fat pads or intrailiac artery. One week later, tumor-bearing mice will be randomized into the same 4 treatment groups. We will euthanize the animals and harvest CD45+ cells from bone marrow, spleen, and blood after two weeks' treatment. We will use imaging mass cytometry to compare the major immune cell subpopulations among different groups. These experiments are expected to elucidate how bEZH2si may alter baseline immune landscape and immune response to tumors. We will pay extra attention to immunosuppressive cells such as MDSCs and Tregs because of previous reports linking them to EZH2i-induced immunosuppression. 15 animals each group and a total of 240 will be required.

(2) *Determine how bEZH2is impacts efficacies of adoptive transfer of antigen-specific T cells and/or immune checkpoint blockade therapies.* Our preliminary data indicate that experimental bone metastases expressing GFP are responsive to T cells extracted from the Jedi mice¹⁹, which target GFP₂₀₀₋₂₀₈ presented on H-2K^d, in conjunction with anti-PD1 (**Fig. 16**). Specifically, one dose of 10⁶ T cells will be transferred retro-orbitally on Day 7 after IIA injection. EZH2 inhibitors will be administered prior to these treatments, on the same day of IIA injection. Four groups without anti-PD-1 treatment will be used: untreated control, ALN (5.15 mg/kg IP daily), GSK126 or Taz (10 mg/kg),⁴¹ GSK126-ALN or Taz-ALN (same regimen as GSK126), GSK126-va-ALN or Taz-va-ALN (same regimen as GSK126). In parallel, four groups with the anti-PD1 treatment will start on the same day and repeat every three days. 15 animals each group and a total of 240 animals will be required. We expect to determine if EZH2i or bEZH2i blunts such responses, which would indicate compromise of anti-tumor immunity.

Potential Caveats and Alternative Strategy. In Aim 3.2, siRNA and EZH2i treatment may not sustain till the end of co-culture assays. If this is the case, we will use EZH2^{fl} mice available from Jacksons Lab (Strain #022616). These mice will be used to derive MSCs, which will then be infected by adenovirus expressing the cre recombinase to induce knockout. We did not observe obvious toxicity reflected by decrease of body weight in our preliminary study (**Fig. 12**). If any unexpected sign of toxicity is detected during the process of above experiments, we will carefully inspect the mechanism of toxicity by pathological examination of key organs such as liver, kidney and spleen.

Breast Cancer subtypes: Since 57% of patients with bone metastasis are hormone receptor (HR)-positive,¹ we have included several ER+ xenograft and PDX models, including MCF-7, ZR-75-1, and HCl011/WHIM9 in this study. Furthermore, we have included several syngeneic triple negative breast cancer (TNBC) models, including 4T1 and EO771 models. We will not be able to use ER+ syngeneic models due to the lack of syngeneic mouse models for this subtype of breast cancer. This has been a long-standing, field-wide challenge. We have been diligently establishing new collaborations to overcome this challenge. In this case, we will employ a series of newly developed Rat-based models that develop ER+ mammary tumors that spontaneously metastasize. This series of models are developed by Dr. Yi Li's group, and are currently being characterized collaboratively by both of our laboratories (please see Dr. Yi Li's support letter).

Statistical methods: Our consultant, Dr. Yi Ding (Ph.D. in Computational Biology) will perform the statistic calculation to ensure the significance of our result. A group size of 10-15 animals usually generates ~80% statistical power to detect a 3- to 5-fold difference in log-transformed BL intensity or a 2- to 3-fold effect in DTC/BMM frequency (counts of events per femur bone) with 5% alpha error by Student's t-test. If the variation is larger than expected, we will repeat the experiment to increase the sample size. In vitro analyses will be performed with at least 5 biological replicates, i.e., independent experiments started in different days, each being an average of at least technical replicates. For biological variables, we will use 8-week old female mice. For analysis of spatial transcriptomic in Aim 3.1, we will use multinomial distribution analysis to calculate the power or required cell sizes. current Visium chip contains 4,992 total spots per capture area. Suppose the tissue area is covered by at least 50% of the spots, it has 0.8 probability to detect at least 20 spots in all cell subpopulations, supposing the rarest cell types are enriched in at least 25 spots (about 1% of total tissue spots). It is worth mentioning that many of the spatial pattern analyses were also previously used in ecological studies. These analyses will be implemented by tools such as Mapboxapi Rstats package.

TIMELINE FOR SPECIFIC AIMS

Aims 1.1 (Y1), 1.2 (Y1-2), 1.3 (Y1-2), Aims 2.1 (Y1-3), 2.2 (Y1-4), Aims 3.1 (Y2-5), 3.2 (Y3-5), 3.3 (Y3-5).

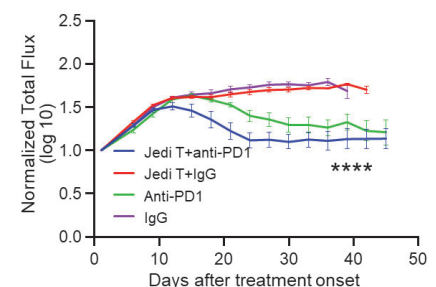


Figure 16. Combined adoptive transfer of Jedi T cells and anti-PD1 treatment effectively regress experimental bone metastases. ****: $p < 0.0001$

PHS Human Subjects and Clinical Trials Information

OMB Number: 0925-0001

Expiration Date: 09/30/2024

Use of Human Specimens and/or Data

Does any of the proposed research in the application involve human specimens and/or data *

☐ Yes

☒ No

Provide an explanation for any use of human specimens and/or data not considered to be human subjects research.

Are Human Subjects Involved

☐ Yes

☒ No

Is the Project Exempt from Federal regulations?

☐ Yes

☐ No

Exemption Number

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ 6

☐ 7

☐ 8

Other Requested Information

VERTEBRATE ANIMALS

Description of Animals and Proposed Procedures.

The proposed experiments will use female athymic nude (for human cell lines), BALB/c (for 4T1 cells), C57BL/6 (for EO771 cells), and SCID/Beige (for PDX cells) mice. Mice are to be used in all specific aims. These mice will be subject to following procedures.

Intra-iliac artery injection will be used to introduce synchronized, experimental metastatic bone colonization. Briefly, cancer cells will be inoculated into external iliac artery by using a 31-G needle. Right after injection, a cotton tip will be used to stop bleeding for 5 mins and the wound will be closed by suture. Animals will be monitored by bioluminescence imaging for bone colonization.

There are no other experimental methods that as accurately recapitulate the proposed methods to address in vivo biology.

Numbers of Mice and Rationale: The numbers below are based on power analyses using specific criteria as discussed in the proposal. **Our consultant, Dr. Yi Ding** (supporting letter attached, M.S. in Applied Statistics and Ph.D. in Computational Biology) will perform the statistic calculation to ensure the significance of our result. A group size of 10-15 animals usually generates ~80% statistical power to detect a 3- to 5-fold difference in log-transformed BL intensity or a 2- to 3-fold effect in DTC/BMM frequency (counts of events per femur bone) with 5% alpha error by Student's t-test. If the variation is larger than expected, we will repeat the experiment to increase sample size. Specific numbers of mice needed for each sub-aim are as follows.

Aims	Number and strain	Experiment
1.3 (1)	48 BALB/c (2 drugs x 4 groups x 6 biological replicates)	In vivo pharmacokinetics studies of bEZH2is.
1.3 (2)	216 BALB/c (2 drugs x 3 groups x 6 time points x 6 biological replicates x 4T1 model)	In vivo biodistribution studies of bEZH2is.
2.1 (1)	540 athymic nu/nu (2 drugs x 9 groups x 15 biological replicates x 2 MCF-7/ZR75-1 models) 540 SCID/Beige (2 drugs x 9 groups x 15 biological replicates x 2 HCl011/WHIM9 models)	In vivo efficacy of bone-targeting EZH2 inhibitors as an adjuvant endocrine therapy to treat bone metastases.
2.1 (2)	540 athymic nu/nu (2 drugs x 3 groups x 6 doses x 15 biological replicates x MCF-7 model)	Comparison of maximal tolerable dosage and minimum effective dosage between wild type GSK126 and bone-targeting GSK126 analogs.
2.2	300 athymic nu/nu (2 drugs x 5 groups x 15 biological replicates x 2 MCF-7/MDA-MB-361 models) 150 BALB/c (2 drugs x 5 groups x 15 biological replicates x 4T1 model) 150 C57BL/6 (2 drugs x 5 groups x 15 biological replicates x EO771 model)	In vivo efficacy of bone-targeting EZH2 inhibitors against secondary metastases from the bone.
3.1	240 athymic nu/nu (2 drugs x 8 groups x 15 biological replicates x MDA-MB-231 model)	Examine global epigenomic impact of bEZH2is on bone metastatic cells.

3.2	120 athymic nu/nu (2 drugs x 4 groups x 15 biological replicates x MDA-MB-231 model)	Determine the potential side effects of bEZH2is on bone homeostasis and fracture healing.
3.3 (1)	240 BALB/c (2 drugs x 8 groups x 15 biological replicates x 4T1 model)	Characterize the shift of immune landscape of animals undergoing bEZH2i treatment with or without tumor burden.
3.3 (2)	240 BALB/c (2 drugs x 8 groups x 15 biological replicates x 4T1 model)	Determine how bEZH2is impact efficacies of adoptive transfer of antigen-specific T cells and/or immune checkpoint blockade therapies.

A total 3324 mice are required.

Age: In all but one experiments, mice will all be used at 8 weeks of age or older since they acquire sexual maturity at 8-10 weeks.

Gender: Because of mammary cancer models, we will use exclusively female mice.

Justifications: The appropriate strain of mice are required in order to address the specific scientific questions. We require BALB/c and C57BL/6 mice as they are immunocompetent and syngeneic hosts for the murine cell lines 4T1 and EO771, respectively. We use athymic nude mice for human cells to avoid rejection of human tissues. We use SCID/Beige mice for PDX cells. There are no other experimental methods that as accurately recapitulate the proposed methods to address in vivo biology.

Veterinary Care: The animals are housed in the Center for Comparative Medicine Texas Mouse Facility (TMF) vivarium and Rice vivarium and will be cared for by experienced technicians and veterinarians with special training and interest in laboratory animals.

Minimization of Pain and Distress: Discomfort, pain, and injury to animals will be limited to that which is unavoidable in the conduct of scientifically valuable research, and analgesic, anesthetic, and tranquilizing drugs will be used routinely to minimize discomfort and pain to animals. Post surgery the animals exhibit no signs of discomfort and can be observed climbing on cage tops and other normal behaviors within an hour of surgery.

Method of Euthanasia. Animals will be sacrificed by CO₂ inhalation followed by decapitation at the termination of an experiment or when discomfort is apparent. This will use automated chambers available in the TMF and is approved in our IACUC protocols. This method is consistent with the recommendations of the Panel of Euthanasia of the American Veterinary Medical Association and in our IACUC approved protocols.

MULTI-PI LEADERSHIP PLAN

The Principal Investigators (PD/PIs) for the proposed application are:

- 1) **Han Xiao, (Contact PI)**, Assistant Professor, Rice University
- 2) **Xiang Zhang, (MPI)**, Professor, Baylor College of Medicine

I. Rationale for multiple Principal Investigators.

This interdisciplinary research project will be conducted using a multiple PI framework. Collectively, the two PIs have substantial expertise in synthetic chemistry, organic chemistry, drug development, antibody engineering, protein engineering, protein modification, bioorthogonal labeling, bone metastasis study, translational studies using mouse models and human clinical specimens, and biochemical techniques to develop novel therapeutic concepts. The Xiao Laboratory focuses on the development and application of methods to prepare bone tumor-targeting epigenetic modulators. The Zhang Laboratory focuses on breast cancer bone metastasis. They pioneered the research on the microenvironment niche in early-stage bone colonization. This experience will also allow the team to test the ability of bone tumor-targeting epigenetic modulators to kill bone metastases *in vitro*, *ex vivo*, and *in vivo*. The two labs are within walking distance. Drs. Xiao and Zhang Laboratories initiated a collaboration two years ago based on the common interest in targeting the bone microenvironment associated with metastatic cancer and had co-authored two papers. Their expertise is highly complementary, and the collaboration will combine innovative technology development with cutting-edge pre-clinical modeling. Therefore, the proposed research will synergistically establish new therapeutic agents and shed light on mechanisms underlying metastasis colonization. The sections below describe their respective responsibilities and contributions to the proposed study.

Han Xiao, Ph.D. (contact PI), is an Assistant Professor of Chemistry, Biosciences, and Bioengineering at Rice University. His research focuses on the development of chemical biology tools as novel therapeutic strategies. He was the first to prepare antibody-bisphosphonate conjugates and tested their efficacy for the treatment of bone metastases *in vivo*. Xiao lab also prepared several small molecule drugs, including monomethyl auristatin F, photosensitizers, for cancer therapy. His group will perform all biochemical approaches proposed, including preparing bone-targeting antibodies, evaluating their activities *in vitro* and *in vivo*.

Specific activities that will be the responsibility of the Rice University-based research team include:

- 1) Prepare bone tumor-targeting EZH2 small molecule inhibitors within Aim 1.1.
- 2) Evaluate bone tumor-targeting EZH2 inhibitors *in vitro* within Aim 1.2.
- 3) Quantify *In vivo* disposition study of bone tumor-targeting EZH2 inhibitors Aim 1.3.
- 4) Demonstrate bone tumor-targeting EZH2 inhibitor as an adjuvant endocrine therapy to treat bone metastases *in vivo* within Aim 2.1.
- 5) Evaluate benefits of bone-targeting EZH2 inhibitor against secondary metastases from the bone within Aim 2.2.
- 6) Determine if bone-targeting EZH2 inhibitors cause immunosuppression when administered as different doses in Aim 3.3.
- 7) Contribute to data analysis as well as to the preparation of manuscripts and presentations.
- 8) Coordinate statistical and bioinformatic analysis for the entire study with Dr. Yi Ding.

Xiang Zhang, Ph.D. (MPI), Professor, McNair Scholar, Baylor College of Medicine—his laboratory investigates early-stage bone colonization of disseminated breast cancer cells and its therapeutic strategies. He has over 12 years of research experience on biological mechanisms and therapeutic strategies of metastasis, the major threat to the lives of solid cancer patients. Dr. Zhang will perform the assays and *in vivo* studies to evaluate the efficacy of bone-targeting antibodies to control macrometastases and eliminate micrometastases in the bone. His lab was the first to discover that EZH2 is up-regulated in the bone microenvironment and drives ER+ BMMs toward a basal and stem-like state. Moreover, EZH2 inhibition reverses endocrine resistant and reduces the stemness and metastasis capacity of cancer cells.

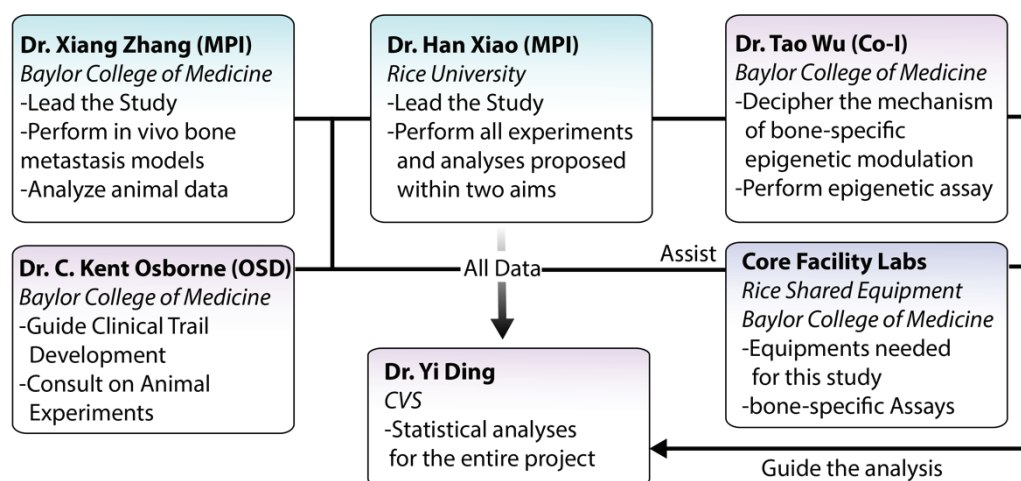
Specific activities that will be the responsibility of the Rice University-based research team include:

- 1) Consult on the *in vitro* experiments to be performed within Aim 1.

- 2) Demonstrate bone tumor-targeting EZH2 inhibitor as an adjuvant endocrine therapy to treat bone metastases in vivo within Aim 2.1.
- 3) Evaluate benefits of bone-targeting EZH2 inhibitor against secondary metastases from the bone within Aim 2.2.
- 4) Elucidate how bone-targeting EZH2 inhibitors reduce further dissemination without directly affecting proliferation or survival of metastatic cells within Aim 3.1.
- 5) Examine the impact of bone-targeting EZH2 inhibitors on osteogenic niche cells as well as other bone resident cells constituting TME in Aim 3.2.
- 6) Determine if bone-targeting EZH2 inhibitors cause immunosuppression when administered as different doses in Aim 3.3.
- 7) Contribute to data analysis as well as to the preparation of manuscripts and presentations.

The team will be supported by: **C. Kent Osborne, M.D.** (OSD; Professor, Lester and Sue Smith Breast Center, Baylor College of Medicine; Founder of the Dan L Duncan Comprehensive Cancer Center)—his research involves the identification of mechanisms of resistance to targeted therapies in breast cancer particularly treatments targeting the estrogen receptor and the HER2 receptor. Dr. Osborne will consult on experimental design, troubleshooting, data analysis, and potential clinical trials (see Letter). **Tao Wu, Ph.D.** (Co-I; Professor, Baylor College of Medicine)—his research involves the deciphering the drivers of epigenetic mutations in cancer and development of new epigenetic therapies for cancer (see Letter). **Yi Ding, Ph.D.** (Lead Data Scientist, CVS) — Dr. Ding will ensure statistical planning and perform all statistical analyses (see Letter). The campus Veterinary Service Center will provide necessary facilities and guidance for our animal work. We also have access to facilities available at Rice University, the Baylor College of Medicine, and Texas Medical Center. The team will communicate via biweekly conference calls and in-person or Zoom meetings monthly.

Below a schematic of activities that will be pursued by each research team:



II. The Governance and Organizational Structure.

Leadership Plan and Communications. Drs. Xiao and Zhang will serve as PIs for the project. In these roles, the two PIs will be responsible for the implementation of the specific aims and ensure that systems are in place to guarantee institutional compliance with US laws, DHHS policies, and NIH policies governing biosafety, human research, data, and facilities. The two laboratories are located within 1 mile in the Texas Medical Center and the two PIs have established track records of collaboration. Drs. Xiao and Zhang interact on a weekly base. They will communicate weekly, either by phone, e-mail, or in person, to discuss experimental design, data analysis, and all other related administrative responsibilities. All PIs will share their research results with members in each group. They will work together to discuss any further changes in the projects and reprogramming of funds as necessary. Publication authorship will be based on the relative scientific contributions of the PIs and key personnel. Dr. Xiao will serve as contact PI and be responsible for the submission of progress reports to NIH and all communication.

- a. **Intellectual Property.** Should intellectual property be developed, the Technology Transfer Offices at the

Baylor College of Medicine and Rice University will be responsible for preparing and negotiating an agreement for intellectual property. An Intellectual Property Committee composed of representatives from each institution will be formed to work together to ensure the intellectual property developed by the PIs is protected according to the policies established in the agreement.

- b. **Conflict resolution.** We do not anticipate conflicts in budgetary issues as these investigators both have long track records of collaborative research, and are dedicated to cancer research. If a conflict does arise, we will first involve our respective department leadership, namely Drs. Matthew Ellis and Anatoly B. Kolomeisky. Both have many years of administrative and leadership experience. We will also discuss any key issues with Dr. Adam Kuspa, Dean of Research at BCM and Dr. Thomas C. Killian, Dean of Wiess School of Natural Sciences at Rice University to seek their advice to resolve any other conflicts.
- c. **Budget allocation.** Funds are allocated to each PI to provide support commensurate with levels of effort appropriate for the work outlined. The contact PI's Institution will subdivide the award funds as described in the budget section, and each PI will be responsible for his own budget. The two PIs will make the decisions on minor changes in research direction, and may reallocate funds and resources between PIs should the need arise.

III. Change in PD/PI Location or Program Leadership.

If one of the PIs moves to a new institution, attempts will be made to transfer the relevant portion of the grant to the new institution. In the event that a PI cannot carry out his duties, a new PI will be recruited as a replacement, subject to the approval of the institution and NIH.

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CONSORTIUM/CONTRACTUAL ARRANGEMENTS

The proposed research includes a consortium arrangement between William Marsh Rice University and the following domestic institution:

Baylor College of Medicine:

Dr. Xiang Zhang will serve as the PI for this site on this multi-PI proposal.

The appropriate programmatic and administrative personnel of each organization involved in this grant application are aware of the agency's consortium agreement policy and are prepared to establish the necessary inter-organizational agreement(s) consistent with that policy.



Anatoly B. Kolomeisky, Ph.D.
Chair, Department of Chemistry

Oct 31, 2022

RE: Multiple-PI R01 Application by Drs. Xiao and Zhang

Dear Colleagues,

As Chair of the Department of Chemistry at Rice University, I am writing to express my strong and enthusiastic support for Dr. Han Xiao's R01 application in collaboration with Dr. Xiang Zhang at Baylor College of Medicine, entitled as "Modulation of Epigenetic Target in the Bone to Treat Breast Cancer Metastasis."

Dr. Xiao came to us with excellent training and preparation for his independent career. He received his undergraduate research training and the undergraduate degree from the University of Science and Technology of China (USTC), where he graduated with a B.S. in chemistry and an honors degree in physical science. He then joined The Scripps Research Institute (TSRI) and earned his Ph.D. under the supervision of Prof. Peter G. Schultz. His graduate work mainly focused on expanding the technique of genetically incorporating noncanonical amino acids (ncAAs) in both prokaryotic and eukaryotic organisms and applying this technique for the preparation of multi-action antibody conjugates. Dr. Xiao's highly productive graduate work is represented by thirteen publications. Dr. Xiao then moved to Stanford University, where he joined the laboratory of Prof. Carolyn R. Bertozzi as a Good Ventures Postdoctoral Fellow of Life Science Research Foundation. At Stanford, he developed the first Glyco-Immune Checkpoint inhibitor, an antibody-sialidase conjugate, enabling precision removal of cell surface glycans that are responsible for suppressing the immune system. Dr. Xiao's postdoctoral work has resulted in one publication and two US patents licensed by Palleon Pharmaceuticals. Dr. Xiao joined our faculty as an Assistant Professor in a tenure-track position in the summer of 2017, and he has already been recognized as a *Norman Hackerman Welch Young Investigator*, a *Cancer Prevention and Research Institute of Texas (CPRIT) Scholar in Cancer Research*, a *Maximizing Investigators' Research Award for Early Stage Investigators (MIRA)*, and *Department of Defense Breast Cancer Breakthrough Award – Level 2*.

Research Activities. Since 2017, Dr. Xiao has established a compelling research program focused on harnessing the power of chemical tools to advance the understanding of complex biological systems and transform treatment of cancers at Rice University. The ability to site-specifically introduce distinct chemical moieties into biomolecules affords a powerful tool to investigate and manipulate biological systems. He has developed innovative solutions for this purpose, including single-atom fluorescence switches, a proximity-based platform for precision antibody labeling (pClick), and a new bone targeting (BonTarg) technology. His work at Rice University already resulted in 25 publications, one PCT patent application, one US patent, and two US provisional patent applications.

In his proposal, Dr. Xiao delineates a plan to engineer epigenetic inhibitors for bone tumor delivery and to treat bone micrometastases and secondary metastases from bone lesions. The successful completion of his proposed study will provide a better therapeutic regimen for the treatment of bone metastatic tumors as well as metastases derived from bone lesions. To ensure fruitful outcomes of the proposed project, Dr. Xiao has successfully established collaborations with Drs. Xiang Zhang (Baylor College of Medicine, a pioneer in studying breast cancer metastases in bones), Jason Yustein (Texas's Children Hospital, an expert in Ewing Sarcoma and metastatic Ewing Sarcoma), Chengcheng Zhang (University of Texas Southwestern Medical Center, a pioneer in studying the functions of immunoinhibitory receptors in cancer), Robin Parihar (Texas Children's Hospital, trained in NK cell therapy), Mostafa Waleed Gaber (Texas's Children Hospital, an expert in cancer imaging and anti-cancer drug delivery).

Education Activities. Dr. Xiao is also an outstanding teacher, and his teaching activities are synergistic with his research. To prepare students at Rice University for careers in chemical biology research, he developed the first Chemical Biology course in the Department (CHEM552) in 2017 and Drug Discovery at the interface of Chemistry and Biology (CHEM554) in 2019. He also taught Organic Chemistry II (CHEM212) in the springs of 2020 and 2021. He has received top teaching evaluations for all these classes. As a laboratory mentor, Dr. Xiao

has mentored six postdoc fellows, nine graduate students, eight undergraduate students, two K-12 teachers, and four visiting students from non-research institution.

Department's Long-term Goals for Dr. Xiao's Career. Dr. Xiao's program is geared toward bridging the fields of chemistry and biology by applying different chemical tools to address fundamental biological questions as well as to develop novel cancer therapy. His research plan is supported by, and integrated into, the research goals of the department. According to our Strategic Plan, "The foundational programs of the Rice Chemistry research fall in three broad interdisciplinary areas: Chemistry of Life, Materials by Chemical Design (chiefly nanomaterials), and Chemistry for Energy and Sustainability." Dr. Xiao has assembled a vital team of collaborators at Baylor College of Medicine, McGovern Medical School at UT Health, Texas Children's Hospital, and MD Anderson Cancer Center; These collaborators will complement the expertise of Dr. Xiao's group and broaden the reach and impact of his group's innovations in cancer research. We will do all that is necessary at the department level to ensure a very positive and career-enhancing experience for him. Dr. Xiao's outstanding research and teaching contributions make him a great asset to our department and a key to our success.

Institutional Commitment. I have read and endorse Dr. Xiao's proposal. The department is fully committed to Xiao's proposed research and Texas Medical Center will provide a collaborative environment for the research he proposed. The following resources and facilities are available to support his research objectives:

1. **Salary support:** Rice University provides Dr. Xiao with nine-month salary support for each academic year.
2. **Laboratory space:** We have allocated a 3,000 sq. ft. lab for Dr. Xiao and his research group.
3. Graduate Student support: We have provided Dr. Xiao with generous graduate students and postdoctoral researchers support through his startup package.
4. Equipment support: We have provided Dr. Xiao with substantial support for equipment for his lab, which has allowed him to set up a state-of-the-art chemical biology facility.
5. Rice shared equipment authority: Dr. Xiao and his research group have full access to the Rice shared equipment authority (<http://sea.rice.edu/>).
6. The department recognizes that Dr. Xiao's research objectives are truly interdisciplinary. We actively encourage and support this interdisciplinary research program and will provide mentoring and collaborative resources essential to this effort. In particular, Dr. Xiao has already initiated collaborative relationships through the Bioscience Research Collaborative (<http://brc.rice.edu/home/>) with the goal of applying chemical tools to biological research.

In closing, let me say that we are very committed to Dr. Xiao's success as a faculty member within the Department of Chemistry at Rice University. Dr. Xiao is an exceptionally creative and driven young scholar. Rice University is fully committed to supporting and promoting the exceptional growth of Dr. Xiao's program; we will fully support the requirements of this application.

Sincerely,



Anatoly B. Kolomeisky, Ph.D.
Chair, Department of Chemistry

Email tolya@rice.edu | Office 713-348-5672
Department of Chemistry - MS 60 | 6100 Main St. | Houston, TX 77005-1892



Tao Wu, Ph. D.
Assistant Professor
Molecular and Human Genetics
Baylor College of Medicine
One Baylor Plaza
Room T905
Houston, TX 77030, USA
Phone: (713) 798-2805
Email: Tao.Wu@bcm.edu

Mar 9th, 2022

Dear Drs. Xiao and Zhang,

I am delighted to collaborate with you on your proposal "Modulation of Epigenetic Target in the Bone to Treat Breast Cancer Metastasis". The proposed project is of great interest for applications in the ongoing studies in my laboratory.

The focus of my laboratory is to define the underlying epigenetic regulatory mechanisms of cellular plasticity and cell fates decision in early embryonic development and cancer progression. Due to our primary interest in cancer epigenetics, my group has developed several analytical tools and expertise to characterize epigenetic alterations, and genomic landscape reprogramming that will be highly relevant to your project, including native ChIP-seq, Cut&Tag, and single-cell ATAC-Seq. Thus, we are in a unique position to collaborate with you and help you study how bone-targeting EZH2 inhibitors can change cellular plasticity and cell fates. I will be happy to provide expertise and share materials with researchers in your laboratory and look forward to working with you on this exciting project.

Your proposal is innovative and significant since there is no EZH2 inhibitor available for tumor-specific delivery. I believe that the success of these proposed studies will lead to the new treatment of these patients with better efficacy and fewer side effects. Based on my extensive experience in epigenetic research, I can provide you with the guidance and advice necessary to support the successful completion of this application. I anticipate this collaboration to be a productive and long-lasting one, and I am looking forward to helping you achieve the overall goals of this study.

Sincerely,

A handwritten signature in black ink, appearing to be "Tao Wu", with a stylized, flowing script.

Tao Wu, Ph.D.
Assistant Professor
Baylor College of Medicine



DAN L DUNCAN
COMPREHENSIVE
CANCER CENTER

November 3, 2022

Xiang H. F. Zhang
Professor
Lester and Sue Smith Breast Center
Molecular and Cellular Biology
Baylor College of Medicine
One Baylor Plaza, BCM600
HOUSTON, TX 77030

Re: Letter of support for your R01 application

Dear Xiang,

As the Founding Director of the Dan L Duncan Comprehensive Cancer Center, it is my great pleasure to participate in your project entitled "Modulation of Epigenetic Target in the Bone to Treat Breast Cancer Metastasis".

As you know, I am a breast cancer oncologist with over 40 years' experience. I have led and participated in numerous clinical trials including some of the early trials that led to current standard of care in breast cancer management. Recently, my team has been focusing on endocrine resistance mechanisms of ER+ breast cancers, and combinatory anti-HER2 treatment to deescalate chemotherapies. I am deeply interested in your proposal to deliver epigenetic inhibitors for better targeting of bone metastases. The preliminary findings support an intriguing possibility that this newly developed agent could overcome poor tissue bioavailability and eliminate minimal residual diseases in the bone more completely. The possibility of specifically targeting EZH2 in the bone cancer niche and only modulate the epigenetics in the cancer niche is highly interesting, and definitely warrants further investigations.

I am happy to contribute my expertise in clinical oncology to the proposed studies. Specifically, I will meet you and Dr. Xiao (the contact PI of this application) periodically, and discuss new findings and emerging opportunities to translate these findings into clinical trials. Based on the preliminary data I have seen and our ongoing discussions, I am optimistic that the new agents you collaboratively developed may eventually lead to shift of paradigm in breast cancer treatment.

I lend my strongest support and wish you best luck for your application.

Respectfully,

A handwritten signature in black ink, appearing to read "C. Kent Osborne".

**C. Kent Osborne, M.D., Founding Director
Dan L Duncan Comprehensive Cancer Center
Baylor College of Medicine**



LESTER AND SUE SMITH
**BREAST
CENTER**

Qian Zhu

Assistant Professor
One Baylor Plaza,
BCM-Jewish Building,
Room 644D
Houston, TX 77030
Qian.Zhu@bcm.edu
Tel: 609-375-7517

October 26, 2022

Dear Xiang,

It is my great pleasure to express my willingness to support your grant application entitled "Modulation of epigenetic target in the bone to treat breast cancer metastasis".

I am a newly appointed assistant professor of computational biology in the Lester Sue and Smith Breast Center at Baylor College of Medicine. My research focuses on developing novel integrative computational methods for single-cell transcriptomics, spatial transcriptomics, and epigenomics. I received rigorous training in multiple related areas including computer science, bioinformatics and biology at Princeton University and Dana-Farber Cancer Institute and have collaborated extensively with basic biologists and clinicians. In the past, I have developed novel approaches to integrate multiple levels of single cell omics as well as analyses of spatial transcriptomics, which have been published in high-impact journals such as *Cell*, *Nature Methods*, *Nature Biotechnology*, and *Genome Biology*. I will be more than happy to apply my expertise and help you evaluate spatial correlation between EZH2 and a number of other pathways and genes characteristic of cancer stem cells in clinical specimens. Of note, the EZH2-related analysis will be conducted on the basis of the matched primary tumor/bone metastasis data that we collaboratively obtained through another project.

I look forward to extending our collaboration to this exciting project.

Sincerely,

A handwritten signature in black ink that reads "Qian Zhu".

Qian Zhu

Assistant Professor
CPRIT Scholar
Lester Sue and Smith Breast Center
Department of Human Genetics
Baylor College of Medicine



Yi Ding, Ph. D.
Lead Data Scientist
CVS Health Corporation*
1 CVS Drive
Woonsocket, RI 02895
florady2014@gmail.com

Oct 20th, 2022

RE: Multiple-PI R01 Application by Drs. Xiao and Zhang

To Whom It May Concern,

I am writing this letter to convey my full support in regard to Drs. Xiao and Zhang's R01 application, entitled as "Modulation of Epigenetic Target in the Bone to Treat Breast Cancer Metastasis." I am happy to serve as a consultant to provide statistic support for data analysis and experimental design.

I obtained training in theory and practice of statistics in the MS Statistics program at Pennsylvania state university and got my PhD degree in Computational Biology at Sanford Burnham Prebys Medical Discovery Institute. During my PhD study, as a computational biologist, I have served several roles, including data analyst, statistician, bioinformatician and biologist. After obtaining my PhD, I worked as a research assistant with professors in the Department of Statistics at Stanford University for a year from 2016 to 2017. That gave me experience in building statistical methods and performing data analysis in biological science. In 2017, I joined IBM Watson Health as a data scientist and had worked on multiple statistical modeling and testing projects in the healthcare area. From 2018 to March 2022, I worked as data scientist at SYSCO on experiment design of campaigns. Since April 2022, I am a lead data scientist at CVS Health.

I am currently employed in a full time at CVS, and will work on this project during my part time. So no compensation is required.

I am looking forward to working with Drs. Xiao and Zhang on the proposed project.

Best,

A handwritten signature in black ink, appearing to read "Yi Ding".

Yi Ding, Ph.D.

* Any opinions expressed are solely my own and do not express the views or opinions of my employer.



November 2, 2022

Xiang H.-F. Zhang, Ph.D. ("Shawn")
Professor of Molecular and Cellular Biology
The William T. Butler, M.D., Endowed Chair for
Distinguished Faculty
Interim Director, Lester and Sue Smith Breast Center
McNair Scholar
Baylor College of Medicine

ARUN SREEKUMAR, PhD

Professor of Cell Biology,
Charles C Bell Jr Endowed Professor of Cell Biology
Co-Director Center for Metabolism and Experimental
Therapeutics,
Co-Leader Cancer Biology Program at Dan L Duncan
Comprehensive Cancer Center,
Senior Advisor Metabolomics Core Laboratory,
Departments of Molecular and Cellular Biology, and
Biochemistry,
One Baylor Plaza, Jewish Building, Suite 120DJ
Houston, Texas 77030
713-798-3305
713-790-1275/fax
sreekuma@bcm.edu

Dear Xiang,

I am writing to confirm my willingness to support your study entitled "Modulation of Epigenetic Target in the Bone to Treat Breast Cancer Metastasis".

As you know, we have established a pipeline to obtain paired primary tumor and bone metastasis samples in collaboration with Dr. Ruth Asirvatham. The first batch contains 11 such pairs. In fact, these samples already play important roles in our existing collaborations, including the manuscript your and my labs are preparing together to describe the unique immune microenvironment of bone metastases. I will be happy to extend our collaboration to the examination of EZH2 as a therapeutic target.

Your previous and proposed studies on EZH2 are fascinating to me. The preliminary evidence that bone-specific inhibition of EZH2 prevents further spread of bone metastasis is convincing to me, and strongly indicate the therapeutic potential. Therefore, I heartedly support your application.

Sincerely,

Sincerely,

A handwritten signature in cursive script, appearing to read "Arun Sreekumar", with a horizontal line underneath.

Arun Sreekumar, PhD
Charles C Bell Jr. Endowed Professor
Department of Molecular and Cellular Biology
Baylor College of Medicine



LESTER & SUE SMITH
**BREAST
CENTER**

Yi Li, Ph.D.

Professor
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Baylor College of Medicine
1 Baylor Plaza, BCM 600
Alkek N1220.06
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Tel: (713) 798-3963
Fax: (713) 798-1659
Email: liyi@bcm.edu

November 4, 2022

Dear Xiang,

It is my great pleasure to write this letter of support for your proposal entitled "Modulation of Epigenetic Target in the Bone to Treat Breast Cancer Metastasis".

As you know, my laboratory has well-established expertise on mammary tumor modeling by intra-nipple injection of oncogenic viral particles. Recently, we have successfully applied this technology to rat models and obtained ER⁺, estrogen responsive mammary tumor models that spontaneously developed bone metastasis. I will be happy to collaborate with you on your project to examine if bEZH2i treatments may achieve therapeutic effect in ER⁺ rat models similar to that in ER⁻ murine models. Given the long-standing collaborative history between our laboratories, I am sure we can once again synergistically combine our strengths.

I look forward to furthering our collaboration on this new project!

Sincerely,

A handwritten signature in black ink, appearing to read "Yi Li".

Yi Li, Ph.D.

RESOURCE SHARING PLAN

1) Data Sharing Plan

Research results will be shared with the academic community and general public through publishing journal articles, presenting at conferences, and publishing patents or books. Rice University and Baylor College of Medicine (BCM) are both committed to disseminating its research and scholarship as widely as possible. The PIs will follow the NIH policies of publication of the data obtained within the sponsored project. Whenever possible, open access to these publications will be provided through the journals themselves and/or through PubMed Central. All researchers in the our laboratories are required to maintain laboratory notebooks. The laboratory notebook documents the design of experiments, development of experimental process, and the results of the research. Laboratory notebooks will be made available to the whole research group. The lab notebooks that are no longer actively in use will be collected and stored in the PIs' offices. Furthermore, lab members will regularly prepare written reports summarizing their research outcomes. These reports are typically transformed into manuscripts submitted to peer-reviewed journals for dissemination to the broader chemical community. Research reports will be made available to the whole research group through Rice BOX or BCM BOX system, both being a secure cloud-based storage system. Specific details of our data format and content are listed below. Research materials generated in the course of this project will also be made freely available to other investigators to the maximum feasible extent, either informally as a courtesy or through formal collaborations.

Chemical Data: Chemical products will be purified, and fully characterized to establish chemical structure/purity using spectroscopic methods. Typical spectroscopic data will include ^1H NMR, ^{13}C NMR, IR rotation and mass spectroscopy. All data will be stored in electronic format (both raw data, such as FID files, and PDF format) in both laboratory hard drives and Rice and BCM cloud-based storage. All spectra are indexed according to the page in the laboratory notebook in which they are recorded for rapid identification. Raw FID files collected can be processed by any number of NMR software programs, and can be made available upon request. The synthesized materials are stored in the cabinets located at Bioscience Research Collaborative 310D, Rice University.

Biological Data: All experimental details will be documented in the laboratory notebooks. Results of the research will be made available in digital form in spreadsheet tables, tab-delimited files, or image files. Images will be saved in standard image formats such as JPEG, TIFF, or PNG. Biomaterials (e.g., plasmids, cell lines, proteins) generated under this project will be stocked in the -80 degree freezer in the lab. All model organisms generated by this project will be distributed freely or deposited into a repository/stock center (e.g., Addgene) making them available to the broader research community immediately after publication. If we assume responsibility for distributing the newly generated model organisms, we fill requests in a timely fashion. In addition, we will provide relevant protocols and published genetic and phenotypic data upon request. Material transfers will be made with no more restrictive terms than in the Simple Letter Agreement (SLA) or the Uniform Biological Materials Transfer Agreement (UBMTA) and without reach through requirements. Should any intellectual property arise which requires a patent, we will ensure that the technology (materials and data) remains widely available to the research community in accordance with the NIH Principles and Guidelines document. Any transcriptomic or epigenomic data generated during the proposed projects will be shared in compliance with the NIH Genomic Data Sharing policy (NOT-OD-14-124).

2) Sharing Model Organisms

Not applicable.

3) Genomic Data Sharing (DGS)

Not applicable.

AUTHENTICATION OF KEY BIOLOGICAL AND/OR CHEMICAL RESOURCES

The methods used for authentication will depend on the key resource type.

Chemicals: Chemicals purchased commercially come with an authentication sheet identifying the purity and contaminant. Otherwise, chemicals will be validated by nuclear magnetic resonance spectroscopy and mass spectrometry.

Cell lines: Key cell lines can be authenticated by chromosomal analysis or short tandem repeat (STR) profiling (MD Anderson Cancer Center facility). Human cell lines acquired from American Type Tissue Collection (ATCC) and other commercial cell line providers assess all of their lots of their cell lines by STR profiling, which allows one to distinguish one DNA sample from another and allows the unique identification of specific cell lines. All of this information is readily available to their customers on the company's website. They will also be checked monthly for mycoplasma.

DNA plasmids: Sequence of DNA plasmids will be validated through DNA sequence. Sizes of plasmids will be checked by enzymatic digestion and gel electrophoresis.

Antibody: Antibodies will be validated by Western blot, ELISA, immunoprecipitation, immunofluorescence, or flow cytometry, depending on the assay proposed. In addition, the sequences of key antibodies or antibody conjugates can be verified by ESI mass-spectrum.

Mice: Key mice cell lines can be validated using short tandem repeat (STR) profiling or single nucleotide polymorphism (SNP) profiling as previously reported¹.

¹. BMC Genomics, 2014, 15, 487. Cytotechnology, 2014, 66, 133.

GENERAL INFORMATION PAGE (subawards at bottom)

Fiscal Year		2023	2024	2025	2026	2027	2028
Faculty Fringes	FF	0.202	0.202	0.202	0.202	0.202	0.202
Faculty Fringes A5 *	FF5	0.202	0.202	0.202	0.202	0.202	0.202
Faculty Fringes A6 *	FF6	0.202	0.202	0.202	0.202	0.202	0.202
Faculty Fringes A7 *	FF7	0.202	0.202	0.202	0.202	0.202	0.202
Faculty Fringes A8 *	FF8	0.202	0.202	0.202	0.202	0.202	0.202
Graduate Tuition Remissior	GS	0.385	0.385	0.385	0.385	0.385	0.385
Staff Fringes	ST	0.256	0.256	0.256	0.256	0.256	0.256
Undergraduate Fringes	UG	0.002	0.002	0.002	0.002	0.002	0.002
Other Fringes B5 *	OT5	0.256	0.256	0.256	0.256	0.256	0.256
Other Fringes B6 *	OT6	0.256	0.256	0.256	0.256	0.256	0.256
Other Fringes B7 *	OT7	0.256	0.256	0.256	0.256	0.256	0.256
Other Fringes B8 *	OT8	0.256	0.256	0.256	0.256	0.256	0.256
Other Fringes B9 *	OT9	0.256	0.256	0.256	0.256	0.256	0.256
Other Fringes B10 *	OT10	0.256	0.256	0.256	0.256	0.256	0.256
Indirect Cost Rate	IC	0.565	0.565	0.565	0.565	0.565	0.565

* Fringe rates for calculating the corresponding budget line in years 1-5 (Senior/Key Person, lines 5-8 and Other Personnel, lines 5-10). Additional information is included in the instruction tab.

	Month*	Day	Fiscal Year	
Start date:	7	1	2023	
*Use calendar month: Jan=1, Feb=2, etc.				
Duration (in whole years)	5			
RATIOS	FY-A	1	FY-B	0
Prorates based on fiscal year beginning July 1.				

Annual Budget Increase Rates	
A. SENIOR/KEY PERSON	0.00%
B. OTHER PERSONNEL	0.00%
D. TRAVEL	0.00%
E. PARTICIPANT SUPPORT COSTS	0.00%
F. OTHER DIRECT COSTS	0.00%

SUBCONTRACTS

Calculates indirect costs on the first \$25,000 for each subaward.

Subcontractor's name		Year 1	Year 2	Year 3	Year 4	Year 5
Baylor Collge of Medicine	Subcontract annual cost (Direct + Indirect)	368,000	368,000	368,000	368,000	368,000
PI: Xiang Zhang	Rice's Indirect Cost	14,125	0	0	0	0
	Subcontract annual cost (Direct + Indirect)					
	Rice's Indirect Cost	0	0	0	0	0
	Subcontract annual cost (Direct + Indirect)					
	Rice's Indirect Cost	0	0	0	0	0
	Subcontract annual cost (Direct + Indirect)					
	Rice's Indirect Cost	0	0	0	0	0
	Subcontract annual cost (Direct + Indirect)					
	Rice's Indirect Cost	0	0	0	0	0
	Subcontract annual cost (Direct + Indirect)					
	Rice's Indirect Cost	0	0	0	0	0
	Subcontract annual cost (Direct + Indirect)					
	Rice's Indirect Cost	0	0	0	0	0
	Subcontract annual cost (Direct + Indirect)					
	Rice's Indirect Cost	0	0	0	0	0
	Subcontract annual cost (Direct + Indirect)					
	Rice's Indirect Cost	0	0	0	0	0
	Subcontract annual cost (Direct + Indirect)					
	Rice's Indirect Cost	0	0	0	0	0
	Subcontract annual cost (Direct + Indirect)					
	Rice's Indirect Cost	0	0	0	0	0
Total Subcontractors	Total Subcontract Costs	368,000	368,000	368,000	368,000	368,000
	Rice's Indirect Costs for all Subs	14,125	0	0	0	0
	MTDC	25,000	0	0	0	0

Period 1 of 5

A. Senior/Key Persons in Budget Period 1

First Name	Last Name	Project Role	Base Salary	Cal. Salary (\$)	Acad. Salary (\$)	Sum. Salary (\$)	Cal. Mons	Acad. Mons	Sum. Mons	Requested Salary (\$)	Fringe Benefits (\$)	Funds Req (\$)
Han	Xiao	PI	128,833					1.20		17,178	3,470	20,648
Total Funds requested for all Senior Key Persons listed on optional lines												

	Total and Rega
B. Other Personnel	

[illegible]

	Total Number Other Personnel
--	------------------------------

3

Total Other Personnel	140,568
------------------------------	----------------

Total Salary, Wages and Fringe Benefits (A+B)	161,216
--	---------

C. Equipment Description

1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11	Total Funds requested for all equipment listed in the optional lines	Total Equipment

D. Travel

1 Domestic Travel Costs (Incl. Canada, Mexico and U.S. Possessions)	2,000
2 Foreign Travel Costs	
Total Travel Cost	2,000

E. Participant/Trainee Support Costs

1 Tuition/Fees/Health Insurance			
2 Stipends			
3 Travel			
4 Subsistence			
5 Other			
Number of Participants/Trainees			
Total Participant/Trainee Support Costs			

F. Other Direct Costs

7. Other Direct Costs		Funds Requested (\$)
1	Materials and Supplies	51,044
2	Publication Costs	
3	Consultant Services	
4	ADP/Computer Services	
5	Subawards/Consortium/Contractual Costs	368,000
6	Equipment or Facility Rental/User Fees	
7	Alterations and Renovations	
8	Tuition Remission	28,490
9	SEA Instrumentation Fee / Professional Service	12,000
10	Animal Housing	5,250
Total Other Direct Costs		464,784

G. Direct Costs

Total Direct Costs (A thru F)		628,000
--------------------------------------	--	----------------

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	Funds Requested (\$)
MTDC	56.50%	256,510	144,928

Cognizant Federal Agency: DHHS. Uyen Tran. Phone: (214) 767-3261

I. Total Direct and Indirect Costs

Total Direct and Indirect Costs		Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G+H)		772,928

**Budget Numbers
to be Used
in Cayuse SP/R&A Form**

Use this number for Initial/Current Budget Period Direct Costs	628,000
--	---------

Use this number for Initial/Current F&A Base	256,510
--	---------

This number should equal the Initial Funds Req. 772,928

Cayuse 424 Budget Form

Period 2 of 5

A. Senior/Key Persons in Budget Period 2

Senior Key Persons in Budget															
First	Last	Project	Base	Cal.	Acad.	Sum.	Cal	Acad.	Sum.	Requested	Fringe				
Pref. Name	MI	Name	Suf.	Role	Salary	Salary (\$)	Salary (\$)	Salary (\$)	Mons	Mons	Mons	Salary (\$)	Benefits (\$)	Funds Req (\$)	
	Han		Xiao		PI	128,833					1.20		17,178	3,470	20,648
Total Funds requested for all Senior Key Persons listed on optional lines															

B. Other Personnel

		Cal	Acad	Sum	Total Senior/Key Persons		
No. of Personnel		Months	Months	Months	Requested Salary (\$)	Fringe Benefits (\$)	Funds Requested (\$)
1	Post Doctoral Associates	12.00			53,000	13,568	66,568
2	Graduate Students	24.00			74,000		74,000
	Undergraduate Students						
	Secretarial/Clerical						
	Technician						
Total Number Other Personnel					Total Other Personnel		
3					140,568		

Total Salary, Wages and Fringe Benefits (A+B) 161,216

C. Equipment Description

	Funds Requested (\$)
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
Total Funds requested for all equipment listed in the optional lines	
Total Equipment	

D. Travel

1	Domestic Travel Costs (Incl. Canada, Mexico and U.S. Possessions)	2,000
2	Foreign Travel Costs	
Total Travel Cost		2,000

E. Participant/Trainee Support Costs

1	Tuition/Fees/Health Insurance	
2	Stipends	
3	Travel	
4	Subsistence	
5	Other	
Number of Participants/Trainees		
Total Participant/Trainee Support Costs		

F. Other Direct Costs

F. Other Direct Costs		Funds Requested (\$)
1	Materials and Supplies	51,044
2	Publication Costs	
3	Consultant Services	
4	ADP/Computer Services	
5	Subawards/Consortium/Contractual Costs	368,000
6	Equipment or Facility Rental/User Fees	
7	Alterations and Renovations	
8	Tuition Remission	28,490
9	SEA Instrumentation Fee / Professional Serv	12,000
10	Animal Housing	5,250
Total Other Direct Costs		464,784

G. Direct Costs

Funds Requested (\$)
Total Direct Costs (A thru F) 628,000

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	Funds Requested (\$)
MTDC	56.50%	231,510	130,803

Cognizant Federal Agency: DHHS, Uyen Tran, Phone: (214) 767-3261

I. Total Direct and Indirect Costs

Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G+H) 758,803

I. Total Direct and Indirect Costs		Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G+H)		758,803

Cayuse 424 Budget Form

Period 4 of 5

A. Senior/Key Persons in Budget Period 4

First Pref. Name	Last MI Name	Suf.	Project Role	Base Salary	Cal. Salary (\$)	Acad. Salary (\$)	Sum. Salary (\$)	Cal. Mons	Acad. Mons	Sum. Mons	Requested Salary (\$)	Fringe Benefits (\$)	Funds Req (\$)
Han		Xiao	PI	128,833					1.20		17,178	3,470	20,648
Total Funds requested for all Senior Key Persons listed on optional lines													

B. Other Personnel

No. of Personnel	Cal	Acad	Sum	Requested Salary (\$)	Fringe Benefits (\$)	Funds Requested (\$)
	Months	Months	Months			
1 Post Doctoral Associates	12.00			53,000	13,568	66,568
2 Graduate Students	24.00			74,000		74,000
Undergraduate Students						
Secretarial/Clerical						
Technician						

Total Number Other Personnel

3

Total Other Personnel	140,568
Total Salary, Wages and Fringe Benefits (A+B)	161,216

C. Equipment Description

1		Funds Requested (\$)
2		
3		
4		
5		
6		
7		
8		
9		
10		
11	Total Funds requested for all equipment listed in the optional lines	Total Equipment

D. Travel

1 Domestic Travel Costs (Incl. Canada, Mexico and U.S. Possessions)	2,000
2 Foreign Travel Costs	
Total Travel Cost	2,000

E. Participant/Trainee Support Costs

1 Tuition/Fees/Health Insurance	
2 Stipends	
3 Travel	
4 Subsistence	
5 Other	
Number of Participants/Trainees	
Total Participant/Trainee Support Costs	

F. Other Direct Costs

1 Materials and Supplies	51,044
2 Publication Costs	
3 Consultant Services	
4 ADP/Computer Services	
5 Subawards/Consortium/Contractual Costs	368,000
6 Equipment or Facility Rental/User Fees	
7 Alterations and Renovations	
8 Tuition Remission	28,490
9 SEA Instrumentation Fee / Professional Serv	12,000
10 Animal Housing	5,250
Total Other Direct Costs	464,784

G. Direct Costs

	Funds Requested (\$)
Total Direct Costs (A thru F)	628,000

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)	Funds Requested (\$)
MTDC	56.50%	231,510	130,803

Cognizant Federal Agency: DHHS, Uyen Tran, Phone: (214) 767-3261

I. Total Direct and Indirect Costs

	Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G+H)	758,803

RESEARCH & RELATED BUDGET - Cumulative Budget

Totals (\$)

Section A, Senior/Key Person

103,240

702,840

Section B, Other Personnel

Total number other Personnel

15

Total Salary, Wages and Fringe Benefits (A + B)

806,080

Section C, Equipment

Section D, Travel

10,000

1. Domestic

10,000

2. Foreign

Section E, Participant/Trainee Support Costs

1. Tuition/Fees/Health Insurance

2. Stipends

3. Travel

4. Subsistence

5. Other

6. Number of Participants/Trainees

Section F, Other Direct Costs

2,323,920

1. Materials and Supplies

255,220

2. Publication Costs

3. Consultant Services

4. ADP/Computer Services

5. Subawards/Consortium/Contractual Costs

1,840,000

6. Equipment of Facility Rental/User Fees

7. Alterations and Renovations

8. Other 1

142,450

9. Other 2

60,000

10. Other 3

26,250

Section G, Direct Costs (A through F)

3,140,000

Use this amt for Project Period/
Project Request Direct Costs

Section H, Indirect Costs

668,141

Section I, Total Direct and Indirect Costs (G+H)

3,808,141

This number should equal the Total
Funds Req.

Section J, Fee

0

Total MTDC for Cayuse SP

1,182,550

Use this number for Project Period/
Project Request F&A Base



Subrecipient Questionnaire

Subrecipient Legal Name: Baylor College of Medicine

Address: One Baylor Plaza, Houston TX 77030-341

EIN#: 1741613878A1

DUNS: 051113330

Years in Business: 51 **Fiscal Year Start Date:** 07/01

Sponsor Classification (check box)

- | | |
|--|--|
| ☐ Public University/Institution of Higher Education | ☒ Not for Profit Entity |
| ☒ Private University | ☐ Foreign Institution of Higher Education |
| ☐ US Based For Profit Corporation (Publicly Held) | ☐ Foreign Based For Profit Corporation (Publicly Held) |
| ☐ US Based For Profit Corporation (Privately Held) | ☐ Foreign Based For Profit Corporation (Privately Held) |
| ☐ US Based Small Business | ☐ Foreign Based Small Business |

Questionnaire	Yes	No	N/A
Does your entity adhere to Cost Accounting Standards Boards (CASB) regulations?	X		
Is your organization required to comply with OMB Uniform Guidance regulations?	X		
Does your entity have written accounting policies and procedures for procurement, payroll, inventory, vendor payments, cash management, and contract administration?	X		
Does your entity have a specified individual/officer who is the custodian of financial records? If yes, please list here:	X		
Are purchasing and payment functions separate?	X		
Does your entity have a federally negotiated Facilities & Administration Rate? If yes, attach a copy of rate agreement.	X		
Are specific officials designated to approve payroll and other major transactions?	X		
Do you have annual financial statements audited by an independent audit firm? If yes, please attach a copy of the statement for most recent year.	X		
Do you have a formal effort reporting policy and procedures to manage after-the-fact effort reporting?	X		
Does your entity maintain an inventory for government property that identifies data such as vendor, purchase date, cost, description, serial number, and disposition?	X		
If foreign, is the entity registered/licensed in the country in which it operates?			X

Subrecipient Authorized Official

Title Leanne Scott, Exec. Dir. Spon. Programs Signature Leanne B. Scott, PhD Date 8/25/2021

Check One



Subrecipient/Consortium



Vendor

Rice OSR Signature:

Date

Digitally signed by Leanne B. Scott,
PhD
DN: cn=Leanne B. Scott, PhD,
o=Baylor College of Medicine,
ou=Sponsored Programs,
email=lspos@bcm.edu, cn=US
Date: 2021.08.25 18:36:40 -05'00'



Subrecipient Questionnaire

Subrecipient Legal Name: Baylor College of Medicine

	Yes	No
Is reporting of executive compensation required (see explanation below)?	☐	☒

The names and total compensation of the five most highly compensated officers of the entity(ies) must be listed if:

(i) the entity in the preceding fiscal year received—

(a) 80 percent or more of its annual gross revenues in Federal awards (federal contracts and subcontracts), loans, grants (and subgrants) and cooperative agreements); AND

(b) \$25,000,000 or more in annual gross revenues from Federal awards; and

(ii) the public does not have access to information about the compensation of the senior executives of the entity through periodic reports filed under section 13(a) or 15(d) of the Securities Exchange Act of 1934 (15 U.S.C. 78m(a), 78o(d)) or section 6104 of the Internal Revenue Code of 1986.

Name of Highly Compensated Officers	Salary
1.	
2.	
3.	
4.	
5.	

Scope of Work

Xiang Zhang, Baylor College of Medicine

The goals of this collaborative project between Labs at Rice University and Baylor College of Medicine are to develop the first bone-targeting EZH2 inhibitors and to study their effects on the survival and progression of breast cancer metastases in bone as well as secondary metastases to multi-organs from bone lesions. In this proposal, we will develop a general strategy of engineering EZH2 inhibitors for bone targeting and demonstrate the utility of these bone-targeting EZH2 inhibitors in the treatment of breast cancer metastases in bone. The strategy adopted in this proposal is innovative, because we currently have no strategy to specifically deliver therapeutic epigenetic modulators to the bone. The bone-targeted reagents developed in this study will reduce the mortality of breast cancer, improve the life of cancer patients with bone and other metastases. Given the P.I.'s extensive experience in molecule engineering and bone metastases studies, combined with the interactive and state-of-the-art research environment of Rice University and Baylor College of Medicine, and support from labs focusing on breast cancer bone metastases and sarcoma, the proposed objectives are achievable.

RESEARCH & RELATED BUDGET - Budget Period 1

OMB Number: 4040-0001

Expiration Date: 12/31/2022

UEI: FXKMA43NTV21

Enter name of Organization: Baylor College of Medicine

Budget Type: ☐ Project ☒ Subaward/Consortium

Budget Period: 1

Start Date: 07/01/2023

End Date: 06/30/2024

A. Senior/Key Person

Prefix	First	Middle	Last	Suffix	Base Salary (\$)	Months			Requested Salary (\$)	Fringe Benefits (\$)	Funds Requested (\$)
						Cal.	Acad.	Sum.			
	Xiang		Zhang	PhD	203,700.00	1.20			20,370.00	5,307.00	25,677.00

Project Role: PD/PI

	Tao		Wu	PhD	128,750.00	0.60			6,438.00	1,635.00	8,073.00
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Project Role: Co-Investigator

Additional Senior Key Persons:

Add Attachment

Delete Attachment

View Attachment

Total Funds requested for all Senior Key Persons in the attached file

Total Senior/Key Person

33,750.00

B. Other Personnel

Number of Personnel	Project Role	Cal.	Months		Sum.	Requested Salary (\$)	Fringe Benefits (\$)	Funds Requested (\$)
			Cal.	Acad.				
2	Post Doctoral Associates	18.00				87,831.00	24,486.00	112,317.00
	Graduate Students							
	Undergraduate Students							
	Secretarial/Clerical							

2 Total Number Other Personnel

Total Other Personnel

112,317.00

Total Salary, Wages and Fringe Benefits (A+B)

146,067.00

C. Equipment Description

List items and dollar amount for each item exceeding \$5,000

Equipment item	Funds Requested (\$)
Additional Equipment:	
Add AttachmentDelete AttachmentView Attachment	
Total funds requested for all equipment listed in the attached file	
Total Equipment	

D. Travel

	Funds Requested (\$)
1. Domestic Travel Costs (Incl. Canada, Mexico and U.S. Possessions)	3,000.00
2. Foreign Travel Costs	
Total Travel Cost	3,000.00

E. Participant/Trainee Support Costs

	Funds Requested (\$)
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other	
Number of Participants/Trainees	Total Participant/Trainee Support Costs

F. Other Direct Costs		Funds Requested (\$)
1. Materials and Supplies		54,233.00
2. Publication Costs		
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Animal Maintenance		7,700.00
9. Laboratory Analysis/Core Charges		19,000.00
10.		
11.		
12.		
13.		
14.		
15.		
16.		
17.		
Total Other Direct Costs		80,933.00

G. Direct Costs	Funds Requested (\$)
Total Direct Costs (A thru F)	230,000.00

H. Indirect Costs				Funds Requested (\$)
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)		
MTDC	60.00	230,000.00		138,000.00
Total Indirect Costs				138,000.00

Cognizant Federal Agency	
(Agency Name, POC Name, and POC Phone Number)	DHHS, Arif Karim, 214-767-3261

I. Total Direct and Indirect Costs	Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G + H)	368,000.00

J. Fee	Funds Requested (\$)

K. Total Costs and Fee	Funds Requested (\$)
Total Costs and Fee (I + J)	368,000.00

L. Budget Justification				
(Only attach one file.)	BCM Budget Justification 10.21.22.pdf	Add Attachment	Delete Attachment	View Attachment

RESEARCH & RELATED BUDGET - Budget Period 2

OMB Number: 4040-0001
Expiration Date: 12/31/2022

UEI: FXXKMA43NTV21

Enter name of Organization: Baylor College of Medicine

Budget Type: ☐ Project ☒ Subaward/Consortium

Budget Period: 2 Start Date: 07/01/2024 End Date: 06/30/2025

A. Senior/Key Person

Prefix	First	Middle	Last	Suffix	Base Salary (\$)	Months			Requested Salary (\$)	Fringe Benefits (\$)	Funds Requested (\$)
						Cal.	Acad.	Sum.			
	Xiang		Zhang	PhD	203,700.00	1.20			20,370.00	5,307.00	25,677.00

Project Role: PD/PI

	Tao		Wu	PhD	128,750.00	0.60			6,438.00	1,635.00	8,073.00
--	-----	--	----	-----	------------	------	--	--	----------	----------	----------

Project Role: Co-Investigator

Additional Senior Key Persons: Total Funds requested for all Senior Key Persons in the attached file

Total Senior/Key Person

B. Other Personnel

Number of Personnel	Project Role	Cal.	Months		Requested Salary (\$)	Fringe Benefits (\$)	Funds Requested (\$)
			Cal.	Sum.			
2	Post Doctoral Associates	18.00			87,831.00	24,486.00	112,317.00
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						

Total Number Other Personnel Total Other Personnel

Total Salary, Wages and Fringe Benefits (A+B)

C. Equipment Description

List items and dollar amount for each item exceeding \$5,000

Equipment item	Funds Requested (\$)
Additional Equipment:	
Add AttachmentDelete AttachmentView Attachment	
Total funds requested for all equipment listed in the attached file	
Total Equipment	

D. Travel

	Funds Requested (\$)
1. Domestic Travel Costs (Incl. Canada, Mexico and U.S. Possessions)	3,000.00
2. Foreign Travel Costs	
Total Travel Cost	3,000.00

E. Participant/Trainee Support Costs

	Funds Requested (\$)
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other	
Number of Participants/Trainees	Total Participant/Trainee Support Costs

F. Other Direct Costs		Funds Requested (\$)
1. Materials and Supplies		54,233.00
2. Publication Costs		
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Animal Maintenance		7,700.00
9. Laboratory Analysis/Core Charges		19,000.00
10.		
11.		
12.		
13.		
14.		
15.		
16.		
17.		
Total Other Direct Costs		80,933.00

G. Direct Costs	Funds Requested (\$)
Total Direct Costs (A thru F)	230,000.00

H. Indirect Costs				Funds Requested (\$)
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)		
MTDC	60.00	230,000.00		138,000.00
Total Indirect Costs				138,000.00

Cognizant Federal Agency	
(Agency Name, POC Name, and POC Phone Number)	DHHS, Arif Karim, 214-767-3261

I. Total Direct and Indirect Costs	Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G + H)	368,000.00

J. Fee	Funds Requested (\$)

K. Total Costs and Fee	Funds Requested (\$)
Total Costs and Fee (I + J)	368,000.00

L. Budget Justification			
(Only attach one file.)	BCM Budget Justification 10.21.22.pdf	Add Attachment	Delete Attachment
			View Attachment

RESEARCH & RELATED BUDGET - Budget Period 3

OMB Number: 4040-0001
Expiration Date: 12/31/2022

UEI: FXXKMA43NTV21

Enter name of Organization: Baylor College of Medicine

Budget Type: ☐ Project ☒ Subaward/Consortium

Budget Period: 3 Start Date: 07/01/2025 End Date: 06/30/2026

A. Senior/Key Person

Prefix	First	Middle	Last	Suffix	Base Salary (\$)	Months			Requested Salary (\$)	Fringe Benefits (\$)	Funds Requested (\$)
						Cal.	Acad.	Sum.			
	Xiang		Zhang	PhD	203,700.00	1.20			20,370.00	5,307.00	25,677.00

Project Role: PD/PI

	Tao		Wu	PhD	128,750.00	0.60			6,438.00	1,635.00	8,073.00
--	-----	--	----	-----	------------	------	--	--	----------	----------	----------

Project Role: Co-Investigator

Additional Senior Key Persons: Total Funds requested for all Senior Key Persons in the attached file

Total Senior/Key Person

B. Other Personnel

Number of Personnel	Project Role	Cal.	Months		Requested Salary (\$)	Fringe Benefits (\$)	Funds Requested (\$)
			Cal.	Sum.			
2	Post Doctoral Associates	18.00			87,831.00	24,486.00	112,317.00
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						

Total Number Other Personnel Total Other Personnel

Total Salary, Wages and Fringe Benefits (A+B)

C. Equipment Description

List items and dollar amount for each item exceeding \$5,000

Equipment item	Funds Requested (\$)
Additional Equipment:	
Add AttachmentDelete AttachmentView Attachment	
Total funds requested for all equipment listed in the attached file	
Total Equipment	

D. Travel

	Funds Requested (\$)
1. Domestic Travel Costs (Incl. Canada, Mexico and U.S. Possessions)	3,000.00
2. Foreign Travel Costs	
Total Travel Cost	3,000.00

E. Participant/Trainee Support Costs

	Funds Requested (\$)
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other	
Number of Participants/Trainees	Total Participant/Trainee Support Costs

F. Other Direct Costs		Funds Requested (\$)
1. Materials and Supplies		54,233.00
2. Publication Costs		
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Animal Maintenance		7,700.00
9. Laboratory Analysis/Core Charges		19,000.00
10.		
11.		
12.		
13.		
14.		
15.		
16.		
17.		
Total Other Direct Costs		80,933.00

G. Direct Costs	Funds Requested (\$)
Total Direct Costs (A thru F)	230,000.00

H. Indirect Costs				Funds Requested (\$)
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)		
MTDC	60.00	230,000.00		138,000.00
Total Indirect Costs				138,000.00

Cognizant Federal Agency	
(Agency Name, POC Name, and POC Phone Number)	DHHS, Arif Karim, 214-767-3261

I. Total Direct and Indirect Costs	Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G + H)	368,000.00

J. Fee	Funds Requested (\$)

K. Total Costs and Fee	Funds Requested (\$)
Total Costs and Fee (I + J)	368,000.00

L. Budget Justification			
(Only attach one file.)	BCM Budget Justification 10.21.22.pdf	Add Attachment	Delete Attachment View Attachment

RESEARCH & RELATED BUDGET - Budget Period 4

OMB Number: 4040-0001
Expiration Date: 12/31/2022

UEI: FXXKMA43NTV21

Enter name of Organization: Baylor College of Medicine

Budget Type: ☐ Project ☒ Subaward/Consortium

Budget Period: 4 Start Date: 07/01/2026 End Date: 06/30/2027

A. Senior/Key Person

Prefix	First	Middle	Last	Suffix	Base Salary (\$)	Months			Requested Salary (\$)	Fringe Benefits (\$)	Funds Requested (\$)
						Cal.	Acad.	Sum.			
	Xiang		Zhang	PhD	203,700.00	1.20			20,370.00	5,307.00	25,677.00

Project Role: PD/PI

	Tao		Wu	PhD	128,750.00	0.60			6,438.00	1,635.00	8,073.00
--	-----	--	----	-----	------------	------	--	--	----------	----------	----------

Project Role: Co-Investigator

Additional Senior Key Persons: Total Funds requested for all Senior Key Persons in the attached file

Total Senior/Key Person

B. Other Personnel

Number of Personnel	Project Role	Cal.	Months		Requested Salary (\$)	Fringe Benefits (\$)	Funds Requested (\$)
			Acad.	Sum.			
2	Post Doctoral Associates	18.00			87,831.00	24,486.00	112,317.00
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						

Total Number Other Personnel

Total Other Personnel

Total Salary, Wages and Fringe Benefits (A+B)

C. Equipment Description

List items and dollar amount for each item exceeding \$5,000

Equipment item	Funds Requested (\$)
Additional Equipment:	
Add AttachmentDelete AttachmentView Attachment	
Total funds requested for all equipment listed in the attached file	
Total Equipment	

D. Travel

	Funds Requested (\$)
1. Domestic Travel Costs (Incl. Canada, Mexico and U.S. Possessions)	3,000.00
2. Foreign Travel Costs	
Total Travel Cost	3,000.00

E. Participant/Trainee Support Costs

	Funds Requested (\$)
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other	
Number of Participants/Trainees	Total Participant/Trainee Support Costs

F. Other Direct Costs		Funds Requested (\$)
1. Materials and Supplies		54,233.00
2. Publication Costs		
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Animal Maintenance		7,700.00
9. Laboratory Analysis/Core Charges		19,000.00
10.		
11.		
12.		
13.		
14.		
15.		
16.		
17.		
Total Other Direct Costs		80,933.00

G. Direct Costs	Funds Requested (\$)
Total Direct Costs (A thru F)	230,000.00

H. Indirect Costs				Funds Requested (\$)
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)		
MTDC	60.00	230,000.00		138,000.00
Total Indirect Costs				138,000.00

Cognizant Federal Agency	
(Agency Name, POC Name, and POC Phone Number)	DHHS, Arif Karim, 214-767-3261

I. Total Direct and Indirect Costs	Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G + H)	368,000.00

J. Fee	Funds Requested (\$)

K. Total Costs and Fee	Funds Requested (\$)
Total Costs and Fee (I + J)	368,000.00

L. Budget Justification			
(Only attach one file.)	BCM Budget Justification 10.21.22.pdf	Add Attachment	Delete Attachment View Attachment

RESEARCH & RELATED BUDGET - Budget Period 5

OMB Number: 4040-0001
Expiration Date: 12/31/2022

UEI: FXXKMA43NTV21

Enter name of Organization: Baylor College of Medicine

Budget Type: ☐ Project ☒ Subaward/Consortium

Budget Period: 5 Start Date: 07/01/2027 End Date: 06/30/2028

A. Senior/Key Person

Prefix	First	Middle	Last	Suffix	Base Salary (\$)	Months			Requested Salary (\$)	Fringe Benefits (\$)	Funds Requested (\$)
						Cal.	Acad.	Sum.			
	Xiang		Zhang	PhD	203,700.00	1.20			20,370.00	5,307.00	25,677.00

Project Role: PD/PI

	Tao		Wu	PhD	128,750.00	0.60			6,438.00	1,635.00	8,073.00
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Project Role: Co-Investigator

Additional Senior Key Persons: Total Funds requested for all Senior Key Persons in the attached file

Total Senior/Key Person

B. Other Personnel

Number of Personnel	Project Role	Cal.	Months		Requested Salary (\$)	Fringe Benefits (\$)	Funds Requested (\$)
			Acad.	Sum.			
2	Post Doctoral Associates	18.00			87,831.00	24,486.00	112,317.00
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						

Total Number Other Personnel

Total Other Personnel

Total Salary, Wages and Fringe Benefits (A+B)

C. Equipment Description

List items and dollar amount for each item exceeding \$5,000

Equipment item	Funds Requested (\$)
Additional Equipment:	Add Attachment Delete Attachment View Attachment
Total funds requested for all equipment listed in the attached file	
Total Equipment	

D. Travel

	Funds Requested (\$)
1. Domestic Travel Costs (Incl. Canada, Mexico and U.S. Possessions)	3,000.00
2. Foreign Travel Costs	
Total Travel Cost	3,000.00

E. Participant/Trainee Support Costs

	Funds Requested (\$)
1. Tuition/Fees/Health Insurance	
2. Stipends	
3. Travel	
4. Subsistence	
5. Other	
Number of Participants/Trainees	Total Participant/Trainee Support Costs

F. Other Direct Costs		Funds Requested (\$)
1. Materials and Supplies		54,233.00
2. Publication Costs		
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Animal Maintenance		7,700.00
9. Laboratory Analysis/Core Charges		19,000.00
10.		
11.		
12.		
13.		
14.		
15.		
16.		
17.		
Total Other Direct Costs		80,933.00

G. Direct Costs	Funds Requested (\$)
Total Direct Costs (A thru F)	230,000.00

H. Indirect Costs				Funds Requested (\$)
Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base (\$)		
MTDC	60.00	230,000.00		138,000.00
Total Indirect Costs				138,000.00

Cognizant Federal Agency	
(Agency Name, POC Name, and POC Phone Number)	DHHS, Arif Karim, 214-767-3261

I. Total Direct and Indirect Costs	Funds Requested (\$)
Total Direct and Indirect Institutional Costs (G + H)	368,000.00

J. Fee	Funds Requested (\$)

K. Total Costs and Fee	Funds Requested (\$)
Total Costs and Fee (I + J)	368,000.00

L. Budget Justification			
(Only attach one file.)	BCM Budget Justification 10.21.22.pdf	Add Attachment	Delete Attachment View Attachment

RESEARCH & RELATED BUDGET - Cumulative Budget

		Totals (\$)
Section A, Senior/Key Person		168,750.00
Section B, Other Personnel		561,585.00
Total Number Other Personnel	10	
Total Salary, Wages and Fringe Benefits (A+B)		730,335.00
Section C, Equipment		
Section D, Travel		15,000.00
1. Domestic	15,000.00	
2. Foreign		
Section E, Participant/Trainee Support Costs		
1. Tuition/Fees/Health Insurance		
2. Stipends		
3. Travel		
4. Subsistence		
5. Other		
6. Number of Participants/Trainees		
Section F, Other Direct Costs		404,665.00
1. Materials and Supplies	271,165.00	
2. Publication Costs		
3. Consultant Services		
4. ADP/Computer Services		
5. Subawards/Consortium/Contractual Costs		
6. Equipment or Facility Rental/User Fees		
7. Alterations and Renovations		
8. Other 1	38,500.00	
9. Other 2	95,000.00	
10. Other 3		
11. Other 4		
12. Other 5		
13. Other 6		
14. Other 7		
15. Other 8		
16. Other 9		
17. Other 10		

Section G, Direct Costs (A thru F)

1,150,000.00

Section H, Indirect Costs

690,000.00

Section I, Total Direct and Indirect Costs (G + H)

1,840,000.00

Section J, Fee

Section K, Total Costs and Fee (I + J)

1,840,000.00

Budget Justification

Baylor College of Medicine

Key Personnel

Xiang Zhang, Ph.D., 10% (1.2 cal mos) is the MPI of this project. He will oversee this project and provide scientific leadership to the research team. Dr. Zhang has been working in the field of cancer metastasis for over 13 years, and published over 50 papers in first-rate journals. He will work with graduate student and research staff on experimental design, analysis and manuscript preparation. He will also teach the incoming postdoctoral fellow experimental skill to fulfill the proposed experiments.

Tao Wu, Ph.D., 5% (0.6 cal mos) will be a Co-Investigator. Dr. Wu's research focuses on how the epigenetic factors dynamically regulate the cell fate decision and the response to external stimulation, specifically underlying cancer therapeutic resistance, which well suits this project. Dr. Wu is currently a tenure-track assistant professor at BCM. His lab has established a full pipeline for epigenomic profiling and analysis.

Other Personnel

Xiaoxin Hao, Ph.D., 100% (12 cal mos), is a Postdoctoral Associate in Zhang Lab. His previous Ph.D work focuses on hematopoiesis. During his training in Zhang Lab, he has acquired strong skills for bone metastasis research. He also has superb skills to conduct most molecular and cellular biology.

Zhan Xu, Ph.D., 50% (6 cal mos) is a Postdoctoral Associate in Zhang Lab. He received his Ph.D training in the field of bone biology in this project. He has all the required skills for implementation of bone-related experiments (e.g., Aim 3.2). In addition, he is also well trained for bone metastasis assays.

Other Significant Contributor

C. Kent Osborne, Ph.D., 0% (0 cal mos) will serve as Other Significant Contributor on the project. Dr. Osborn is the funding director of Dan L. Duncan Comprehensive Cancer Center, Dudley and Tina Sharp Chair for Cancer Research, and the Professor of Medicine at Baylor College of Medicine. He is a clinical oncologist with over 40 years of experience and a research focus on breast cancer. Dr. Osborne will provide advice for future translation of the proposed studies as needed and there is no measurable effort for his contribution.

Qian Zhu, Ph.D., 0% (0 cal mos) will serve as an Other Significant Contributor on the project. Dr. Zhu is a leading computational biologist specialized in integrated single cell omics and spatial transcriptomics. His previous works established new methodologies for discovery of spatial features of gene/pathway expression. He will help perform analysis described in Aim 3.1. Since the dataset will be obtained as a part of another project and the proposed analyses can be done using the same pipeline, he is not requesting salary coverage from this application.

Fringe Benefits

The fringe benefits are specifically identified to each employee and are charged individually as direct cost.

Other Direct Costs

Materials and Supplies: The amount requested in each of the following categories has been derived by cost accounting using actual expenditures (from purchase orders) over the past year on a project involving similar quantity of experiments and personnel.

Lab supplies (glassware, rubber items, plastic, paper supplies) for all routine molecular and cellular experiments: \$5,000.

Antibodies for immunofluorescence staining, western blot, and flow cytometry: \$8,000.

Chemicals including buffers, ingredients of buffers, culture mediums, fluorescence dyes, reagents for molecular experiments: \$4,233.

Kits for molecular experiments including but not limited to DNA/RNA/Protein purification, RT-PCR, and DNA transduction: \$5,000.

Animal Purchase: We request funds to cover the purchase cost of 400 animals per year (see Research Strategy), including NSG, athymic nude and C57/B6 mice. The average cost of one mouse is about \$80

based on the actual expenditure in the past year). Therefore, the total cost for each year is expected to be \$32,000.

Travel

Funds are requested for travel of both the PI and postdoctoral associate to attend one national meeting per year (e.g., San Antonio Breast Cancer Symposium): \$3,000 each year.

Animal Maintenance cost at BCM is 0.77/cage/day. For the 400 purchased mice each year, based on the actual records of the past year and the expected total number of mice, we expect to maintain 100 cages with each cage for 100 days for each year. Based on these numbers, the cost of animal per diem maintenance will be \$7,700 each year.

Laboratory Analysis/Core Charges: Based on the actual expenditure of the past year, funds will cover the cost of service from the following:

Pathology Core at BCM for regular paraffin-embedding and microtome sectioning: \$1,000

Integrated Microscopy Core: \$1,000

Flow Cytometry Core: \$5,000

Next generation sequencing Core: \$12,000

Subtotal: \$19,000 each year.

Indirect Costs

BCM's facilities and administrative cost rates are established in accordance with the cost principles outlined in OMB PART 200—UNIFORM ADMINISTRATIVE REQUIREMENTS, COST PRINCIPLES, AND AUDIT REQUIREMENTS FOR FEDERAL AWARDS and agreed upon in negotiation with the federal government. The Indirect Cost Rate at Baylor College of Medicine is 60% from July 1, 2022 to June 30, 2026 for On-Campus Research.

COLLEGES AND UNIVERSITIES RATE AGREEMENT

EIN: 1741613878A1

DATE:05/25/2022

ORGANIZATION:

Baylor College of Medicine

One Baylor Plaza

Room T119, MSBCM200

Houston, TX 77030-3498

FILING REF.: The preceding
agreement was dated
12/13/2018

The rates approved in this agreement are for use on grants, contracts and other agreements with the Federal Government, subject to the conditions in Section III.

SECTION I: Facilities And Administrative Cost Rates

RATE TYPES: FIXED FINAL PROV. (PROVISIONAL) PRED. (PREDETERMINED)

EFFECTIVE PERIOD

<u>TYPE</u>	<u>FROM</u>	<u>TO</u>	<u>RATE(%)</u>	<u>LOCATION</u>	<u>APPLICABLE TO</u>
PRED.	07/01/2019	06/30/2022	60.00	On Campus	Organized Research
PRED.	07/01/2018	06/30/2022	35.00	On Campus	Instruction
PRED.	07/01/2018	06/30/2022	33.00	On Campus	Other Sponsored Activities
PRED.	07/01/2018	06/30/2022	43.80	On Campus	(1) Children's Nutrition Research Center
PRED.	07/01/2018	06/30/2022	26.00	Off Campus	All Programs
PRED.	07/01/2022	06/30/2026	60.00	On Campus	Organized Research
PRED.	07/01/2022	06/30/2026	35.00	On Campus	Instruction
PRED.	07/01/2022	06/30/2026	33.00	On Campus	Other Sponsored Activities
PRED.	07/01/2022	06/30/2026	43.80	On Campus	(1) Children's Nutrition Research Center
PRED.	07/01/2022	06/30/2026	26.00	Off Campus	All Programs

ORGANIZATION: Baylor College of Medicine

AGREEMENT DATE: 5/25/2022

<u>TYPE</u>	<u>FROM</u>	<u>TO</u>	<u>RATE(%)</u>	<u>LOCATION</u>	<u>APPLICABLE TO</u>
PROV.	07/01/2026	Until Amended			Use same rates and conditions as those cited for fiscal year ending June 30, 2026.

(1) This rate is applicable to the Children's Nutrition Research Center (CNRC) and is applied to the MTDC of the Grants performed at the CNRC other than the Center Grant funded by the Department of Agriculture.

*BASE

Modified total direct costs, consisting of all direct salaries and wages, applicable fringe benefits, materials and supplies, services, travel and up to the first \$25,000 of each subaward (regardless of the period of performance of the subawards under the award). Modified total direct costs shall exclude equipment, capital expenditures, charges for patient care, rental costs, tuition remission, scholarships and fellowships, participant support costs and the portion of each subaward in excess of \$25,000. Other items may only be excluded when necessary to avoid a serious inequity in the distribution of indirect costs, and with the approval of the cognizant agency for indirect costs.

ORGANIZATION: Baylor College of Medicine

AGREEMENT DATE: 5/25/2022

SECTION II: SPECIAL REMARKS

TREATMENT OF FRINGE BENEFITS:

The fringe benefits are specifically identified to each employee and are charged individually as direct costs. The directly claimed fringe benefits are listed below.

TREATMENT OF PAID ABSENCES

Vacation, holiday, sick leave pay and other paid absences are included in salaries and wages and are claimed on grants, contracts and other agreements as part of the normal cost for salaries and wages. Separate claims are not made for the cost of these paid absences.

OFF CAMPUS DEFINITION:

For all activities performed in facilities not owned by the institution and to which rent is directly allocated to the project(s), the off-campus rate will apply. Actual costs will be apportioned between on-campus and off-campus components. Each portion will bear the appropriate rate.

FRINGE BENEFITS:

FICA	Disability Insurance
Retirement	Termination Vacation Pay
Health Insurance	Unemployment Insurance
Dental Insurance	Partial Parking or Transportation Subsidy
Life Insurance	

The Baylor College of Medicine receives awards which are performed by university employees in the following Hospitals: St. Luke's, Methodist, Harris County, Texas Children's and The Institute for Rehabilitation and Research. The university may combine its off-campus rate with the hospital's affiliate rate for the purpose of seeking reimbursement from these awards. The affiliate rates are listed in the respective hospital negotiation rate agreements.

Per 2 CFR 200.414(g) - A rate extension has been granted.

The next indirect cost proposal based on actual costs for the fiscal year ending 06/30/2025, is due by 12/31/2025.

EQUIPMENT DEFINITION:

Equipment means tangible personal property (including information technology systems) having a useful life of more than one year and a per-unit acquisition cost which equals or exceeds \$5,000.

ORGANIZATION: Baylor College of Medicine

AGREEMENT DATE: 5/25/2022

SECTION III: GENERAL

A. LIMITATIONS:

The rates in this Agreement are subject to any statutory or administrative limitations and apply to a given grant, contract or other agreement only to the extent that funds are available. Acceptance of the rates is subject to the following conditions: (1) Only costs incurred by the organization were included in its facilities and administrative cost pools as finally accepted; such costs are legal obligations of the organization and are allowable under the governing cost principles; (2) The same costs that have been treated as facilities and administrative costs are not claimed as direct costs; (3) Similar types of costs have been accorded consistent accounting treatment; and (4) The information provided by the organization which was used to establish the rates is not later found to be materially incomplete or inaccurate by the Federal Government. In such situations the rate(s) would be subject to renegotiation at the discretion of the Federal Government.

B. ACCOUNTING CHANGES:

This Agreement is based on the accounting system purported by the organization to be in effect during the Agreement period. Changes to the method of accounting for costs which affect the amount of reimbursement resulting from the use of this Agreement require prior approval of the authorized representative of the cognizant agency. Such changes include, but are not limited to, changes in the charging of a particular type of cost from facilities and administrative to direct. Failure to obtain approval may result in cost disallowances.

C. FIXED RATES:

If a fixed rate is in this Agreement, it is based on an estimate of the costs for the period covered by the rate. When the actual costs for this period are determined, an adjustment will be made to a rate of a future year(s) to compensate for the difference between the costs used to establish the fixed rate and actual costs.

D. USE BY OTHER FEDERAL AGENCIES:

The rates in this Agreement were approved in accordance with the authority in Title 2 of the Code of Federal Regulations, Part 200 (2 CFR 200), and should be applied to grants, contracts and other agreements covered by 2 CFR 200, subject to any limitations in A above. The organization may provide copies of the Agreement to other Federal Agencies to give them early notification of the Agreement.

E. OTHER:

If any Federal contract, grant or other agreement is reimbursing facilities and administrative costs by a means other than the approved rate(s) in this Agreement, the organization should (1) credit such costs to the affected programs, and (2) apply the approved rate(s) to the appropriate base to identify the proper amount of facilities and administrative costs allocable to these programs.

BY THE INSTITUTION:

Baylor College of Medicine

(INSTITUTION)



(SIGNATURE)

Jamie Bailey

(NAME)

Chief Financial Officer

(TITLE)

6/2/22

(DATE)

ON BEHALF OF THE FEDERAL GOVERNMENT:

DEPARTMENT OF HEALTH AND HUMAN SERVICES

(AGENCY)

Arif M. Karim - S^s Digitally signed by Arif M. Karim -
Date: 2022.05.27 11:46:11 -05'00'

(SIGNATURE)

Arif Karim

(NAME)

Director, Cost Allocation Services

(TITLE)

5/25/2022

(DATE) 1523

HHS REPRESENTATIVE: Denise Shirlee

Telephone: (214) 767-3261

Department of Health and Human Services

Part 1. Overview Information

Participating Organization(s)

National Institutes of Health ([NIH \(http://www.nih.gov\)](http://www.nih.gov))

Components of Participating Organizations

NATIONAL INSTITUTES OF HEALTH ([NIH \(\)](#))

National Eye Institute ([NEI \(https://www.nei.nih.gov/\)](https://www.nei.nih.gov/))

National Heart, Lung, and Blood Institute ([NHLBI \(https://www.nhlbi.nih.gov/\)](https://www.nhlbi.nih.gov/))

National Human Genome Research Institute ([NHGRI \(https://www.genome.gov/\)](https://www.genome.gov/))

National Institute on Aging ([NIA \(https://www.nia.nih.gov/\)](https://www.nia.nih.gov/))

National Institute on Alcohol Abuse and Alcoholism ([NIAAA \(https://www.niaaa.nih.gov/\)](https://www.niaaa.nih.gov/))

National Institute of Allergy and Infectious Diseases ([NIAID \(https://www.niaid.nih.gov/\)](https://www.niaid.nih.gov/))

National Institute of Arthritis and Musculoskeletal and Skin Diseases ([NIAMS \(https://www.niams.nih.gov/\)](https://www.niams.nih.gov/))

National Institute of Biomedical Imaging and Bioengineering ([NIBIB \(https://www.nibib.nih.gov/\)](https://www.nibib.nih.gov/))

Eunice Kennedy Shriver National Institute of Child Health and Human Development ([NICHD \(https://www.nichd.nih.gov/\)](https://www.nichd.nih.gov/))

National Institute on Deafness and Other Communication Disorders ([NIDCD \(https://www.nidcd.nih.gov/\)](https://www.nidcd.nih.gov/))

National Institute of Dental and Craniofacial Research ([NIDCR \(https://www.nidcr.nih.gov/\)](https://www.nidcr.nih.gov/))

National Institute of Diabetes and Digestive and Kidney Diseases ([NIDDK \(https://www.niddk.nih.gov/\)](https://www.niddk.nih.gov/))

National Institute on Drug Abuse ([NIDA \(https://www.drugabuse.gov/\)](https://www.drugabuse.gov/))

National Institute of Environmental Health Sciences ([NIEHS \(https://www.niehs.nih.gov/\)](https://www.niehs.nih.gov/))

National Institute of General Medical Sciences ([NIGMS \(https://www.nigms.nih.gov/\)](https://www.nigms.nih.gov/))

National Institute of Mental Health ([NIMH \(https://www.nimh.nih.gov/index.shtml\)](https://www.nimh.nih.gov/index.shtml))

National Institute of Neurological Disorders and Stroke ([NINDS \(https://www.ninds.nih.gov/\)](https://www.ninds.nih.gov/))

National Institute of Nursing Research ([NINR \(https://www.ninr.nih.gov/\)](https://www.ninr.nih.gov/))

National Institute on Minority Health and Health Disparities ([NIMHD \(https://www.nimhd.nih.gov/\)](https://www.nimhd.nih.gov/))

National Library of Medicine ([NLM \(https://www.nlm.nih.gov/\)](https://www.nlm.nih.gov/))

National Center for Complementary and Integrative Health ([NCCIH \(https://nccih.nih.gov/\)](https://nccih.nih.gov/))

Division of Program Coordination, Planning and Strategic Initiatives, Office of Research Infrastructure Programs ([ORIP \(https://orip.nih.gov/\)](https://orip.nih.gov/))

National Cancer Institute ([NCI \(https://www.cancer.gov/\)](https://www.cancer.gov/))

Note: Not all NIH Institutes and Centers (ICs) participate in Parent Announcements. Applicants should carefully note which ICs participate in this announcement and view their respective areas of research interest at the R01 IC-Specific Scientific Interests and Contact website. ICs that do not participate in this announcement will not consider applications for funding.

Funding Opportunity Title

NIH Research Project Grant (Parent R01 Clinical Trial Not Allowed)

Activity Code

[R01 \(//grants.nih.gov/grants/funding/ac_search_results.htm?text_curr=r01&Search.x=0&Search.y=0&Search_Type=Activity\)](https://grants.nih.gov/grants/funding/ac_search_results.htm?text_curr=r01&Search.x=0&Search.y=0&Search_Type=Activity) Research Project Grant

Announcement Type

Reissue of [PA-19-056 \(https://grants.nih.gov/grants/guide/pa-files/pa-19-056.html\)](https://grants.nih.gov/grants/guide/pa-files/pa-19-056.html)

Related Notices

See [Notices of Special Interest \(https://grants.nih.gov/grants/guide/NOSIs_targetingList.cfm?GuideDocID=33280\)](https://grants.nih.gov/grants/guide/NOSIs_targetingList.cfm?GuideDocID=33280) associated with this funding opportunity

See [Notices of Special Interest \(//grants.nih.gov/grants/guide/notice-files/NOT-HL-22-006.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-HL-22-006.html) related to this funding opportunity.

February 16, 2022 - Notice of NIDA Participation in NOT-MH-21-105, "Notice of Special Interest: Advancing Health Communication Research on HIV Prevention, Treatment and Cure". See Notice [NOT-DA-21-079 \(//grants.nih.gov/grants/guide/notice-files/NOT-DA-21-079.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-DA-21-079.html)

December 28, 2021 - Notice of Special Interest: Promoting Research on COVID-19 and Rheumatic, Musculoskeletal and Skin Diseases). See Notice [NOT-AR-22-012 \(//grants.nih.gov/grants/guide/notice-files/NOT-AR-22-012.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-AR-22-012.html)

December 28, 2021 - Notice of Special Interest (NOSI): Advancing Innovative Non-invasive Physiologic Monitoring To Improve Fetal Assessment. See Notice [NOT-HD-21-053 \(//grants.nih.gov/grants/guide/notice-files/NOT-HD-21-053.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-HD-21-053.html)

November 23, 2021 - Notice of Special Interest (NOSI): Basic and Translational Oral Health Research Related to HIV/AIDS. See Notice [NOT-DE-21-019 \(//grants.nih.gov/grants/guide/notice-files/NOT-DE-21-019.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-DE-21-019.html)

October 28, 2021 - Reminder: FORMS-G Grant Application Forms & Instructions Must be Used for Due Dates On or After January 25, 2022 - New Grant Application Instructions Now Available. See Notice [NOT-OD-22-018 \(https://grants.nih.gov/grants/guide/notice-files/NOT-OD-22-018.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-OD-22-018.html).

September 13, 2021 - Updates to the Non-Discrimination Legal Requirements for NIH Recipients. See Notice [NOT-OD-21-181 \(https://grants.nih.gov/grants/guide/notice-files/NOT-OD-21-181.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-OD-21-181.html).

August 5, 2021 - New NIH "FORMS-G" Grant Application Forms and Instructions Coming for Due Dates on or after January 25, 2022. See Notice [NOT-OD-21-169 \(https://grants.nih.gov/grants/guide/notice-files/NOT-OD-21-169.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-OD-21-169.html).

August 5, 2021 - Update: Notification of Upcoming Change in Federal-wide Unique Entity Identifier Requirements. See Notice [NOT-OD-21-170 \(https://grants.nih.gov/grants/guide/notice-files/NOT-OD-21-170.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-OD-21-170.html)

April 20, 2021 - Expanding Requirement for eRA Commons IDs to All Senior/Key Personnel. See Notice [NOT-OD-21-109 \(https://grants.nih.gov/grants/guide/notice-files/NOT-OD-21-109.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-OD-21-109.html)

See [Notices of Special Interest \(//grants.nih.gov/grants/guide/notice-files/NOT-MH-21-330.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-MH-21-330.html) associated with this funding opportunity.

- **October 26, 2021** - Notice of Special Interest (NOSI): IDEA2Health: Innovative Data Evaluation and Analysis to Health. See Notice [NOT-HL-22-001 \(//grants.nih.gov/grants/guide/notice-files/NOT-HL-22-001.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-HL-22-001.html)
- **October 26, 2021** - Notice of NHLBI Participation in NOT-AT-21-002 "Notice of Special Interest (NOSI): Promoting Research on Interoception and Its Impact on Health and Disease (R01). See Notice [NOT-HL-21-037 \(//grants.nih.gov/grants/guide/notice-files/NOT-HL-21-037.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-HL-21-037.html)
- **October 18, 2021** - Notice of Special Interest (NOSI): Somatic Cell Gene Editing Therapies To Improve Transplantation Outcomes. See Notice [NOT-AI-21-080 \(//grants.nih.gov/grants/guide/notice-files/NOT-AI-21-080.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-AI-21-080.html)
- **August 23, 2021** - NOSI: Maximizing the Scientific Value of Secondary Analyses of Existing Cohorts and Datasets in Order to Address Research Gaps and Foster Additional Opportunities in Aging Research. See Notice [NOT-AG-21-020 \(//grants.nih.gov/grants/guide/notice-files/NOT-AG-21-020.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-AG-21-020.html).
- **July 13, 2021** - NOSI: Academy of Finland (AKA) –Notice of NHLBI Participation in NOT-HD-21-027. See Notice [NOT-HL-21-021 \(https://grants.nih.gov/grants/guide/notice-files/NOT-HL-21-021.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-HL-21-021.html).
- **November 13, 2020** - NOSI: Academy of Finland (AKA) – National Institutes of Health (NIH) Partnership Program. See Notice [NOT-OD-21-021 \(//grants/guide/notice-files/NOT-OD-21-021.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-OD-21-021.html).
- **October 5, 2020** - Consolidated Notice on NIMH Clinical Trials Policies. See Notice [NOT-MH-20-105 \(//grants.nih.gov/grants/guide/notice-files/NOT-MH-20-105.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-MH-20-105.html).

- **September 24, 2020** - Notice of Change to NIEHS target FOAs for NOT-HL-20-788. See Notice [NOT-HL-20-015 \(/grants/guide/notice-files/NOT-HL-20-015.html\)](#).
- **August 26, 2020** - Notice of Correction to Eligibility in NIH Funding Opportunity Announcements. See Notice [NOT-OD-20-171 \(/grants/guide/notice-files/NOT-OD-20-171.html\)](#).
- **August 26, 2020** - Notice of NIGMS Participation in NOT-ES-20-018. See Notice [NOT-GM-20-046 \(/grants/guide/notice-files/NOT-GM-20-046.html\)](#).

Funding Opportunity Announcement (FOA) Number

PA-20-185

Companion Funding Opportunity

[PA-20-183 \(/grants.nih.gov/grants/guide/pa-files/PA-20-183.html\)](#) - NIH Research Project Grant (Parent R01 Clinical Trial Required)

[PA-20-184 \(/grants.nih.gov/grants/guide/pa-files/PA-20-184.html\)](#) Research Project Grant (Parent R01 Basic Experimental Studies with Humans Required)

Number of Applications

See [Section III. 3. Additional Information on Eligibility](#).

Catalog of Federal Domestic Assistance (CFDA) Number(s)

93.213; 93.393, 93.394, 93.395, 93.396, 93.399; 93.172; 93.837, 93.838, 93.839, 93.840, 93.233; 93.866; 93.273; 93.855; 93.173; 93.847; 93.113; 93.853; 93.879; 93.351; 93.867; 93.846; 93.286; 93.865; 93.279; 93.859; 93.242; 93.121; 93.307; 93.361

Funding Opportunity Purpose

The NIH Research Project Grant supports a discrete, specified, circumscribed project in areas representing the specific interests and competencies of the investigator(s). The proposed project must be related to the programmatic interests of one or more of the participating NIH Institutes and Centers (ICs) based on their scientific missions.

This Funding Opportunity Announcement does not accept applications proposing clinical trial(s).

Key Dates

Posted Date

May 05, 2020

Open Date (Earliest Submission Date)

May 05, 2020

Letter of Intent Due Date(s)

Not applicable

Application Due Date(s)

[Standard dates \(/grants.nih.gov/grants/guide/url_redirect.htm?id=11111\)](#) apply.

The first application due date for this FOA is June 5, 2020.

All applications are due by 5:00 PM local time of applicant organization. All [types of non-AIDS applications](#) allowed for this funding opportunity announcement are due on the listed date(s).

Applicants are encouraged to apply early to allow adequate time to make any corrections to errors found in the application during the submission process by the due date.

AIDS Application Due Date(s)

[Standard AIDS dates \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11112\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11112) apply.

All applications are due by 5:00 PM local time of applicant organization. All [types of AIDS and AIDS-related applications](#) allowed for this funding opportunity announcement are due on the listed date(s).

The first AIDS application due date for this FOA is September 7, 2020.

Applicants are encouraged to apply early to allow adequate time to make any corrections to errors found in the application during the submission process by the due date.

Scientific Merit Review

[Standard dates \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11113\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11113) apply

Advisory Council Review

[Standard dates \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11113\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11113) apply

Earliest Start Date

[Standard dates \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11113\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11113) apply

Expiration Date

May 08, 2023

Due Dates for E.O. 12372

Not Applicable

Required Application Instructions

It is critical that applicants follow the instructions in the Research (R) Instructions in the [SF424 \(R&R\) Application Guide \(//grants.nih.gov/grants/guide/url_redirect.htm?id=12000\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=12000), except where instructed to do otherwise (in this FOA or in a Notice from [NIH Guide for Grants and Contracts \(//grants.nih.gov/grants/guide/\)](#)).

Conformance to all requirements (both in the Application Guide and the FOA) is required and strictly enforced. Applicants must read and follow all application instructions in the Application Guide as well as any program-specific instructions noted in [Section IV](#). When the program-specific instructions deviate from those in the Application Guide, follow the program-specific instructions.

Applications that do not comply with these instructions may be delayed or not accepted for review.

There are several options available to submit your application through Grants.gov to NIH and Department of Health and Human Services partners. You **must** use one of these submission options to access the application forms for this opportunity.

1. Use the NIH ASSIST system to prepare, submit and track your application online.

Apply Online Using ASSIST

2. Use an institutional system-to-system (S2S) solution to prepare and submit your application to Grants.gov and [eRA Commons \(http://public.era.nih.gov/commons/\)](http://public.era.nih.gov/commons/) to track your application. Check with your institutional officials regarding availability.
3. Use [Grants.gov \(https://www.grants.gov/web/grants/applicants/download-application-package.html#search=true&oppNum=PA-20-185\)](https://www.grants.gov/web/grants/applicants/download-application-package.html#search=true&oppNum=PA-20-185) Workspace to prepare and submit your application and [eRA Commons \(http://public.era.nih.gov/commons/\)](http://public.era.nih.gov/commons/) to track your application.

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Part 2. Full Text of Announcement

Section I. Funding Opportunity Description

The NIH Research Project Grant supports a discrete, specified, circumscribed project in scientific areas that represent the investigators' specific interests and competencies and that fall within the mission of the participating NIH Institutes and Centers (ICs). The R01 is the original, and historically the oldest, grant mechanism used by the NIH to support health-related research and development.

Research grant applications are assigned to participating ICs based on receipt and referral guidelines and applications may be assigned to multiple participating ICs with related research interests. Applicants are encouraged to identify a participating IC that supports their area of research via the [R01 IC-Specific Scientific Interests and Contact \(https://grants.nih.gov/grants/guide/contacts/parent-R01-CT-NotAllowed.html\)](https://grants.nih.gov/grants/guide/contacts/parent-R01-CT-NotAllowed.html) website and contact Scientific/Research staff from relevant ICs to inquire about their interest in supporting the proposed research project.

This Funding Opportunity Announcement does not accept applications proposing clinical trial(s).

For specific information about the mission of each NIH IC, visit the [List of NIH Institutes, Centers, and Offices \(http://www.nih.gov/institutes-nih/list-nih-institutes-centers-offices\)](http://www.nih.gov/institutes-nih/list-nih-institutes-centers-offices) website.?

See [Section VIII. Other Information](#) for award authorities and regulations.

Section II. Award Information

Funding Instrument

Grant: A support mechanism providing money, property, or both to an eligible entity to carry out an approved project or activity.

Application Types Allowed

Revision

New

Renewal

Resubmission

The [OER Glossary \(https://grants.nih.gov/grants/guide/url_redirect.htm?id=11116\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11116) and the SF424 (R&R) Application Guide provide details on these application types. Only those application types listed here are allowed for this FOA.

Clinical Trial?

Not Allowed: Only accepting applications that do not propose clinical trials

[Need help determining whether you are doing a clinical trial? \(https://grants.nih.gov/grants/guide/url_redirect.htm?id=82370\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=82370)

Funds Available and Anticipated Number of Awards

The number of awards is contingent upon NIH appropriations and the submission of a sufficient number of meritorious applications

Award Budget

Application budgets are not limited but need to reflect the actual needs of the proposed project.

Award Project Period

The scope of the proposed project should determine the project period. The maximum project period is 5 years.

NIH grants policies as described in the [NIH Grants Policy Statement \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11120\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11120) will apply to the applications submitted and awards made from this FOA.

Section III. Eligibility Information

1. Eligible Applicants

Eligible Organizations

Higher Education Institutions

- Public/State Controlled Institutions of Higher Education
- Private Institutions of Higher Education

The following types of Higher Education Institutions are always encouraged to apply for NIH support as Public or Private Institutions of Higher Education:

- Hispanic-serving Institutions
- Historically Black Colleges and Universities (HBCUs)
- Tribally Controlled Colleges and Universities (TCCUs)
- Alaska Native and Native Hawaiian Serving Institutions
- Asian American Native American Pacific Islander Serving Institutions (AANAPISIs)

Nonprofits Other Than Institutions of Higher Education

- Nonprofits with 501(c)(3) IRS Status (Other than Institutions of Higher Education)
- Nonprofits without 501(c)(3) IRS Status (Other than Institutions of Higher Education)

For-Profit Organizations

- Small Businesses
- For-Profit Organizations (Other than Small Businesses)

Local Governments

- State Governments
- County Governments
- City or Township Governments
- Special District Governments
- Indian/Native American Tribal Governments (Federally Recognized)
- Indian/Native American Tribal Governments (Other than Federally Recognized)

Federal Governments

- Eligible Agencies of the Federal Government
- U.S. Territory or Possession

Other

- Independent School Districts
- Public Housing Authorities/Indian Housing Authorities
- Native American Tribal Organizations (other than Federally recognized tribal governments)
- Faith-based or Community-based Organizations
- Regional Organizations
- Non-domestic (non-U.S.) Entities (Foreign Institutions)

Foreign Institutions

Non-domestic (non-U.S.) Entities (Foreign Institutions) **are** eligible to apply.

Non-domestic (non-U.S.) components of U.S. Organizations **are** eligible to apply.

Foreign components, as [defined in the NIH Grants Policy Statement \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11118\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11118), are allowed.

Required Registrations

Applicant organizations

Applicant organizations must complete and maintain the following registrations as described in the SF 424 (R&R) Application Guide to be eligible to apply for or receive an award. All registrations must be completed prior to the application being submitted. Registration can take 6 weeks or more, so applicants should begin the registration process as soon as possible. The [NIH Policy on Late Submission of Grant Applications \(//grants.nih.gov/grants/guide/notice-files/NOT-OD-15-039.html\)](https://grants.nih.gov/grants/guide/notice-files/NOT-OD-15-039.html) states that failure to complete registrations in advance of a due date is not a valid reason for a late submission.

- [Dun and Bradstreet Universal Numbering System \(DUNS\) \(http://fedgov.dnb.com/webform\)](http://fedgov.dnb.com/webform) - All registrations require that applicants be issued a DUNS number. After obtaining a DUNS number, applicants can begin both SAM and eRA Commons registrations. The same DUNS number must be used for all registrations, as well as on the grant application.
- [System for Award Management \(SAM\) \(https://www.sam.gov/portal/public/SAM/\)](https://www.sam.gov/portal/public/SAM/) – Applicants must complete and maintain an active registration, which requires renewal at least annually. The renewal process may require as much time as the initial registration. SAM registration includes the assignment of a Commercial and Government Entity (CAGE) Code for domestic organizations which have not already been assigned a CAGE Code.
 - [NATO Commercial and Government Entity \(NCAGE\) Code \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11176\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11176) – Foreign organizations must obtain an NCAGE code (in lieu of a CAGE code) in order to register in SAM.
- [eRA Commons \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11123\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11123) - Applicants must have an active DUNS number to register in eRA Commons. Organizations can register with the eRA Commons as they are working through their SAM or Grants.gov registration, but all registrations must be in place by time of submission. eRA Commons requires organizations to identify at least one Signing Official (SO) and at least one Program Director/Principal Investigator (PD/PI) account in order to submit an application.
- Grants.gov – Applicants must have an active DUNS number and SAM registration in order to complete the Grants.gov registration.

Program Directors/Principal Investigators (PD(s)/PI(s))

All PD(s)/PI(s) must have an eRA Commons account. PD(s)/PI(s) should work with their organizational officials to either create a new account or to affiliate their existing account with the applicant organization in eRA Commons. If the PD/PI is also the organizational Signing Official, they must have two distinct eRA Commons accounts, one for each role. Obtaining an eRA Commons account can take up to 2 weeks.

Eligible Individuals (Program Director/Principal Investigator)

Any individual(s) with the skills, knowledge, and resources necessary to carry out the proposed research as the Program Director(s)/Principal Investigator(s) (PD(s)/PI(s)) is invited to work with his/her organization to develop an application for support. Individuals from underrepresented racial and ethnic groups as well as individuals with disabilities are always encouraged to apply for NIH support.

For institutions/organizations proposing multiple PDs/PIs, visit the Multiple Program Director/Principal Investigator Policy and submission details in the Senior/Key Person Profile (Expanded) Component of the SF424 (R&R) Application Guide.

2. Cost Sharing

This FOA does not require cost sharing as defined in the [NIH Grants Policy Statement. \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11126\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11126)

3. Additional Information on Eligibility

Number of Applications

Applicant organizations may submit more than one application, provided that each application is scientifically distinct.

The NIH will not accept duplicate or highly overlapping applications under review at the same time. This means that the NIH will not accept:

- A new (A0) application that is submitted before issuance of the summary statement from the review of an overlapping new (A0) or resubmission (A1) application.
- A resubmission (A1) application that is submitted before issuance of the summary statement from the review of the previous new (A0) application.

- An application that has substantial overlap with another application pending appeal of initial peer review (see [NOT-OD-11-101](https://grants.nih.gov/grants/guide/notice-files/NOT-OD-11-101.html) (<https://grants.nih.gov/grants/guide/notice-files/NOT-OD-11-101.html>))

Section IV. Application and Submission Information

1. Requesting an Application Package

The application forms package specific to this opportunity must be accessed through ASSIST, Grants.gov Workspace or an institutional system-to-system solution. Links to apply using ASSIST or Grants.gov Workspace are available in [Part 1](#) of this FOA. See your administrative office for instructions if you plan to use an institutional system-to-system solution.

2. Content and Form of Application Submission

It is critical that applicants follow the instructions in the Research (R) Instructions in the [SF424 \(R&R\) Application Guide](https://grants.nih.gov/grants/guide/url_redirect.htm?id=12000) (https://grants.nih.gov/grants/guide/url_redirect.htm?id=12000) except where instructed in this funding opportunity announcement to do otherwise. Conformance to the requirements in the Application Guide is required and strictly enforced. Applications that are out of compliance with these instructions may be delayed or not accepted for review.

Page Limitations

All page limitations described in the SF424 Application Guide and the [Table of Page Limits](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11133) (https://grants.nih.gov/grants/guide/url_redirect.htm?id=11133) must be followed.

Instructions for Application Submission

The following section supplements the instructions found in the SF424 (R&R) Application Guide and should be used for preparing an application to this FOA.

SF424(R&R) Cover

All instructions in the SF424 (R&R) Application Guide must be followed.

SF424(R&R) Project/Performance Site Locations

All instructions in the SF424 (R&R) Application Guide must be followed.

SF424(R&R) Other Project Information

All instructions in the SF424 (R&R) Application Guide must be followed.

SF424(R&R) Senior/Key Person Profile

All instructions in the SF424 (R&R) Application Guide must be followed.

R&R or Modular Budget

All instructions in the SF424 (R&R) Application Guide must be followed.

R&R Subaward Budget

All instructions in the SF424 (R&R) Application Guide must be followed.

PHS 398 Cover Page Supplement

All instructions in the SF424 (R&R) Application Guide must be followed.

PHS 398 Research Plan

All instructions in the SF424 (R&R) Application Guide must be followed, with the following additional instructions:

Resource Sharing Plan: Individuals are required to comply with the instructions for the Resource Sharing Plans as provided in the SF424 (R&R) Application Guide.

The following modifications also apply:

Appendix:

Only limited Appendix materials are allowed. Follow all instructions for the Appendix as described in the SF424 (R&R) Application Guide.

PHS Human Subjects and Clinical Trials Information

When involving human subjects research, clinical research, and/or NIH-defined clinical trials (and when applicable, clinical trials research experience) follow all instructions for the PHS Human Subjects and Clinical Trials Information form in the SF424 (R&R) Application Guide, with the following additional instructions:

If you answered “Yes” to the question “Are Human Subjects Involved?” on the R&R Other Project Information form, you must include at least one human subjects study record using the **Study Record: PHS Human Subjects and Clinical Trials Information** form or **Delayed Onset Study** record.

Study Record: PHS Human Subjects and Clinical Trials Information

All instructions in the SF424 (R&R) Application Guide must be followed.

Delayed Onset Study

Note: Delayed onset does NOT apply to a study that can be described but will not start immediately (i.e., delayed start). All instructions in the SF424 (R&R) Application Guide must be followed.

PHS Assignment Request Form

All instructions in the SF424 (R&R) Application Guide must be followed.

Foreign Institutions

Foreign (non-U.S.) institutions must follow policies described in the [NIH Grants Policy Statement \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11137\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11137), and procedures for foreign institutions.

3. Unique Entity Identifier and System for Award Management (SAM)

See Part 1. Section III.1 for information regarding the requirement for obtaining a unique entity identifier and for completing and maintaining active registrations in System for Award Management (SAM), NATO Commercial and Government Entity (NCAGE) Code (if applicable), eRA Commons, and Grants.gov

4. Submission Dates and Times

[Part I. Overview Information](#) contains information about Key Dates and times. Applicants are encouraged to submit applications before the due date to ensure they have time to make any application corrections that might be necessary for successful submission. When a submission date falls on a weekend or [Federal holiday \(https://grants.nih.gov/grants/guide/url_redirect.html?id=82380\)](https://grants.nih.gov/grants/guide/url_redirect.html?id=82380), the application deadline is automatically extended to the next business day.

Organizations must submit applications to [Grants.gov \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11128\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11128) (the online portal to find and apply for grants across all Federal agencies). Applicants must then complete the submission process by tracking the status of the application in the [eRA Commons \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11123\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11123), NIH's electronic system for grants administration. NIH and Grants.gov systems check the application against many of the application instructions upon submission. Errors must be corrected and a changed/corrected application must be submitted to Grants.gov on or before the application due date and time. If a Changed/Corrected application is submitted after the deadline, the application will be considered late. Applications that miss the due date and time are subjected to the NIH Policy on Late Application Submission.

Applicants are responsible for viewing their application before the due date in the eRA Commons to ensure accurate and successful submission.

Information on the submission process and a definition of on-time submission are provided in the SF424 (R&R) Application Guide.

5. Intergovernmental Review (E.O. 12372)

This initiative is not subject to [intergovernmental review. \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11142\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11142)

6. Funding Restrictions

All NIH awards are subject to the terms and conditions, cost principles, and other considerations described in the [NIH Grants Policy Statement \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11120\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11120).

Pre-award costs are allowable only as described in the [NIH Grants Policy Statement \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11143\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11143).

7. Other Submission Requirements and Information

Applications must be submitted electronically following the instructions described in the SF424 (R&R) Application Guide. Paper applications will not be accepted.

Applicants must complete all required registrations before the application due date. [Section III. Eligibility Information](#) contains information about registration.

For assistance with your electronic application or for more information on the electronic submission process, visit [How to Apply – Application Guide \(https://grants.nih.gov/grants/how-to-apply-application-guide.html\)](https://grants.nih.gov/grants/how-to-apply-application-guide.html). If you encounter a system issue beyond your control that threatens your ability to complete the submission process on-time, you must follow the [Dealing with System Issues](#)

(<https://grants.nih.gov/grants/how-to-apply-application-guide/due-dates-and-submission-policies/dealing-with-system-issues.htm>) guidance. For assistance with application submission, contact the Application Submission Contacts in [Section VII](#).

Important reminders:

All PD(s)/PI(s) must include their eRA Commons ID in the Credential field of the Senior/Key Person Profile Component of the SF424(R&R) Application Package. Failure to register in the Commons and to include a valid PD/PI Commons ID in the credential field will prevent the successful submission of an electronic application to NIH. See [Section III](#) of this FOA for information on registration requirements.

The applicant organization must ensure that the DUNS number it provides on the application is the same number used in the organization's profile in the eRA Commons and for the System for Award Management. Additional information may be found in the SF424 (R&R) Application Guide.

See [more tips \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11146\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11146) for avoiding common errors.

Upon receipt, applications will be evaluated for completeness and compliance with application instructions by the Center for Scientific Review, NIH. Applications that are incomplete or non-compliant will not be reviewed.

Requests of \$500,000 or more for direct costs in any year

Applicants requesting \$500,000 or more in direct costs in any year (excluding consortium F&A) must contact a Scientific/ Research Contact at least 6 weeks before submitting the application and follow the Policy on the Acceptance for Review of Unsolicited Applications that Request \$500,000 or More in Direct Costs as described in the SF424 (R&R) Application Guide.

Post Submission Materials

Applicants are required to follow the instructions for post-submission materials, as described in [the policy \(//grants.nih.gov/grants/guide/url_redirect.htm?id=82299\)](#). Any instructions provided here are in addition to the instructions in the policy.

Section V. Application Review Information

1. Criteria

Only the review criteria described below will be considered in the review process. Applications submitted to the NIH in support of the [NIH mission \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11149\)](#) are evaluated for scientific and technical merit through the NIH peer review system.

Overall Impact

Reviewers will provide an overall impact score to reflect their assessment of the likelihood for the project to exert a sustained, powerful influence on the research field(s) involved, in consideration of the following review criteria and additional review criteria (as applicable for the project proposed).

Scored Review Criteria

Reviewers will consider each of the review criteria below in the determination of scientific merit, and give a separate score for each. An application does not need to be strong in all categories to be judged likely to have major scientific impact. For example, a project that by its nature is not innovative may be essential to advance a field.

Significance

Does the project address an important problem or a critical barrier to progress in the field? Is the prior research that serves as the key support for the proposed project rigorous? If the aims of the project are achieved, how will scientific knowledge, technical capability, and/or clinical practice be improved? How will successful completion of the aims change the concepts, methods, technologies, treatments, services, or preventative interventions that drive this field?

Investigator(s)

Are the PD(s)/PI(s), collaborators, and other researchers well suited to the project? If Early Stage Investigators or those in the early stages of independent careers, do they have appropriate experience and training? If established, have they demonstrated an ongoing record of accomplishments that have advanced their field(s)? If the project is collaborative or multi-PD/PI, do the investigators have complementary and integrated expertise; are their leadership approach, governance and organizational structure appropriate for the project?

Innovation

Does the application challenge and seek to shift current research or clinical practice paradigms by utilizing novel theoretical concepts, approaches or methodologies, instrumentation, or interventions? Are the concepts, approaches or methodologies, instrumentation, or interventions novel to one field of research or novel in a broad sense? Is a refinement, improvement, or new application of theoretical concepts, approaches or methodologies, instrumentation, or interventions proposed?

Approach

Are the overall strategy, methodology, and analyses well-reasoned and appropriate to accomplish the specific aims of the project? Have the investigators included plans to address weaknesses in the rigor of prior research that serves as the key support for the proposed project? Have the investigators presented strategies to ensure a robust and unbiased approach, as appropriate for the work proposed? Are potential problems, alternative strategies, and benchmarks for success presented? If the project is in the early stages of development, will the strategy establish feasibility and will particularly risky aspects be managed? Have the investigators presented adequate plans to address relevant biological variables, such as sex, for studies in vertebrate animals or human subjects?

If the project involves human subjects and/or NIH-defined clinical research, are the plans to address 1) the protection of human subjects from research risks, and 2) inclusion (or exclusion) of individuals on the basis of sex/gender, race, and ethnicity, as well as the inclusion or exclusion of individuals of all ages (including children and older adults), justified in terms of the scientific goals and research strategy proposed?

Environment

Will the scientific environment in which the work will be done contribute to the probability of success? Are the institutional support, equipment and other physical resources available to the investigators adequate for the project proposed? Will the project benefit from unique features of the scientific environment, subject populations, or collaborative arrangements?

Additional Review Criteria

As applicable for the project proposed, reviewers will evaluate the following additional items while determining scientific and technical merit, and in providing an overall impact score, but will not give separate scores for these items.

Protections for Human Subjects

For research that involves human subjects but does not involve one of the categories of research that are exempt under 45 CFR Part 46, the committee will evaluate the justification for involvement of human subjects and the proposed protections from research risk relating to their participation according to the following five review criteria: 1) risk to subjects, 2) adequacy of protection against risks, 3) potential benefits to the subjects and others, 4) importance of the knowledge to be gained, and 5) data and safety monitoring for clinical trials.

For research that involves human subjects and meets the criteria for one or more of the categories of research that are exempt under 45 CFR Part 46, the committee will evaluate: 1) the justification for the exemption, 2) human subjects involvement and characteristics, and 3) sources of materials. For additional information on review of the Human Subjects section, please refer to the [Guidelines for the Review of Human Subjects \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11175\)](https://grants.nih.gov/grants/guide/redirect.htm?id=11175).

Inclusion of Women, Minorities, and Individuals Across the Lifespan

When the proposed project involves human subjects and/or NIH-defined clinical research, the committee will evaluate the proposed plans for the inclusion (or exclusion) of individuals on the basis of sex/gender, race, and ethnicity, as well as the inclusion (or exclusion) of individuals of all ages (including children and older adults) to determine if it is justified in terms of the scientific goals and research strategy proposed. For additional information on review of the Inclusion section, please refer to the [Guidelines for the Review of Inclusion in Clinical Research \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11174\)](https://grants.nih.gov/grants/guide/redirect.htm?id=11174).

Vertebrate Animals

The committee will evaluate the involvement of live vertebrate animals as part of the scientific assessment according to the following criteria: (1) description of proposed procedures involving animals, including species, strains, ages, sex, and total number to be used; (2) justifications for the use of animals versus alternative models and for the appropriateness of the species proposed; (3) interventions to minimize discomfort, distress, pain and injury; and (4) justification for euthanasia method if NOT consistent with the AVMA Guidelines for the Euthanasia of Animals. Reviewers will assess the use of chimpanzees as they would any other application proposing the use of vertebrate animals. For additional information on review of the Vertebrate Animals section, please refer to the [Worksheet for Review of the Vertebrate Animal Section \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11150\)](https://grants.nih.gov/grants/guide/redirect.htm?id=11150).

Biohazards

Reviewers will assess whether materials or procedures proposed are potentially hazardous to research personnel and/or the environment, and if needed, determine whether adequate protection is proposed.

Resubmissions

For Resubmissions, the committee will evaluate the application as now presented, taking into consideration the responses to comments from the previous scientific review group and changes made to the project.

Renewals

For Renewals, the committee will consider the progress made in the last funding period.

Revisions

For Revisions, the committee will consider the appropriateness of the proposed expansion of the scope of the project. If the Revision application relates to a specific line of investigation presented in the original application that was not recommended for approval by the committee, then the committee will consider whether the responses to comments from the previous scientific review group are adequate and whether substantial changes are clearly evident.

Additional Review Considerations

As applicable for the project proposed, reviewers will consider each of the following items, but will not give scores for these items, and should not consider them in providing an overall impact score.

Applications from Foreign Organizations

Reviewers will assess whether the project presents special opportunities for furthering research programs through the use of unusual talent, resources, populations, or environmental conditions that exist in other countries and either are not readily available in the United States or augment existing U.S. resources.

Select Agent Research

Reviewers will assess the information provided in this section of the application, including 1) the Select Agent(s) to be used in the proposed research, 2) the registration status of all entities where Select Agent(s) will be used, 3) the procedures that will be used to monitor possession use and transfer of Select Agent(s), and 4) plans for appropriate biosafety, biocontainment, and security of the Select Agent(s).

Resource Sharing Plans

Reviewers will comment on whether the following Resource Sharing Plans, or the rationale for not sharing the following types of resources, are reasonable: (1) [Data Sharing Plan \(//grants.nih.gov/grants/guide/redirect.htm?id=11151\)](https://grants.nih.gov/grants/guide/redirect.htm?id=11151); (2) [Sharing Model Organisms \(//grants.nih.gov/grants/guide/redirect.htm?id=11152\)](https://grants.nih.gov/grants/guide/redirect.htm?id=11152); and (3) [Genomic Data Sharing Plan \(GDS\) \(//grants.nih.gov/grants/guide/redirect.htm?id=11153\)](https://grants.nih.gov/grants/guide/redirect.htm?id=11153).

Authentication of Key Biological and/or Chemical Resources:

For projects involving key biological and/or chemical resources, reviewers will comment on the brief plans proposed for identifying and ensuring the validity of those resources.

Budget and Period of Support

Reviewers will consider whether the budget and the requested period of support are fully justified and reasonable in relation to the proposed research.

2. Review and Selection Process

Applications will be evaluated for scientific and technical merit by (an) appropriate Scientific Review Group(s), in accordance with [NIH peer review policy and procedures \(//grants.nih.gov/grants/guide/redirect.htm?id=11154\)](https://grants.nih.gov/grants/guide/redirect.htm?id=11154), using the stated [review criteria \(file:///C:/Users/mckenziene/AppData/Local/Microsoft/Windows/INetCache/Content.Outlook/13V4QPZR/Research%20Draft.doc#_1_ Criteria\)](https://grants.nih.gov/grants/guide/redirect.htm?id=11154). Assignment to a Scientific Review Group will be shown in the eRA Commons.

As part of the scientific peer review, all applications will receive a written critique.

Applications may undergo a selection process in which only those applications deemed to have the highest scientific and technical merit (generally the top half of applications under review) will be discussed and assigned an overall impact score.

Applications will be assigned on the basis of established PHS referral guidelines to the appropriate NIH Institute or Center. Applications will compete for available funds with all other recommended applications. Following initial peer review, recommended applications will receive a second level of review by the appropriate national Advisory Council or Board. The following will be considered in making funding decisions:

- Scientific and technical merit of the proposed project as determined by scientific peer review.
- Availability of funds.
- Relevance of the proposed project to program priorities.

3. Anticipated Announcement and Award Dates

After the peer review of the application is completed, the PD/PI will be able to access his or her Summary Statement (written critique) via the [eRA Commons \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11123\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11123). Refer to Part 1 for dates for peer review, advisory council review, and earliest start date.

Information regarding the disposition of applications is available in the [NIH Grants Policy Statement \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11156\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11156).

Section VI. Award Administration Information

1. Award Notices

If the application is under consideration for funding, NIH will request "just-in-time" information from the applicant as described in the [NIH Grants Policy Statement \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11157\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11157).

A formal notification in the form of a Notice of Award (NoA) will be provided to the applicant organization for successful applications. The NoA signed by the grants management officer is the authorizing document and will be sent via email to the grantee's business official.

Awardees must comply with any funding restrictions described in [Section IV.5. Funding Restrictions](#). Selection of an application for award is not an authorization to begin performance. Any costs incurred before receipt of the NoA are at the recipient's risk. These costs may be reimbursed only to the extent considered allowable pre-award costs.

Any application awarded in response to this FOA will be subject to terms and conditions found on the [Award Conditions and Information for NIH Grants \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11158\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11158) website. This includes any recent legislation and policy applicable to awards that is highlighted on this website.

Institutional Review Board or Independent Ethics Committee Approval: Grantee institutions must ensure that all protocols are reviewed by their IRB or IEC. To help ensure the safety of participants enrolled in NIH-funded studies, the awardee must provide NIH copies of documents related to all major changes in the status of ongoing protocols.

2. Administrative and National Policy Requirements

All NIH grant and cooperative agreement awards include the [NIH Grants Policy Statement \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11120\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11120) as part of the NoA. For these terms of award, see the [NIH Grants Policy Statement Part II: Terms and Conditions of NIH Grant Awards, Subpart A: General \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11157\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11157) and [Part II: Terms and Conditions of NIH Grant Awards, Subpart B: Terms and Conditions for Specific Types of Grants, Grantees, and Activities \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11159\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11159). More information is provided at [Award Conditions and Information for NIH Grants \(//grants.nih.gov/grants/guide/url_redirect.htm?id=11158\)](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11158).

Recipients of federal financial assistance (FFA) from HHS must administer their programs in compliance with federal civil rights laws that prohibit discrimination on the basis of race, color, national origin, disability, age and, in some circumstances, religion, conscience, and sex. This includes ensuring programs are accessible to persons with limited English proficiency. The HHS Office for Civil Rights provides guidance on complying with civil rights laws enforced by HHS. Please see <https://www.hhs.gov/civil-rights/for-providers/provider-obligations/index.html> (<https://www.hhs.gov/civil-rights/for-providers/provider-obligations/index.html>) and <http://www.hhs.gov/ocr/civilrights/understanding/section1557/index.html> (<http://www.hhs.gov/ocr/civilrights/understanding/section1557/index.html>).

HHS recognizes that research projects are often limited in scope for many reasons that are nondiscriminatory, such as the principal investigator's scientific interest, funding limitations, recruitment requirements, and other considerations. Thus, criteria in research protocols that target or exclude certain populations are warranted where nondiscriminatory justifications establish that such criteria are appropriate with respect to the health or safety of the subjects, the scientific study design, or the purpose of the research. For additional guidance regarding how the provisions apply to NIH grant programs, please contact the Scientific/Research Contact that is identified in Section VII under Agency Contacts of this FOA.

- Recipients of FFA must ensure that their programs are accessible to persons with limited English proficiency. HHS provides guidance to recipients of FFA on meeting their legal obligation to take reasonable steps to provide meaningful access to their programs by persons with limited English proficiency. Please see <https://www.hhs.gov/civil-rights/for-individuals/special-topics/limited-english-proficiency/fact-sheet-guidance/index.html> (<https://www.hhs.gov/civil-rights/for-individuals/special-topics/limited-english-proficiency/fact-sheet-guidance/index.html>) and <https://www.lep.gov> (<https://www.lep.gov>). For further guidance on providing culturally and linguistically appropriate services, recipients should review the National Standards for Culturally and Linguistically Appropriate Services in Health and Health Care at <https://minorityhealth.hhs.gov/omh/browse.aspx?lvl=2&lvlid=53> (<https://minorityhealth.hhs.gov/omh/browse.aspx?lvl=2&lvlid=53>).
- Recipients of FFA also have specific legal obligations for serving qualified individuals with disabilities. Please see <http://www.hhs.gov/ocr/civilrights/understanding/disability/index.html> (<http://www.hhs.gov/ocr/civilrights/understanding/disability/index.html>).
- HHS funded health and education programs must be administered in an environment free of sexual harassment. Please see <https://www.hhs.gov/civil-rights/for-individuals/sex-discrimination/index.html> (<https://www.hhs.gov/civil-rights/for-individuals/sex-discrimination/index.html>); <https://www2.ed.gov/about/offices/list/ocr/docs/shguide.html>; and <https://www.eeoc.gov/eeoc/publications/upload/fs-sex.pdf> (<https://www.eeoc.gov/eeoc/publications/upload/fs-sex.pdf>). For information about NIH's commitment to supporting a safe and respectful work environment, who to contact with questions or concerns, and what NIH's expectations are for institutions and the individuals supported on NIH-funded awards, please see <https://grants.nih.gov/grants/policy/harassment.htm> (<https://grants.nih.gov/grants/policy/harassment.htm>).
- Recipients of FFA must also administer their programs in compliance with applicable federal religious nondiscrimination laws and applicable federal conscience protection and associated anti-discrimination laws. Collectively, these laws prohibit exclusion, adverse treatment, coercion, or other discrimination against persons or entities on the basis of their consciences, religious beliefs, or moral convictions. Please see <https://www.hhs.gov/conscience/conscience-protections/index.html> (<https://www.hhs.gov/conscience/conscience-protections/index.html>) and <https://www.hhs.gov/conscience/religious-freedom/index.html> (<https://www.hhs.gov/conscience/religious-freedom/index.html>).

Please contact the HHS Office for Civil Rights for more information about obligations and prohibitions under federal civil rights laws at <https://www.hhs.gov/ocr/about-us/contact-us/index.html> (<https://www.hhs.gov/ocr/about-us/contact-us/index.html>) or call 1-800-368-1019 or TDD 1-800-537-7697.

In accordance with the statutory provisions contained in Section 872 of the Duncan Hunter National Defense Authorization Act of Fiscal Year 2009 (Public Law 110-417), NIH awards will be subject to the Federal Awardee Performance and Integrity Information System (FAPIIS) requirements. FAPIIS requires Federal award making officials to review and consider information about an applicant in the designated integrity and performance system (currently FAPIIS) prior to making an award. An applicant, at its option, may review information in the designated integrity and performance systems accessible through FAPIIS and comment on any information about itself that a Federal agency previously entered and is currently in FAPIIS. The Federal awarding agency will consider any comments by the applicant, in addition to other information in FAPIIS, in making a judgement about the applicant's integrity, business ethics, and record of performance under Federal awards when completing the review of risk posed by applicants as described in 45 CFR Part 75.205 "Federal awarding agency review of risk posed by applicants." This provision will apply to all NIH grants and cooperative agreements except fellowships.

Cooperative Agreement Terms and Conditions of Award

Not Applicable

3. Reporting

When multiple years are involved, awardees will be required to submit the [Research Performance Progress Report \(RPPR\)](https://grants.nih.gov/grants/rppr/index.htm) (<https://grants.nih.gov/grants/rppr/index.htm>) annually and financial statements as required in the [NIH Grants Policy Statement](https://grants.nih.gov/grants/guide/uri_redirect.htm?id=11161). (https://grants.nih.gov/grants/guide/uri_redirect.htm?id=11161)

A final RPPR, invention statement, and the expenditure data portion of the Federal Financial Report are required for closeout of an award, as described in the [NIH Grants Policy Statement](https://grants.nih.gov/grants/guide/uri_redirect.htm?id=11161) (https://grants.nih.gov/grants/guide/uri_redirect.htm?id=11161).

The Federal Funding Accountability and Transparency Act of 2006 (Transparency Act), includes a requirement for awardees of Federal grants to report information about first-tier subawards and executive compensation under Federal assistance awards issued in FY2011 or later. All awardees of applicable NIH grants and cooperative agreements are required to report to the Federal Subaward Reporting System (FSRS) available at www.fsrs.gov (https://grants.nih.gov/grants/guide/uri_redirect.htm?id=11170) on all subawards over \$25,000. See the [NIH Grants Policy Statement](https://grants.nih.gov/grants/guide/uri_redirect.htm?id=11171) (https://grants.nih.gov/grants/guide/uri_redirect.htm?id=11171) for additional information on this reporting requirement.

In accordance with the regulatory requirements provided at 45 CFR 75.113 and Appendix XII to 45 CFR Part 75, recipients that have currently active Federal grants, cooperative agreements, and procurement contracts from all Federal awarding agencies with a

cumulative total value greater than \$10,000,000 for any period of time during the period of performance of a Federal award, must report and maintain the currency of information reported in the System for Award Management (SAM) about civil, criminal, and administrative proceedings in connection with the award or performance of a Federal award that reached final disposition within the most recent five-year period. The recipient must also make semiannual disclosures regarding such proceedings. Proceedings information will be made publicly available in the designated integrity and performance system (currently FAPIIS). This is a statutory requirement under section 872 of Public Law 110-417, as amended (41 U.S.C. 2313). As required by section 3010 of Public Law 111-212, all information posted in the designated integrity and performance system on or after April 15, 2011, except past performance reviews required for Federal procurement contracts, will be publicly available. Full reporting requirements and procedures are found in Appendix XII to 45 CFR Part 75 – Award Term and Conditions for Recipient Integrity and Performance Matters.

Section VII. Agency Contacts

We encourage inquiries concerning this funding opportunity and welcome the opportunity to answer questions from potential applicants.

Application Submission Contacts

eRA Service Desk (Questions regarding ASSIST, eRA Commons, application errors and warnings, documenting system problems that threatens submission by the due date, and post-submission issues)

Finding Help Online: <http://grants.nih.gov/support/> ([//grants.nih.gov/support/](http://grants.nih.gov/support/)) (preferred method of contact)

Telephone: 301-402-7469 or 866-504-9552 (Toll Free)

General Grants Information (Questions regarding application instructions, application processes, and NIH grant resources)

Email: GrantsInfo@nih.gov (<mailto:GrantsInfo@nih.gov>) (preferred method of contact)

Telephone: 301-945-7573

Grants.gov Customer Support (Questions regarding Grants.gov registration and Workspace)

Contact Center Telephone: 800-518-4726

Email: support@grants.gov (<mailto:support@grants.gov>)

Scientific/Research Contact(s)

Participating NIH Institutes and Centers are listed in “Components of Participating Organizations” in [Part 1. Overview](https://grants.nih.gov/grants/guide/pa-files/PA-16-160.html#Components_of_Participating) (https://grants.nih.gov/grants/guide/pa-files/PA-16-160.html#Components_of_Participating). Scientific/Research Contact information is listed on the [R01 IC-Specific Scientific Interests and Contact](https://grants.nih.gov/grants/guide/contacts/parent-R01-CT-NotAllowed.html) (<https://grants.nih.gov/grants/guide/contacts/parent-R01-CT-NotAllowed.html>) [website](https://grants.nih.gov/grants/guide/contacts/parent-R01-CT-NotAllowed.html)

Peer Review Contact(s)

Examine your eRA Commons account for review assignment and contact information (information appears two weeks after the submission due date).

Financial/Grants Management Contact(s)

Participating NIH Institutes and Centers are listed in “Components of Participating Organizations” in [Part 1. Overview](https://grants.nih.gov/grants/guide/pa-files/PA-16-160.html#Components_of_Participating) (https://grants.nih.gov/grants/guide/pa-files/PA-16-160.html#Components_of_Participating). Financial/Grants Management Contact information is listed on the [R01 IC-Specific Scientific Interests and Contact](https://grants.nih.gov/grants/guide/contacts/parent-R01-CT-NotAllowed.html) (<https://grants.nih.gov/grants/guide/contacts/parent-R01-CT-NotAllowed.html>) [website](https://grants.nih.gov/grants/guide/contacts/parent-R01-CT-NotAllowed.html)

Section VIII. Other Information

Recently issued trans-NIH [policy notices](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11163) ([//grants.nih.gov/grants/guide/url_redirect.htm?id=11163](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11163)) may affect your application submission. A full list of policy notices published by NIH is provided in the [NIH Guide for Grants and Contracts](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11164) ([//grants.nih.gov/grants/guide/url_redirect.htm?id=11164](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11164)). All awards are subject to the terms and conditions, cost principles, and other considerations described in the [NIH Grants Policy Statement](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11120) ([//grants.nih.gov/grants/guide/url_redirect.htm?id=11120](https://grants.nih.gov/grants/guide/url_redirect.htm?id=11120)).

Authority and Regulations

Awards are made under the authorization of Sections 301 and 405 of the Public Health Service Act as amended (42 USC 241 and 284) and under Federal Regulations 42 CFR Part 52 and 45 CFR Part 75.

[Weekly TOC for this Announcement](https://grants.nih.gov/grants/guide/WeeklyIndex.cfm?05-08-20) ([/grants/guide/WeeklyIndex.cfm?05-08-20](https://grants.nih.gov/grants/guide/WeeklyIndex.cfm?05-08-20))

[NIH Funding Opportunities and Notices](https://grants.nih.gov/grants/guide/index.html) ([/grants/guide/index.html](https://grants.nih.gov/grants/guide/index.html))



(<http://www.hhs.gov/>)

Department of Health
and Human Services (HHS)



(<http://www.usa.gov/>)

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Note: For help accessing PDF, RTF, MS Word, Excel, PowerPoint, Audio or Video files, see [Help Downloading Files \(/grants/edocs.htm\)](#).